Numerical analysis of superconducting phases in the extended Hubbard model with non-local pairing

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Abstract

[To be continued...]

- Appendix on s-wave SC ghost simulations on HM
- Appendix on triplet pairing in MFT self-consistent scheme
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List of symbols and abbreviations

AF	Antiferromagnetic
BCS	Bardeen-Cooper-Schrieffer (theory)
BZ	Brillouin zone
β	Inverse temperature
$\Delta^{(\gamma)}$	γ-wave component of the gap function
DoF	Degree of freedom
EHM	Extended Hubbard model
EV(s)	Expectation value(s)
FS	Fermi surface
HF	Hartree-Fock
HFP(s)	Hartree-Fock parameter(s)
HM	Hubbard model
L_ℓ	Number of lattice sites in direction ℓ
iHF	Iterative Hartree-Fock
LRT	Linear Response Theory
MBZ	Magnetic Brillouin zone
MFT	Mean-Field Theory
NBZ	Nematic Brillouin zone
NN(s)	Nearest neighbor(s)
PH	Particle-hole
PHS	Particle-hole symmetry
RWC(s)	Relevant Wick's contraction(s)
RMP(s)	Renormalized model parameter(s)
SC	Superconductor
SDW	Spin-density wave
SSB	Spontaneous symmetry breaking
T_{c}	Critical temperature

Introduction

This thesis project is about my favorite ice cream flavor. [To be continued. . .]

Chapter 1

Introduction to Hartree-Fock methods

The Hartree-Fock (HF) method is a variational approach to determine the ground-state wavefunction of a many-body hamiltonian. The wavefunction is determined the one minimising energy, limiting search to a specific subspace of states. The general idea is to approximate the real hamiltonian with the best non-interacting fermionic hamiltonian.

To start, we need to define what Hilbert subspace we work with. Then we need to identify a way of parametrising the possible non-interacting hamiltonians and their respective ground-states, and finally a method to choose the best approximation to our original model. This is done in Sec. 1.2, largely based on [1, 2].

Variational algorithms generally work by numeric minimisation of some given functional. This is not the case. We implement HF theory starting from a theoretical minimisation procedure, with some derivative being set to zero, and then prove that this procedure leads to solving a self-consistency problem. We end up with a set of equations where the right-hand side depends non-trivially on the left-hand side, and through an iterative approach we determine the solution. This procedure is sketched in detail in Sec. 1.3.

1.1 Prerequisites and generalities

Let us start by discussing the properties of the many-body states we use to approximate the real ground-state of the system. We introduce Slater determinants to grasp the general idea behind HF variational method, and then extend to a larger set of states.

1.1.1 Slater determinants

A many-body wavefunction describing a collective electronic state needs to be antisymmetric under interchange of coordinates and spins $(\mathbf{r}_i, \sigma_i)$, $(\mathbf{r}_j, \sigma_j)$ of any couple of electrons. The simplest way to build a many-body wavefunction satisfying this rule is "to antisymmetrize" a set of orthonormal N one-body spin-orbitals, being N the total number of electrons and D the total number of spin-orbitals. Let $\{\psi_i\}$ be said set,

$$\frac{1}{2\mathcal{V}} \sum_{\sigma \in \{\uparrow,\downarrow\}} \int \mathrm{d}\mathbf{r}\ \psi_i(\mathbf{r},\sigma)\psi_j(\mathbf{r},\sigma) = \delta_{ij}$$

being $\mathcal{V}$ the total volume.

Let S_N be the $N!$ elements symmetric group of permutations of N objects and $\mathrm{P} \in \mathrm{S}_N$ a specific permutation of a given N-uple,

$$\mathrm{P} \quad : \quad (x_1, x_2, \cdots, x_N) \to \left(x_{\mathrm{P}(1)}, x_{\mathrm{P}(2)}, \cdots, x_{\mathrm{P}(N)}\right)$$

Let F_{P} be the operator that implements the permutation P on the N variables of a function f,

$$F_{\mathrm{P}}\left[f(x_1, x_2, \cdots, x_N)\right] = f\left(x_{\mathrm{P}(1)}, x_{\mathrm{P}(2)}, \cdots, x_{\mathrm{P}(N)}\right)$$

We define the antisymmetrization operator:

$$\mathcal{A}[f] \equiv \frac{1}{\sqrt{N!}} \sum_{\mathrm{P}\in \mathrm{S}_N} (-1)^{\sigma(\mathrm{P})} F_{\mathrm{P}}[f]$$

being $\sigma(\mathrm{P})$ the "parity factor" of the permutation $\mathrm{P} \in \mathcal{S}_N$. The factor $\sigma(\mathrm{P})$ counts the number of exchanges needed to obtain P: this way, an even number of exchanges gives a total sign $+1$, an odd number gives -1. By construction, the function $\mathcal{A}[f]$ is antisymmetric under exchange of a pair of coordinates.

Consider now the many-body wavefunction

$$\Psi(\mathbf{r}_1, \sigma_1; \mathbf{r}_2, \sigma_2; \cdots ; \mathbf{r}_N, \sigma_N) = \mathcal{A}\left[\prod_{i=1}^{N} \psi_i(\mathbf{r}_i, \sigma_i)\right]$$

This class of wavefunctions, obtained as antisymmetric combinations of product states, are representable as so-called Slater determinants,

$$\Psi(\mathbf{r}_1, \sigma_1; \mathbf{r}_2, \sigma_2; \cdots ; \mathbf{r}_N, \sigma_N) = \frac{1}{\sqrt{N!}} \det \begin{bmatrix} \psi_1(\mathbf{r}_1, \sigma_1) & \psi_1(\mathbf{r}_2, \sigma_2) & \cdots & \psi_1(\mathbf{r}_N, \sigma_N) \\ \psi_2(\mathbf{r}_1, \sigma_1) & \psi_2(\mathbf{r}_2, \sigma_2) & \cdots & \psi_2(\mathbf{r}_N, \sigma_N) \\ \vdots & \vdots & & \vdots \\ \psi_N(\mathbf{r}_1, \sigma_1) & \psi_N(\mathbf{r}_2, \sigma_2) & \cdots & \psi_N(\mathbf{r}_N, \sigma_N) \end{bmatrix}$$

This representation includes automatically Pauli exclusion principle: determinants of matrices with two equal rows or two equal columns are null. Thus, all spin-orbitals must differ, as well as no single spin-orbital ψ_i can be doubly occupied (which is, all spin-coordinate couples must be different).

Any normalised combinations of Slater determinants gives an antisymmetric wavefunction, although not necessarily representable as a single Slater determinant. In HF theory, we search for the best possible choice of orthonormal one-body spin orbitals $\{\psi_1, \psi_2, \cdots, \psi_N\}$ to approximate the true ground-state of the model through a single Slater determinant. The space of possible "Slater-like" many-body wavefunctions is the parametric space where we perform the variational constrained search.

The EHM is formulated in second-quantisation. It is then useful to represent Slater-like states in this language. Let us map all the spin-orbitals from the discussion above to pairs of annihilation-creation fermionic operators,

$$\psi_i \quad \rightarrow \quad \hat{c}_i, \hat{c}_i^\dagger$$

Any Slater state can be expressed in second-quantisation as:

$$|\Psi\rangle = \prod_{i=1}^{N} \hat{c}_i^\dagger |\Omega\rangle \tag{1.1}$$

being $|\Omega\rangle$ the vacuum. Antisymmetry is guaranteed by Fermi anticommutation rules.

1.1.2 Wick's theorem applied to Slater determinants

The concept of vacuum state needs to be interpreted in relation to an appropriate choice of second-quantisation annihilation and creation operators. For a given set of fermion operators, the vacuum is that state mapped to zero by any annihilation operator. As we will see, to correctly define the vacuum state is crucial in order to apply Wick's theorem and develop an HF approximation of EHM. Before moving to generalisation, let us discuss a playground setup.

Consider some unspecified model whose ground state at null temperature is given in Eq. (1.1) for a given set of electronic operators $\hat{c}_i$, $\hat{c}_i^\dagger$. Let $\hat{c}_i$ with $N < i < D$ identify all spin-orbitals not already included in the definition of $|\Psi\rangle$ of Eq. (1.1) (thus, non populated). Then it holds:

$$\begin{aligned} 0 \leq i \leq N \qquad & \hat{c}_i^\dagger |\Psi\rangle = 0 \\ i > N \qquad & \hat{c}_i |\Psi\rangle = 0 \end{aligned}$$

Let:

$$\hat{a}_i \equiv \begin{cases} \hat{c}_i^\dagger & \text{for } 0 \leq i \leq N \\ \hat{c}_i & \text{for } i > N \end{cases}$$

These new operators satisfy a Fermi algebra $\{\hat{a}_i, \hat{a}_j^\dagger\} = \delta_{ij}$ and annihilate the "vacuum" state $|\Psi\rangle$. In other words, any Slater state like Eq. (1.1) can be seen as a vacuum state for a suitably chosen set of fermionic operators.

This property is handy for evaluation of expectation values (EV). Let $\hat{O}^{(\ell)}$ be a linear combination of $\hat{c}_i$ or $\hat{c}_i^\dagger$,

$$\hat{O}^{(\ell)} \equiv \sum_{i=1}^{D} \left(u_i^{(\ell)} \hat{c}_i + v_i^{(\ell)} \hat{c}_i^\dagger \right) \qquad u_i^{(\ell)}, v_i^{(\ell)} \in \mathbb{C} \tag{1.2}$$

Consider the product operator:

$$\hat{\mathcal{O}} \equiv \hat{O}^{(1)} \hat{O}^{(2)} \cdots \hat{O}^{(2k)} \qquad k \in \mathbb{N}$$

Inserting the above decomposition, we get a large sum of many terms, each containing some sequence of annihilations and creations. Using linearity we can concentrate on one specific term of these, derive results and then recompose them for the original operator. Consider one of these terms,

$$\hat{\mathcal{O}} = \cdots + (\text{coefficient}) \times \hat{C}_1 \hat{C}_2 \cdots \hat{C}_{2k} + \cdots$$

with $\hat{C}_i$ some fermionic operator, either $\hat{c}_j$ or $\hat{c}_j^\dagger$ for some j. Now, Wick's theorem states that

$$\begin{aligned} \hat{C}_1 \hat{C}_2 \cdots \hat{C}_{2k} = \mathrm{N} &\left\{ \hat{C}_1 \hat{C}_2 \cdots \hat{C}_{2k} \right\} \\ &+ \sum_{(\alpha\beta)} (-1)^{\zeta_{\alpha\beta}} D_{\alpha\beta} \mathrm{N} \left\{ \prod_{i \neq \alpha,\beta} \hat{C}_i \right\} \\ &+ \sum_{(\alpha\beta)(\gamma\delta)} (-1)^{\zeta_{\alpha\beta}} D_{\alpha\beta} (-1)^{\zeta_{\gamma\delta}} D_{\gamma\delta} \mathrm{N} \left\{ \prod_{i \neq \alpha,\beta,\gamma,\delta} \hat{C}_i \right\} \\ &+ \cdots \end{aligned}$$

where $\mathrm{N}\{\cdots\}$ indicates normal ordering, $D_{\alpha\beta}$ is the contraction of the pair $\hat{C}_\alpha \hat{C}_\beta$ and $\zeta_{\alpha\beta}$ is a symmetry factor included to take care of the number of fermionic algebra. In this context contractions are just expectation values over $|\Psi\rangle$. Sums are performed over pairs of indices, implicitly taken to be different from each other. To take averages of this last operator over $|\Psi\rangle$ is a not-so-hard task, given the simple expression of the state in terms of electronic annihilations and creations. However it can be made simpler: if we change basis and express all operators in terms of $\hat{a}$, $\hat{a}^\dagger$,

$$\hat{C}_i \quad \rightarrow \quad \hat{A}_i$$

the above decomposition holds identically, with slight change of notation. Now, taking a zero temperature average on $|\Psi\rangle$ (the vacuum with respect to the new operators) all normal-ordered terms disappear,

$$\langle \Psi | \mathrm{N}\{\hat{A}_i \hat{A}_j \cdots\} | \Psi \rangle = 0$$

thus in the decomposition above only fully contracted terms survive. Going back to $\hat{c}$ operators, we can re-express fully contracted terms in the original basis. Thus:

$$\left\langle \hat{C}_1 \hat{C}_2 \cdots \hat{C}_{2k} \right\rangle = \sum (\text{total sign factor}) \times (\text{product of complete contractions})$$

where sum is intended, implicitly, over all possible ways of choosing k couples of indices from a starting set of $2k$ elements.

Taking another step back, due to linearity we can apply this result to all terms of $\mathcal{O}$ and conclude that the EV of $\mathcal{O}$ decomposes as a sum of 2-points EVs. **This is the defining property**

of gaussian states. In this procedure, the only key property of Slater states we used was the fact that they are vacuum to some fermionic basis. Finite-temperature extension of Wick's theorem as well as more complicated change of basis are possible, but the concept remains the same: on a Slater state, which is essentially a non-interacting state for some set of fermions, Wick's theorem is particularly well-behaved.

From the point above we can conclude that:

$$\langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta \rangle = \langle \hat{c}^\dagger_\alpha \hat{c}_\delta \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\gamma \rangle - \langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle \langle \hat{c}^\dagger_\gamma \hat{c}_\delta \rangle$$

being the average intended on a Slater state. In this context, only contractions with an equal number of creation and annihilation operators contribute.

1.1.3 Generalisation to Bogoliubov-Valatin states

In HF theory we look for a good approximation of the true many-body wavefunction constraining search to a class of states. Introducing Slater determinants was useful to grasp the concrete idea behind the method. **However, in order to treat some physical phenomena we need to extend the variational procedure to a other sets of states. For example, Bardeen-Cooper-Shrieffer (BCS) theory of superconductivity gives a many-body ground-state which is not representable as a Slater determinant of "standard" electronic operators.** In this section we extend the notions introduced before and set grounds for the generalised Hartree-Fock-Bogoliubov (HFB) method we use in this thesis.

Consider the Nambu spinor

$$\hat{\Phi} \equiv \begin{bmatrix} \hat{\mathbf{c}} \\ \hat{\mathbf{c}}^\dagger \end{bmatrix}$$

being $\hat{\mathbf{c}}$ the vector of all annihilation operators, ordered, and $\hat{\mathbf{c}}^\dagger$ of all creation operators. Let us assume the vectors have D entries. Consider an unitary map M, such that

$$\hat{\Gamma} \equiv M\hat{\Phi} \quad \text{being} \quad \hat{\Gamma} \equiv \begin{bmatrix} \hat{\boldsymbol{\gamma}} \\ \hat{\boldsymbol{\gamma}}^\dagger \end{bmatrix}$$

The new Nambu vectors $\hat{\boldsymbol{\gamma}}$ and $\hat{\boldsymbol{\gamma}}^\dagger$ contain our new, generalised fermionic operators in second quantisation. Let:

$$M \equiv \begin{bmatrix} U & V \\ U^\dagger & V^\dagger \end{bmatrix}$$

If M is unitary, it follows:

$$\begin{aligned} \mathbb{I}_{2D\times 2D} &= MM^\dagger \\ &= \begin{bmatrix} U & V \\ U^\dagger & V^\dagger \end{bmatrix} \begin{bmatrix} U & U^\dagger \\ V^\dagger & V \end{bmatrix} \\ &= \begin{bmatrix} UU^\dagger + VV^\dagger & U^2 + V^2 \\ (U^\dagger)^2 + (V^\dagger)^2 & U^\dagger U + V^\dagger V \end{bmatrix} \end{aligned}$$

thus, among other conditions, $UU^\dagger + VV^\dagger = \mathbb{I}_{D\times D}$.

The new operators in $\hat{\Gamma}$ satisfy Fermi anticommutation rules. In fact, writing the mapping explicitly for any of the new operators and some index $i \in [1, D]$,

$$\hat{\gamma}_i \equiv \sum_{j=1}^{D} U_{ij}\hat{c}_j + \sum_{j=1}^{D} V_{ij}\hat{c}^\dagger_j \qquad \hat{\gamma}^\dagger_i \equiv \sum_{j=1}^{D} U^*_{ji}\hat{c}^\dagger_j + \sum_{j=1}^{D} V^*_{ji}\hat{c}_j$$

and substituting in the anti-commutator

$$\begin{aligned}\{\hat{\gamma}_i, \hat{\gamma}_j^\dagger\} &= \sum_{i'=1}^{D} U_{ii'} \sum_{j'=1}^{D} U^*_{j'j} \{\hat{c}_{i'}, \hat{c}^\dagger_{j'}\} + \sum_{i'=1}^{D} V_{ii'} \sum_{j'=1}^{D} V^*_{j'j} \{\hat{c}^\dagger_{i'}, \hat{c}_{j'}\} \\ &= \sum_{i'=1}^{D} U_{ii'} \sum_{j'=1}^{D} U^*_{j'j} \delta_{i'j'} + \sum_{i'=1}^{D} V_{ii'} \sum_{j'=1}^{D} V^*_{j'j} \delta_{i'j'} \\ &= \left[UU^\dagger + VV^\dagger\right]_{ij} \\ &= \delta_{ij}\end{aligned}$$

being M unitary.

Assume now $|\Psi\rangle$ to be the vacuum state for the new operators,

$$\forall i \quad : \quad \hat{\gamma}_i |\Psi\rangle = 0$$

Suppose we want to evaluate, with identical meaning of the notation as in previous section, something like

$$\left\langle \hat{C}_1 \hat{C}_2 \cdots \hat{C}_{2k} \right\rangle$$

We can apply the result obtained above. Indeed, inverse maps linking old and new operators read:

$$\hat{c}_i \equiv \sum_{j=1}^{D} U_{ij} \hat{\gamma}_j + \sum_{j=1}^{D} U^*_{ji} \hat{\gamma}^\dagger_j \qquad \hat{c}^\dagger_i \equiv \sum_{j=1}^{D} V^*_{ji} \hat{\gamma}^\dagger_j + \sum_{j=1}^{D} V_{ij} \hat{\gamma}_j$$

In Sec. 1.1.2 we argued that, when averaging a product of operators like the one of Eq. (1.2), only 2-points EVs contribute. This time the idea is the same, substituting $\mathcal{O}^{(\ell)} \to \hat{C}_\ell$ and $\hat{c}_i, \hat{c}^\dagger_i \to \hat{\gamma}_i, \hat{\gamma}^\dagger_i$. Thus, also here we can decompose the product operator, apply Wick's theorem, discard all non-fully-contracted terms and recompose the results. All we used was just the unitary mapping between $\hat{\Phi}$ and $\hat{\Gamma}$, the fact that $\gamma_i, \hat{\gamma}^\dagger_i$ obey Fermi anticommutation rules and "see" $|\Psi\rangle$ as a vacuum state.

As in Sec. 1.1.2, we conclude that:

$$\langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta \rangle = \underbrace{\langle \hat{c}^\dagger_\alpha \hat{c}_\delta \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\gamma \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle \langle \hat{c}^\dagger_\gamma \hat{c}_\delta \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle \langle \hat{c}_\gamma \hat{c}_\delta \rangle}_{\text{Cooper}} \tag{1.3}$$

The subscripts indicate the name we will give to these terms throughout the text. Note that, differently from "pure Slater states" EVs with two annihilations or two creations do not vanish, in general.

All of this holds when averaging is performed on a state, $|\Psi\rangle$, which is vacuum for a set of operators linked to fermionic Nambu spinor $\hat{\Phi}$ by some unitary operation. This set of states is the domain of our variational constrained search. This kind of unitary mapping is often referred to as Bogoliubov-Valatin transformation, and the method hereby presented as Hartree-Fock-Bogoliubov (HFB). For simplicity we will stick, from now on, to the name "HF method" referring to this discussion. **This argument has been carried out at zero temperature, but can be extended to finite temperature using appropriate path-integral techniques**.

1.2 Sketch of HF variational method

We have identified the features of the states we use to approximate the model ground-state. Now, let us get to the core of HF method and give a technical introduction on how it works. Consider the generic quartic hamiltonian:

$$\hat{H} = \sum_{\alpha,\beta} A_{\alpha\beta} \hat{c}^\dagger_\alpha \hat{c}_\beta + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta} \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta \tag{1.4}$$

where

$$A_{\alpha\beta} = A^*_{\beta\alpha} \qquad B_{\alpha\beta\gamma\delta} = B^*_{\delta\gamma\beta\alpha} \tag{1.5}$$

to ensure hermiticity. Moreover, the model must show the following symmetry:

$$B_{\alpha\beta\gamma\delta} = B_{\beta\alpha\delta\gamma} \tag{1.6}$$

because exchanging the position of the two annihilations and the two creations, the operator must remain the same.

Our aim is to approximate the above hamiltonian with something like:

$$\hat{H} \simeq \hat{H}_{\text{HF}} \equiv E_0 + \sum_{\alpha,\beta} k_{\alpha\beta}\hat{\gamma}^\dagger_\alpha \hat{\gamma}_\beta \tag{1.7}$$

Old and new operators must be linked by an unitary map: with identical notation as in previous sections, $\hat{\Gamma} = M\hat{\Phi}$. To find the best approximation to the original hamiltonian means to identify:

- the offset E_0;
- the fermionic operators $\hat{\gamma}_\alpha$, $\hat{\gamma}^\dagger_\alpha$;
- the $k_{\alpha\beta}$ coefficients.

Note that, in terms of algebraic solution, E_0 is irrelevant. To determine it correctly is only relevant in order to get the correct approximation of the actual ground-state energy. Let us discuss briefly the features of $\hat{H}_{\text{HF}}$ and then propose a solution.

1.2.1 Properties of the target quadratic hamiltonian

In this short section we enumerate some useful properties of $\hat{H}_{\text{HF}}$. We will use also the following notation:

$$\hat{a}^-_\alpha \equiv \hat{c}_\alpha \qquad \hat{a}^+_\alpha \equiv \hat{c}^\dagger_\alpha$$

Due to linearity of the unitary mapping, the operator $\hat{H}_{\text{HF}}$ of Eq. (1.7) can also be expressed as the most general quadratic hamiltonians for electronic operators:

$$\hat{H}_{\text{HF}} = E_0 + \sum_{\alpha,\beta}\sum_{p,q} h^{pq}_{\alpha\beta}\hat{a}^p_\alpha \hat{a}^q_\beta \tag{1.8}$$

By changing basis from $\hat{\gamma}$ to $\hat{c}$ operators, we moved all our ignorance into the new $h^{pq}_{\alpha\beta}$ fields.

We can infer some mathematical properties of the fields $h^{pq}_{\alpha\beta}$. First, note that the hamiltonian can be represented through Nambu spinors as

$$\begin{aligned}\hat{H}_{\text{HF}} &= E_0 + \hat{\Phi}^\dagger \mathbf{h}\hat{\Phi} \\ &= E_0 + \begin{bmatrix}\hat{\mathbf{c}} \\ \hat{\mathbf{c}}^\dagger\end{bmatrix}^\dagger \begin{bmatrix} h^{+-} & h^{++} \\ h^{--} & h^{-+}\end{bmatrix}\begin{bmatrix}\hat{\mathbf{c}} \\ \hat{\mathbf{c}}^\dagger\end{bmatrix}\end{aligned}$$

Our hamiltonian must be hermitian. We conclude that:

$$\hat{H}^\dagger_{\text{HF}} = \hat{H}_{\text{HF}} \qquad \Longrightarrow \qquad h^{++} = \left[h^{--}\right]^\dagger$$

which reads, element-wise:

$$h^{++}_{\alpha\beta} = \left(h^{--}_{\beta\alpha}\right)^* \tag{1.9}$$

Same-sign off-diagonal fields are linked.

We can set, without loss of generality,

$$\operatorname{Tr} h^{+-} + \operatorname{Tr} h^{-+} = 0$$

This passage is guaranteed by the fact that if $\mathbf{h}$ is not traceless, we can redefine it by subtracting its trace and including the latter in the energy shift E_0. Then

$$\begin{aligned}
\hat{H}_{\mathrm{HF}} &= E_0 + \sum_{\alpha,\beta}\left[h^{+-}_{\alpha\beta}\hat{c}^\dagger_\alpha\hat{c}_\beta + h^{-+}_{\alpha\beta}\hat{c}_\alpha\hat{c}^\dagger_\beta\right] + (\text{same sign terms}) \\
&= E_0 + \sum_{\alpha,\beta}\left[h^{+-}_{\alpha\beta}\left(\delta_{\alpha\beta} - \hat{c}_\beta\hat{c}^\dagger_\alpha\right) + h^{-+}_{\alpha\beta}\left(\delta_{\alpha\beta} - \hat{c}^\dagger_\beta\hat{c}_\alpha\right)\right] + (\text{same sign terms}) \\
&= E_0 - \sum_{\alpha,\beta}\left[h^{+-}_{\beta\alpha}\hat{c}_\alpha\hat{c}^\dagger_\beta + h^{-+}_{\beta\alpha}\hat{c}^\dagger_\alpha\hat{c}_\beta\right] + (\text{same sign terms})
\end{aligned}$$

In second row we got rid of the traces sum and exchanged indices $\alpha \leftrightarrow \beta$. Confronting last and first row we conclude:

$$h^{+-}_{\beta\alpha} = -h^{-+}_{\alpha\beta} \tag{1.10}$$

thus, also on-diagonal fields are linked. We are allowed to determine, say, just h^{+-}: its counterpart h^{-+} is completely determined. These properties were all we needed to get to the main argument of the method.

1.2.2 Self-consistent variational solution

We follow and extend the variational scheme proposed by Fabrizio [2]. Let:

$$\Delta^{pq}_{\alpha\beta} \equiv \left\langle \hat{a}^p_\alpha \hat{a}^q_\beta \right\rangle$$

thus, explicitly

$$\Delta^{+-}_{\alpha\beta} = \left\langle \hat{c}^\dagger_\alpha\hat{c}_\beta\right\rangle \qquad \Delta^{-+}_{\alpha\beta} = \left\langle \hat{c}_\alpha\hat{c}^\dagger_\beta\right\rangle \qquad \Delta^{++}_{\alpha\beta} = \left\langle \hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\right\rangle \qquad \Delta^{--}_{\alpha\beta} = \left\langle \hat{c}_\alpha\hat{c}_\beta\right\rangle \tag{1.11}$$

The first two are "normal densities"; the second two are "anomalous densities". Conjugation of the last two gives a symmetry relation similar to Eq. (1.9):

$$\left(\Delta^{++}_{\alpha\beta}\right)^\dagger = \Delta^{--}_{\beta\alpha} \tag{1.12}$$

The HF ground-state energy is given by:

$$\begin{aligned}
E_{\mathrm{HF}} &= \left\langle \hat{H}_{\mathrm{HF}} \right\rangle \\
&= E_0 + \sum_{\alpha,\beta}\sum_{p,q} h^{pq}_{\alpha\beta}\Delta^{pq}_{\alpha\beta}
\end{aligned}$$

Here $\Delta^{pq}_{\alpha\beta}$ are functions of the fields $h^{pq}_{\alpha\beta}$. Derive energy:

$$\begin{aligned}
\frac{\partial E_{\mathrm{HF}}}{\partial h^{rs}_{\mu\nu}} &= \frac{\partial}{\partial h^{rs}_{\mu\nu}}\sum_{\alpha,\beta}\sum_{p,q} h^{pq}_{\alpha\beta}\Delta^{pq}_{\alpha\beta} \\
&= \Delta^{rs}_{\mu\nu} + \sum_{\alpha,\beta}\sum_{p,q} h^{pq}_{\alpha\beta}\frac{\partial \Delta^{pq}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}}
\end{aligned}$$

Note that by construction E_{HF} is greater than the actual ground-state energy, since our approximated wavefunction is not the actual ground-state wavefunction. Now, from Hellman-Feynman theorem [1], we know that for a parametric hamiltonian it holds:

$$\begin{aligned}
\frac{\partial E_{\mathrm{HF}}}{\partial h^{rs}_{\mu\nu}} &= \langle\Psi|\frac{\partial \hat{H}_{\mathrm{HF}}}{\partial h^{rs}_{\mu\nu}}|\Psi\rangle \\
&= \langle\Psi|\frac{\partial}{\partial h^{rs}_{\mu\nu}}\sum_{\alpha,\beta}\sum_{p,q} h^{pq}_{\alpha\beta}\hat{a}^p_\alpha\hat{a}^q_\beta|\Psi\rangle \\
&= \left\langle \hat{a}^r_\mu\hat{a}^s_\nu\right\rangle \\
&= \Delta^{rs}_{\mu\nu}
\end{aligned}$$

The last two results imply:

$$\sum_{\alpha,\beta}\sum_{p,q} h^{pq}_{\alpha\beta}\frac{\partial \Delta^{pq}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} = 0 \tag{1.13}$$

This must hold for any possible choice of fields $h^{pq}_{\alpha\beta}$. Note that we did not impose the energy derivative to be zero: it is not the HF energy we are minimising.

Let us get back to the actual hamiltonian $\hat{H}$. To reduce it to a quadratic form is equivalent to assume that its ground-state has the properties described in previous sections. Then the following decomposition must hold:

$$\begin{aligned}
E &= \langle \hat{H} \rangle \\
&= \sum_{\alpha,\beta} A_{\alpha\beta}\langle \hat{c}^\dagger_\alpha \hat{c}_\beta\rangle + \frac{1}{2}\sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta}\langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta\rangle \\
&= \sum_{\alpha,\beta} A_{\alpha\beta}\langle \hat{a}^+_\alpha \hat{a}^-_\beta\rangle + \frac{1}{2}\sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta}\left(\langle \hat{a}^+_\alpha \hat{a}^+_\beta\rangle\langle \hat{a}^-_\gamma \hat{a}^-_\delta\rangle - \langle \hat{a}^+_\alpha \hat{a}^-_\gamma\rangle\langle \hat{a}^+_\beta \hat{a}^-_\delta\rangle + \langle \hat{a}^+_\alpha \hat{a}^-_\delta\rangle\langle \hat{a}^+_\beta \hat{a}^-_\gamma\rangle\right) \\
&= \sum_{\alpha,\beta} A_{\alpha\beta}\Delta^{+-}_{\alpha\beta} + \frac{1}{2}\sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta}\left(\Delta^{++}_{\alpha\beta}\Delta^{--}_{\gamma\delta} - \Delta^{+-}_{\alpha\gamma}\Delta^{+-}_{\beta\delta} + \Delta^{+-}_{\alpha\delta}\Delta^{+-}_{\beta\gamma}\right) \\
&= \sum_{\alpha,\beta} A_{\alpha\beta}\Delta^{+-}_{\alpha\beta} + \frac{1}{2}\sum_{\alpha,\beta,\gamma,\delta} \left(B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta}\right)\Delta^{+-}_{\alpha\beta}\Delta^{+-}_{\gamma\delta} + \frac{1}{2}\sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta}\Delta^{++}_{\alpha\beta}\Delta^{--}_{\gamma\delta}
\end{aligned}$$

In second passage we inserted the the relations (1.11). Take the derivative:

$$\begin{aligned}
\frac{\partial E}{\partial h^{rs}_{\mu\nu}} = \sum_{\alpha,\beta} A_{\alpha\beta}\frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} &+ \frac{1}{2}\sum_{\alpha,\beta,\gamma,\delta}\left(B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta}\right)\left(\frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}}\Delta^{+-}_{\gamma\delta} + \Delta^{+-}_{\alpha\beta}\frac{\partial \Delta^{+-}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}}\right) \\
&+ \frac{1}{2}\sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta}\left(\frac{\partial \Delta^{++}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}}\Delta^{--}_{\gamma\delta} + \Delta^{++}_{\alpha\beta}\frac{\partial \Delta^{--}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}}\right)
\end{aligned}$$

Using Eq. (1.6) we end up with:

$$\begin{aligned}
\frac{\partial E}{\partial h^{rs}_{\mu\nu}} = \sum_{\alpha,\beta}&\left[A_{\alpha\beta} + \sum_{\gamma,\delta}\left(B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta}\right)\Delta^{+-}_{\gamma\delta}\right]\frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} \\
&+ \sum_{\alpha,\beta,\gamma,\delta}\frac{B_{\alpha\beta\gamma\delta}\Delta^{--}_{\gamma\delta}}{2}\frac{\partial \Delta^{++}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} + \sum_{\alpha,\beta,\gamma,\delta}\frac{B_{\alpha\beta\gamma\delta}\Delta^{++}_{\alpha\beta}}{2}\frac{\partial \Delta^{--}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}}
\end{aligned}$$

Let now:

$$C_{\alpha\beta} \equiv \frac{1}{2}\left[A_{\alpha\beta} + \sum_{\gamma,\delta}\left(B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta}\right)\Delta^{+-}_{\gamma\delta}\right] \qquad D_{\alpha\beta} \equiv \sum_{\gamma,\delta}\frac{B_{\alpha\beta\gamma\delta}\Delta^{--}_{\gamma\delta}}{2} \tag{1.14}$$

From Fermi anticommutation rules, we have:

$$\Delta^{+-}_{\alpha\beta} = \delta_{\alpha\beta} - \Delta^{-+}_{\beta\alpha} \qquad \Longrightarrow \qquad \frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} = -\frac{\partial \Delta^{-+}_{\beta\alpha}}{\partial h^{rs}_{\mu\nu}}$$

which implies:

$$\begin{aligned}
\sum_{\alpha,\beta} 2C_{\alpha\beta}\frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} &= \sum_{\alpha,\beta}\left(C_{\alpha\beta}\frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} + C_{\alpha\beta}\frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}}\right) \\
&= \sum_{\alpha,\beta}\left(C_{\alpha\beta}\frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} - C_{\beta\alpha}\frac{\partial \Delta^{-+}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}}\right)
\end{aligned}$$

Moreover, using Eqns. (1.5), (1.12),

$$\sum_{\alpha,\beta} \frac{B_{\alpha\beta\gamma\delta}\Delta^{++}_{\alpha\beta}}{2} = \sum_{\alpha,\beta} \left(\frac{B_{\delta\gamma\beta\alpha}\Delta^{--}_{\beta\alpha}}{2} \right)^*$$

Using these two relations, we conclude:

$$\sum_{\alpha,\beta,\gamma,\delta} \frac{B_{\alpha\beta\gamma\delta}\Delta^{++}_{\alpha\beta}}{2} \frac{\partial \Delta^{--}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}} = \sum_{\gamma,\delta} D^*_{\delta\gamma} \frac{\partial \Delta^{--}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}}$$

Recollect everything and impose the solution to be a stationary point of energy,

$$\frac{\partial E}{\partial h^{rs}_{\mu\nu}} \overset{!}{=} 0$$

The notation "$\overset{!}{=}$" indicates, throughout the thesis, that we are imposing some condition. In the end we have:

$$\frac{\partial E}{\partial h^{rs}_{\mu\nu}} = \sum_{\alpha,\beta} \left(C_{\alpha\beta} \frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} - C_{\beta\alpha} \frac{\partial \Delta^{-+}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} + D_{\alpha\beta} \frac{\partial \Delta^{++}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} + D^*_{\beta\alpha} \frac{\partial \Delta^{--}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} \right) \overset{!}{=} 0 \tag{1.15}$$

Now compare Eqns. (1.13) and (1.15). We recognise $h^{+-}_{\alpha\beta} = C_{\alpha\beta}$ and $h^{++}_{\alpha\beta} = D_{\alpha\beta}$; the remaining fields are determined by the symmetry relations (1.10) and (1.9). Explicitly, the fields are self-consistently determined by the following two equations:

$$h^{+-}_{\alpha\beta} = \frac{1}{2} \left[A_{\alpha\beta} + \sum_{\gamma,\delta} \left(B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta} \right) \Delta^{+-}_{\gamma\delta} \right] \tag{1.16}$$

$$h^{++}_{\alpha\beta} = \sum_{\gamma,\delta} \frac{B_{\alpha\beta\gamma\delta}\Delta^{--}_{\gamma\delta}}{2} \tag{1.17}$$

The right-hand side of both equations depends on the fields themselves: $\Delta^{pq}_{\alpha\beta}$ are defined as expectation values of two-points fermionic operators over the ground-state $|\Psi\rangle$. For a given set of fields $h^{pq}_{\alpha\beta}$, the latter is determined (up to some unitary transformation) as the ground-state of a fermionic non-interacting model. In other words, being $\mathbf{h}$ the tensor containing all fields,

$$\Delta^{pq}_{\alpha\beta}[\mathbf{h}] = \langle \Psi[\mathbf{h}] | \hat{a}^p_\alpha \hat{a}^q_\beta | \Psi[\mathbf{h}] \rangle \tag{1.18}$$

This set of equations defines our computational problem. We are searching a set of parameters $\{\Delta^{pq}_{\alpha\beta}\}$ such that, determining the fields via Eqns. (1.16), (1.17), then Eq. (1.18) is respected. **The great advantage of this scheme is that, with an appropriate choice of the Hilbert space basis (for example, moving to reciprocal space for a translationally-invariant model), we can implement easily model symmetries.**

The list of parameters $\{\Delta^{pq}_{\alpha\beta}\}$ to be determined self-consistently may be not-so-short: by looking at Eq. (1.18), the indices α, β, p, q are spanning a large Hilbert space. However, this equation can be manipulated and transformed, reducing the set of parameters to be determined self-consistently to a very manageable number. For each phase we define these last as "Hartree-Fock parameters" (HFPs): a set of objects self-consistently determined by a set equivalent, but transformed, version of Eq. (1.18).

1.2.3 Connection with mean-field theories

This short last section concludes the theoretical introduction to HF methods, and bridges the gap between HF approaches and mean-field theory (MFT). **In general, in MFT approaches we substitute the two-body (or more) interactions of the hamiltonian with a single-body interaction with a mean field, determined locally via self-consistency equations.**

This is, for example, the Weiß mean-field solution to the quantum Ising model. The HF method we sketched here does not this. In the context of Hubbard models, the analogous of Weiß MFT would be dynamical mean-field theory (DMFT) [8].

However, there are some points of connection. Let us sketch first the general scheme behind MFT. Suppose we have some operator $\hat{A} = \hat{B}\hat{C}$. Let

$$\delta\hat{O} \equiv \hat{O} - \langle\hat{O}\rangle \qquad \hat{O} \in \{\hat{A}, \hat{B}, \hat{C}\}$$

Then we have:

$$\begin{aligned}
\hat{A} &= \left(\langle\hat{B}\rangle + \delta\hat{B}\right)\left(\langle\hat{C}\rangle + \delta\hat{C}\right) \\
&= \langle\hat{B}\rangle\langle\hat{C}\rangle + \langle\hat{B}\rangle\delta\hat{C} + \delta\hat{B}\langle\hat{C}\rangle + \delta\hat{B}\delta\hat{C} \\
&= \langle\hat{B}\rangle\langle\hat{C}\rangle + \langle\hat{B}\rangle\left(\hat{C} - \langle\hat{C}\rangle\right) + \left(\hat{B} - \langle\hat{B}\rangle\right)\langle\hat{C}\rangle + \delta\hat{B}\delta\hat{C} \\
&= -\langle\hat{B}\rangle\langle\hat{C}\rangle + \langle\hat{B}\rangle\hat{C} + \hat{B}\langle\hat{C}\rangle + \delta\hat{B}\delta\hat{C}
\end{aligned}$$

Up to now, this relation is exact. MFT relies on the idea that the last term, of order $\mathcal{O}(\delta^2)$, can be neglected. This amounts to say that the operator fluctuations about its average are "small". This gives the approximation:

$$\hat{A} \simeq \langle\hat{B}\rangle\hat{C} + \hat{B}\langle\hat{C}\rangle - \langle\hat{B}\rangle\langle\hat{C}\rangle$$

Within the assumption that independently $\hat{B}$ and $\hat{C}$ fluctuations are "small", we are able to simplify $\hat{A}$ with a sum of three terms: the product of each operator with the averaged value of the other, minus the product of the averages.

In the context of HF methods, we are in an slightly different situation. We are not making assumptions on the fluctuations of the quartic sectors of the hamiltonian. We are constraining the space of wavefunctions and minimising energy there. Now, Eqns. (1.16), (1.17), although general and correct, are a little unpractical to use. To derive them was useful in order to prove that the self-consistent method is a variational search in disguise. We here sketch a more concrete method to obtain the approximated hamiltonian and connect it with MFT interpretation. What we derive here is a mere reformulation of the self-consistent result obtained above, presented in a somewhat lighter form.

Let us restart from the interacting hamiltonian of Eq. (1.4). In this thesis we deal with the EHM, thus (limitedly to this section) let us restrict to the two-body generic quartic interaction:

$$\hat{V} = \sum_{\alpha,\beta} V_{\alpha\beta}\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\hat{c}_\beta\hat{c}_\alpha$$

We work with HF states. Thus Eq. (1.3) holds exactly (adapted for two indices instead of four)

$$\langle\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\hat{c}_\beta\hat{c}_\alpha\rangle = \underbrace{\langle\hat{c}^\dagger_\alpha\hat{c}_\alpha\rangle\langle\hat{c}^\beta_\beta\hat{c}_\beta\rangle}_{\text{Hartree}} - \underbrace{\langle\hat{c}^\dagger_\alpha\hat{c}_\beta\rangle\langle\hat{c}^\dagger_\beta\hat{c}_\alpha\rangle}_{\text{Fock}} + \underbrace{\langle\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\rangle\langle\hat{c}_\beta\hat{c}_\alpha\rangle}_{\text{Cooper}}$$

This gives:

$$\begin{aligned}
\langle\hat{V}\rangle &= \sum_{\alpha,\beta} V_{\alpha\beta}\langle\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\hat{c}_\beta\hat{c}_\alpha\rangle \\
&= \sum_{\alpha,\beta} V_{\alpha\beta}\left(\langle\hat{c}^\dagger_\alpha\hat{c}_\alpha\rangle\langle\hat{c}^\dagger_\beta\hat{c}_\beta\rangle - \langle\hat{c}^\dagger_\alpha\hat{c}_\beta\rangle\langle\hat{c}^\dagger_\beta\hat{c}_\alpha\rangle + \langle\hat{c}^\dagger_\alpha\hat{c}_\beta\rangle\langle\hat{c}^\dagger_\beta\hat{c}_\alpha\rangle\right)
\end{aligned}$$

from which we have the equations:

$$\frac{\partial\langle\hat{V}\rangle}{\partial\langle\hat{c}^\dagger_\alpha\hat{c}_\alpha\rangle} = \sum_\beta\langle\hat{c}^\dagger_\beta\hat{c}_\beta\rangle \qquad \frac{\partial\langle\hat{V}\rangle}{\partial\langle\hat{c}^\dagger_\alpha\hat{c}_\beta\rangle} = \sum_\beta\langle\hat{c}^\dagger_\beta\hat{c}_\alpha\rangle \qquad \eta_{\alpha\beta} = \frac{\partial\langle\hat{V}\rangle}{\partial\langle\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\rangle}$$

Now, our aim is to approximate $\hat{V}$ as:

$$\hat{V} \simeq \sum_{\alpha\beta}\left[\lambda_\alpha\hat{c}^\dagger_\alpha\hat{c}_\alpha + \mu_{\alpha\beta}\hat{c}^\dagger_\alpha\hat{c}_\beta + \eta_{\alpha\beta}\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta + \text{h.c.}\right] + (\text{some constant})$$

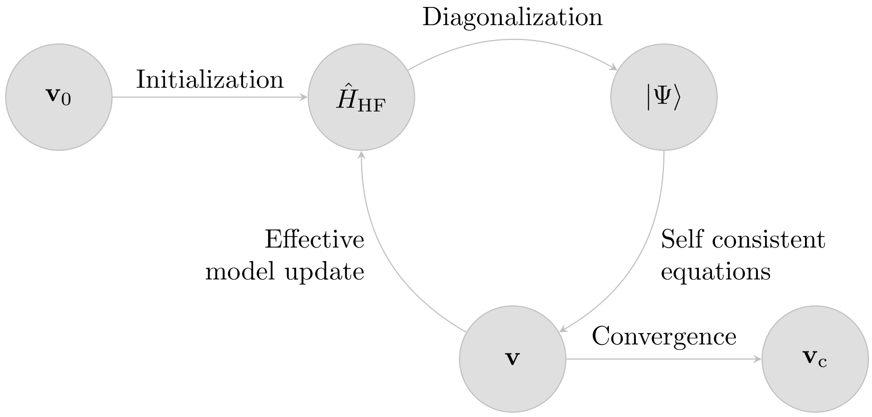


Figure 1.1 Schematisation of the self-consistent Hartree Fock method described in Sec. 1.3. The effective model is solved, the solution is used to estimate self-consistently the Hartree Fock vector $\mathbf{v}$, which is then used to update iteratively the model itself. The notation $\mathbf{v}_0$ indicates the initialiser HFPs vector, $\mathbf{v}_\mathrm{c}$ the converged HFPs vector.

Thus:

$$\langle \hat{V} \rangle \simeq \sum_{\alpha\beta} \left[\lambda_\alpha \langle \hat{c}^\dagger_\alpha \hat{c}_\alpha \rangle + \mu_{\alpha\beta} \langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle + \eta_{\alpha\beta} \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle \right]$$

It holds:

$$\lambda_\alpha = \frac{\partial \langle \hat{V} \rangle}{\partial \langle \hat{c}^\dagger_\alpha \hat{c}_\alpha \rangle} \qquad \mu_{\alpha\beta} = \frac{\partial \langle \hat{V} \rangle}{\partial \langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle} \qquad \eta_{\alpha\beta} = \frac{\partial \langle \hat{V} \rangle}{\partial \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle}$$

1.3 Sketch of the iHF algorithm

In this section we give a practical sketch of the algorithm we use to compute Eq. (1.18) for all given phases. We refer to this technique as “iterative Hartree Fock” (iHF). Let $\mathbf{v}$ be the vector containing all HFPs. Suppose we have N HFPs v_i: then

$$\mathbf{v} \in \mathbb{C}^N \qquad \mathbf{v} \equiv \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_N \end{bmatrix}$$

In order to get a physical solution, each HFP must satisfy a self-consistency equation similar to Eq. (1.18). Each equation has the structure:

$$v_i = (\text{Some non-linear function of } \mathbf{v})$$

Let A be the non-linear operator collecting all these functions. The problem of finding a self-consistent solution can be reformulated as follows: $\mathbf{v}$ represents a physical solution if it is a fixed point of the map A,

$$\mathrm{A} \colon \mathbb{C}^N \to \mathbb{C}^N \qquad \mathrm{A}(\mathbf{v}) = \mathbf{v} \quad \Longrightarrow \quad \mathbf{v} \text{ is a solution.} \tag{1.19}$$

We implement a rather easy scheme: first, we obtain an analytic expression for A. Then we choose an initialiser $\mathbf{v}_0$ and apply A to it, obtaining a new HFPs vector. Iterating this procedure, we “follow” the non-linear map A until we claim to have converged. Let us explain in detail:

1. Start with a generic quartic hamiltonian:

$$\hat{H} = \sum_{\alpha\beta} A_{\alpha\beta} \hat{c}^\dagger_\alpha \hat{c}_\beta + \sum_{\alpha\beta\gamma\delta} B_{\alpha\beta\gamma\delta} \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta$$

Choose the model parameters and set an electronic density;

2. Following the discussion of Sec. 1.2, approximate analytically $\hat{H}$ with a quadratic hamiltonian

$$\hat{H} \simeq \hat{H}_{\mathrm{HF}} \equiv \sum_{\alpha\beta} \left[h^{+-}_{\alpha\beta} \hat{c}^{\dagger}_{\alpha} \hat{c}_{\beta} - h^{+-}_{\beta\alpha} \hat{c}_{\alpha} \hat{c}^{\dagger}_{\beta} + h^{++}_{\alpha\beta} \hat{c}^{\dagger}_{\alpha} \hat{c}^{\dagger}_{\beta} + \left(h^{++}_{\beta\alpha} \right)^{*} \hat{c}_{\alpha} \hat{c}_{\beta} \right]$$

with the fields given by Eqns. (1.16), (1.17);

3. Impose selection rules on the fields $h^{+-}_{\alpha\beta}$, $h^{++}_{\alpha\beta}$, eventually eliminating those not respecting the symmetry sector we are working in. As an example: if we choose not to break total charge conservation, set $\forall \alpha, \beta \colon h^{++}_{\alpha\beta} = 0$;

4. Choose a starting value for the HFPs vector $\mathbf{v}_0$;

5. Set $\mathbf{v} = \mathbf{v}_0$ and enter the iterative scheme of Fig. 1.1:
    a) Get a numeric approximation of the hamiltonian for the current value of $\mathbf{v}$, $\hat{H}_{\mathrm{HF}}[\mathbf{v}]$;
    b) Determine the value $\mu[\mathbf{v}]$ giving the desired density;
    c) Diagonalise the effective hamiltonian $\hat{H}_{\mathrm{HF}}[\mathbf{v}] - \mu[\mathbf{v}]\hat{N}$, and obtain the approximated ground-state $|\Psi[\mathbf{v}]\rangle$;
    d) Compute $\mathrm{A}(\mathbf{v})$ using self-consistency equations;
    e) If the algorithm converged, exit; otherwise, update the value of $\mathbf{v}$ repeat from point a).

We voluntarily left unspecified three details:

- How is the chemical potential μ determined?
- What criterion do we use to claim that the algorithm "converged"?
- How is the HFPs vector $\mathbf{v}$ updated?

The sections that follows deal with these questions.

1.3.1 Fixed density simulations and the role of μ

In this thesis all derivations and simulations are carried out at fixed density. This requires a way to fix μ during iterations of the algorithm. This must be done iteratively: changing at each step the HFPs vector, we are actively modifying the approximate hamiltonian. Chemical potential must be determined accordingly to maintain the required density.

For a given choice of HFPs vector and chemical potential, the ground state $|\Psi[\mathbf{v}, \mu]\rangle$ is uniquely determined; then we compute density as

$$n(\mathbf{v}, \mu) \equiv \frac{1}{\mathcal{V}} \sum_{\alpha} \langle \Psi[\mathbf{v}, \mu] | \hat{c}^{\dagger}_{\alpha} \hat{c}_{\alpha} | \Psi[\mathbf{v}, \mu] \rangle$$

being $\mathcal{V}$ the volume. Suppose we fix density to a value $n_0 \in [0, 1]$. At step i, naming the current HFPs vector $\mathbf{v}_i$, μ_i (the chemical potential at current step) is determined as

$$\mu_i \equiv \min_{\mu} |n(\mathbf{v}_i, \mu) - n_0| \tag{1.20}$$

The density n must be a monotonically increasing function of μ: then we can translate the above equation, which would inspire a minimisation procedure, to a much simpler node-search rule:

$$\mu_i = \mu \,|_{\delta n(\mathbf{v}_i, \mu) = 0} \quad \text{being} \quad \delta n(\mathbf{v}_i, \mu) \equiv n(\mathbf{v}_i, \mu) - n_0$$

In order to where the function changes sign (which is, the position of μ_i) we use a basic bisection algorithm:

1. Set an arbitrary tolerance $\Delta n > 0$ and a shifting parameter $\Delta\mu > 0$;

2. Choose starting lower and upper bounds $\mu_{\mathrm{L}} < \mu_{\mathrm{U}}$;

3. Compute $\delta n(\mathbf{v}_i, \mu_\mathrm{L})$. Evaluate:
   - if $|\delta n(\mathbf{v}_i, \mu_\mathrm{L})| \leq \Delta n$, set $\mu_i = \mu_\mathrm{L}$ and halt;
   - if $\delta n(\mathbf{v}_i, \mu_\mathrm{L}) < -\Delta n$ do nothing;
   - if $\delta n(\mathbf{v}_i, \mu_\mathrm{L}) > \Delta n$, shift the lower bound down, $\mu_\mathrm{L} \to \mu_\mathrm{L} - \Delta\mu$ and repeat from step 3;
4. Compute $\delta n(\mathbf{v}_i, \mu_\mathrm{U})$. Evaluate:
   - if $|\delta n(\mathbf{v}_i, \mu_\mathrm{U})| \leq \Delta n$, set $\mu_i = \mu_\mathrm{U}$ and halt;
   - if $\delta n(\mathbf{v}_i, \mu_\mathrm{U}) > \Delta n$ do nothing;
   - if $\delta n(\mathbf{v}_i, \mu_\mathrm{U}) < -\Delta n$, shift the upper bound up, $\mu_\mathrm{U} \to \mu_\mathrm{U} + \Delta\mu$ and repeat from step 4;
5. At this point, the the lower bound underestimates density while the upper bound overestimates. Start iterations:
   a) Compute the midpoint,
   $$\mu_\mathrm{M} = \frac{\mu_\mathrm{L} + \mu_\mathrm{U}}{2}$$
   b) Evaluate:
      - If $|\delta n(\mathbf{v}_i, \mu_\mathrm{M})| \leq \Delta n$, set $\mu_i = \mu_\mathrm{M}$ and halt;
      - If $\delta n(\mathbf{v}_i, \mu_\mathrm{M}) > \Delta n$, set $\mu_\mathrm{U} = \mu_\mathrm{M}$ and repeat from point a);
      - If $\delta n(\mathbf{v}_i, \mu_\mathrm{M}) < -\Delta n$, set $\mu_\mathrm{L} = \mu_\mathrm{M}$ and repeat from point a);

The bottleneck of the procedure is our ability to compute expectation values. This depends on how much we are able to simplify analytically the computations for each phase.

As will turn out in the derivation, chemical potential will be renormalised by interaction. This means that applying self-consistent HF methods to the EHM will produce an effective hamiltonian containing a shift to the chemical potential,

$$\mu \to \mu + F[\mathbf{v}] \equiv \tilde{\mu}[\mathbf{v}]$$

being F some function of the HFPs. The chemical potential μ is an example of Renormalised Model Parameter (RMP). As we will see, also the hopping t will turn out to be renormalised by interactions.

Note that at step i of the iHF algorithm, our bisection procedure determines $\tilde{\mu}_i$, not μ_i. This consideration is relevant when dealing with ground state energy estimation. Let us abstract from the EHM and iHF, and consider a generic hamiltonian $\hat{H}$. To find the ground state $|\Psi\rangle$ means precisely to solve

$$\frac{\delta}{\delta \langle\Psi|} \langle\Psi|\hat{H}|\Psi\rangle = 0$$

Now, working with a fixed particles number N means we are performing a constrained minimisation (which is, taking the functional derivative) on the composite operator

$$\hat{H} - \mu(\hat{N} - N) \equiv \hat{K} + \mu N$$

being μ the Lagrange multiplier, $\hat{N}$ the number operator and $\hat{K} \equiv \hat{H} - \mu\hat{N}$. Finding the constrained ground state parametrised by a vector $\mathbf{v}$ now means to solve

$$\mathbf{v} \implies |\Psi[\mathbf{v}]\rangle \quad : \quad \begin{cases} \left.\dfrac{\delta}{\delta \langle\Psi|} \langle\Psi|\hat{K} + \mu N|\Psi\rangle \right|_{|\Psi\rangle = |\Psi[\mathbf{v}]\rangle} = 0 \\ \langle\Psi[\mathbf{v}]|\hat{N}|\Psi[\mathbf{v}]\rangle = N \end{cases}$$

In the first line μN is derived to zero; the constraint is imposed in the second line. Energy is given by:

$$\begin{aligned} \langle\hat{H}\rangle &= \langle\Psi[\mathbf{v}]|\hat{K}|\Psi[\mathbf{v}]\rangle + \mu N \\ &= \Omega[\mathbf{v}] + (\tilde{\mu}[\mathbf{v}] - F[\mathbf{v}])N \end{aligned}$$

being Ω the expectation value (EV) of $\hat{K}$. To correctly compute energy, we need to add $+\mu N$; since we determine $\tilde{\mu}[\mathbf{v}] \neq \mu$, everything needs to be expressed in terms of it. Using this relation, when the algorithm has converged, we can estimate ground-state energy.

1.3.2 The reject-mix-iterate convergence scheme

Let us discuss briefly the convergence criterion of our algorithm and the iteration mechanism. The iHF scheme we design is a sequential loop that runs up to a logical halt condition. In order to specify such condition, we need to define our notion of "convergence". Recall the definition of $\mathrm{A}(\mathbf{v})$, the non-linear map in HFPs space whose component are given by a set of self-consistency equations: we claim to have reached convergence when $\mathbf{v}$ comes "close enough" to a fixed point of the map A, as in Eq. (1.19). Explicitly, we stop whenever the following logic condition is satisfied for each component of $\mathbf{v} = (v_1, v_2, \cdots)$:

$$\forall v_j \quad : \quad |v_j - \mathrm{A}_j(\mathbf{v})| \leq \Delta v_j$$

being $\mathrm{A}(\mathbf{v})$ the map image of the HFP vector $\mathbf{v}$, and A_j its j-th component. The tolerance Δv_j is arbitrary. If just one component of $\mathbf{v}$ is mapped "too far" (with respect to Δv_j) the step is rejected and iterations continue. In this text we deal in with quantities of order 10^{-1}, so convergence values of order $10^{-3} \div 10^{-4}$ will be in general sufficient.

At step i, we check the distance of the two vectors $\mathrm{A}(\mathbf{v}_i)$ and $\mathbf{v}_i$ and if possible, halt. If instead convergence is not reached, we need repeat the cycle with a new initialiser $\mathbf{v}_{i+1}$. A very simple choice would be to set $\mathbf{v}_{i+1} = \mathbf{v}_i$. We do something slightly different:

$$\mathbf{v}_{i+1} \equiv g\mathrm{A}\left(\mathbf{v}_i\right) + (1-g)\mathbf{v}_i \tag{1.21}$$

The "mixing parameter" g is a simple but useful knob to tune algorithmic evolution. Its usefulness comes from this observation: even if a fixed point exists, we don't know how easy it is to actually get there following A. Setting $g = 1$, the new initialiser is just the image of the previous initialiser. By moving "slower" ($g < 1$) with an accurate optimisation of the mixing, it is possible to reach convergence in a controlled way.

Let us give some concrete justification about the relevance of controlling g. When it comes to HF iterative schemes of this kind, it can sometimes happen for the algorithm to remain stuck in infinite loops. An example of this behaviour is given in Sec. 3.6.3. A typical situation is the following: during algorithm evolution we reach a couple of points $\mathbf{v}^{(a)}$, $\mathbf{v}^{(b)}$ such that

$$\cdots \xrightarrow{\mathrm{A},g} \mathbf{v}^{(a)} \xrightarrow{\mathrm{A},g} \mathbf{v}^{(b)} \xrightarrow{\mathrm{A},g} \mathbf{v}^{(a)} \xrightarrow{\mathrm{A},g} \mathbf{v}^{(b)} \xrightarrow{\mathrm{A},g} \cdots$$

With the notation "$\xrightarrow{\mathrm{A},g}$" we intend one algorithmic step at mixing parameter g. We are stuck forever in this loop. Now, to predict if couples like $(\mathbf{v}^{(a)}, \mathbf{v}^{(b)})$ exist, how they depend on g, where they are located in HFPs space... is impossible at this level. Thus there is no immediate way to avoid these loops *a priori*. One workaround we found, however, was to just make g smaller. We gained empirical evidence that to "slow down" in HFPs space makes numeric instabilities less frequent. The trade-off, however, is that if we move too slow, runtime could get very large. Note that inaccurate setup of the algorithm could lead to resources-intensive runs also in "ordinary" situations, when we are not stuck in a loop.

To keep the number of iterations under control, we also defined an upper bound p to the total number of steps. If after p steps we algorithm failed to converge, it halts. In that case, $\mathbf{v}$ is saved as a "`NaN`" (Not a number) vector and we move to another simulation. Introducing this simple rule, runtime is controlled and we gain an easy method to identify the regions of parameter space requiring more care – just read from the raw data where `NaN`s have been saved and repeat simulations fine-tuning the parameters. This basically means to lower g and increase p. These considerations end this technical introductory chapter on iHF methods.

Chapter 2

Phase transitions and SSB in EHM

The general idea of this project is to apply HF methods to the EHM, derive analytically the approximated quadratic hamiltonian of Eq. (1.7) and then apply the iHF procedure sketched in Sec. 1.3 to extract the self-consistent values of the various HFPs. One of the advantages of using iHF methods is that we can impose rather simply the symmetries of the HF approximated wavefunction.

This is most useful when dealing with competing phases. A phase transition is basically described by some spontaneous symmetry breaking (SSB) and the rising to a non-zero value of an order parameter. We will deal with three phases:

1. the normal phase, where no symmetry is broken;

2. the antiferromagnetic phase, where we allow for spin density to be non-uniform across sites;

3. the superconducting phase, where we do not conserve particle number.

Each phase is discussed in a separate chapter. Many other "phases" are possible, based on which symmetries are respected and which are not.

For each phase, we will extract the best approximation to the true ground state wavefunction. In computational terms, we will find three solutions to the general self-consistency problem of Eq. (1.18). In the end we want to compare the free energy of these three solutions, assessing which one is the most stable. Any HF approximation, being by construction not the true ground state, overestimates energy. Thus, comparing energies of our solutions we get a hint of the features of the true ground state.

2.1 The EHM symmetries

In this section we present the symmetries of the extended Hubbard model introduced in Eq. (**??**)

$$\hat{H}_{\text{EHM}} = \underbrace{-t\sum_{\langle ij\rangle}\sum_{\sigma}\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}}_{\hat{H}_t} \underbrace{+U\sum_{i}\hat{n}_{i\uparrow}\hat{n}_{i\downarrow}}_{\hat{H}_U} \underbrace{-V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{n}_{i\sigma}\hat{n}_{j\sigma'}}_{\hat{H}_V}$$

The symmetries of the model can be divided in two groups: the ones related to the geometry of the square lattice, and those related to the algebraic properties of $\hat{H}_{\text{EHM}}$. We start by the lattice symmetries.

2.1.1 Lattice symmetries

We study the EHM on the planar square lattice, with one orbital per site. We indicate with L_x the number of sites in direction x, and L_y in direction y. The hamiltonian of Eq. (**??**) respects three fundamental lattice symmetries:

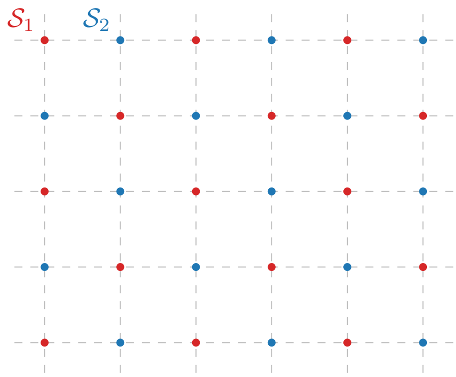


Figure 2.1 | Square lattice $\mathcal{L}$ bipartition in two sublattices $\mathcal{S}_1$ (red), $\mathcal{S}_2$ (blue).

1. Discrete translations symmetry in both the x and y directions. We refer to the groups of one-site translations as as $\mathrm{T}_1^{(x)}$, $\mathrm{T}_1^{(x)}$. It holds:

$$\mathrm{T}_n^{(\ell)} \supset \mathrm{T}_{kn}^{(\ell)} \qquad k \in \mathbb{N} \tag{2.1}$$

   A shift in direction ℓ by kn sites can be decomposed in k shift by n sites. Invariance by n-sites shifts, thus, implies invariance also for kn-sites shifts.

2. Discrete $\pi/2$ rotations symmetry. This symmetry is described by the four-fold cyclic point group C_4, containing rotations by angle $\pi/2, \pi, 3\pi/2$ and identity. A similar property as for the translations holds for cyclic point groups,

$$\mathrm{C}_n \supset \mathrm{C}_{n/k} \qquad k, n/k \in \mathbb{N} \tag{2.2}$$

   Invariance for rotations of angle $2\pi/n$ implies invariance for rotations of angle $k \times 2\pi/n$.

3. Inversion symmetry. We define I_2 as the inversion group in two-dimensional space, containing the reflection operator and identity. This point may appear redundant, because

$$\mathrm{I}_2 \equiv \mathrm{C}_2 \tag{2.3}$$

   In 2D space, to rotate by π around the origin and to mirror the position of a point are the same operation.

The planar square lattice is bipartite. Let us give a definition. Consider a graph $\mathcal{L}$ and the set of its vertices $V(\mathcal{L}) = \{v_i\}$. Let $E(\mathcal{L})$ be the set of graph edges,

$$E(\mathcal{L}) = \{(v, w) \mid v, w \in V(\mathcal{L}) \land v,\ w \text{ are connected}\}$$

The graph is said to be bipartite if two sub-graphs $\mathcal{S}_1$, $\mathcal{S}_2$ exist such that:

- The vertices of $\mathcal{L}$ split in the two sub-graphs,

$$V(\mathcal{L}) = V(\mathcal{S}_1) \cup V(\mathcal{S}_2)$$

- Vertices of one sub-graph are linked only to vertices of the other sub-graph. In other words

$$v, w \in \mathcal{S}_i \quad \Longrightarrow \quad (v, w) \notin E(\mathcal{L})$$

The planar square lattice is bipartite: it can be divided in two square lattices, as represented in Fig. 2.1. Two sites are "connected" if linked by a non-zero hopping operator. Which is, if they are NNs. Note that one-site translations leave the hamiltonian invariant but exchange the two sublattices, while two-sites translations don't.

2.1.2 Spin and charge symmetries

The conventional HM shows a global U(2) symmetry, extended to $\mathrm{SU}(2) \times \mathrm{SU}(2)/\mathbb{Z}_2 = \mathrm{SO}(4)$ symmetry at half-filling on a bipartite lattice [12], such as the square lattice. For the EHM, we limit to prove that we can identify a $\mathrm{U}(2) \subset \mathrm{SO}(4)$ global symmetry.

Gauge symmetry. The EHM is symmetric under the global transformation

$$\hat{c}_{i\sigma} \to e^{-i\eta}\hat{c}_{i\sigma} \qquad \hat{c}^\dagger_{i\sigma} \to e^{i\eta}\hat{c}^\dagger_{i\sigma} \qquad \eta \in \mathbb{R} \tag{2.4}$$

To prove this, it suffices to note that

$$\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma'} \quad \to \quad e^{i\eta}\hat{c}^\dagger_{i\sigma}e^{-i\eta}\hat{c}_{j\sigma'} = \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma'} \tag{2.5}$$

As a consequence, any product of fermionic operators with an equal number of annihilations and creations is symmetric. Hence the kinetic part $\hat{H}_t$ and the interactions $\hat{H}_U$, $\hat{H}_V$ are invariant. As a consequence, the total number of particles (charge)

$$\hat{N} = \sum_{i\sigma} \hat{c}^\dagger_{i\sigma}\hat{c}_{i\sigma}$$

is a good quantum number. The symmetry is described by the transformation of Eq. (2.4), which is a pure phase shift by η. The relative group is $\mathrm{U}^{\mathrm{c}}(1)$, with a superscript "c" indicating "charge".

Global spin symmetry The EHM is described by three operators: NN hopping, independent of σ; local repulsion between opposite-spin densities; non-local attraction, independent of σ. If we rotate all spins by an equal amount, the hamiltonian must remain identical. An identical observation holds for the number operator, $\hat{N}$. Thus $\hat{H}_{\mathrm{EHM}} - \mu\hat{N}$ must be exhibit a global SU(2) symmetry, related to the rotation in 3D space of all spins altogether.

Let us prove this rigorously. Let the on-site spinor:

$$\hat{\Psi}_i \equiv \begin{bmatrix} \hat{c}_{i\uparrow} \\ \hat{c}_{i\downarrow} \end{bmatrix}$$

and the Pauli matrices

$$\tau^x \equiv \begin{bmatrix} & 1 \\ 1 & \end{bmatrix} \qquad \tau^y \equiv \begin{bmatrix} & -i \\ i & \end{bmatrix} \qquad \tau^z \equiv \begin{bmatrix} 1 & \\ & -1 \end{bmatrix}$$

We define the local spin operator as:

$$\hat{S}^\alpha_i \equiv \frac{1}{2}\hat{\Psi}^\dagger_i \tau^\alpha \hat{\Psi}_i \qquad \left[\hat{S}^\alpha_j, \hat{S}^\beta_j\right] = i\sum_\gamma \epsilon_{\alpha\beta\gamma}\hat{S}^\gamma_j$$

with $\epsilon_{\alpha\beta\gamma}$ the Levi-Civita tensor. It can be checked by explicit calculation that the $\mathfrak{su}(2)$ spin algebra is actually respected. For example, consider $\alpha = x$, $\beta = y$:

$$\begin{aligned}
\left[\hat{S}^x_j, \hat{S}^y_j\right] &= \frac{1}{4}\left[\hat{\Psi}^\dagger_j \begin{bmatrix} & 1 \\ 1 & \end{bmatrix} \hat{\Psi}_j, \hat{\Psi}^\dagger_j \begin{bmatrix} & -i \\ i & \end{bmatrix} \hat{\Psi}_j\right] \\
&= \frac{1}{4}\left[\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow} + \hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow}, -i\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow} + i\hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow}\right] \\
&= \frac{i}{4}\left[\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow}, \hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow}\right] - \frac{i}{4}\left[\hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow}, \hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow}\right] \\
&= \frac{i}{2}\left[\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow}, \hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow}\right] \\
&= \frac{i}{2}\left(\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow}\hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow} - \hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow}\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow}\right) \\
&= \frac{i}{2}\left(-\hat{c}^\dagger_{j\uparrow}\hat{c}^\dagger_{j\downarrow}\hat{c}_{j\downarrow}\hat{c}_{j\uparrow} + \hat{c}^\dagger_{j\uparrow}\hat{c}_{j\uparrow} + \hat{c}^\dagger_{j\downarrow}\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\uparrow}\hat{c}_{j\downarrow} - \hat{c}^\dagger_{j\downarrow}\hat{c}_{j\downarrow}\right) \\
&= \frac{i}{2}\left(\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\uparrow} - \hat{c}^\dagger_{j\downarrow}\hat{c}_{j\downarrow}\right) \\
&= i\hat{S}^z_j
\end{aligned}$$

We used Fermi anticommutation rules, $\{\hat{c}_{i\sigma}, \hat{c}^\dagger_{j\sigma'}\} = \delta_{ij}\delta_{\sigma\sigma'}$. Identical derivations can be carried out for all other components.

Let the raising and lowering operators:

$$\hat{S}^+_j \equiv \frac{\hat{S}^x_j + i\hat{S}^y_j}{2} \qquad \hat{S}^-_j \equiv \frac{\hat{S}^x_j - i\hat{S}^y_j}{2}$$

and the global spin operators

$$\hat{S}^\alpha \equiv \sum_{i\in\mathcal{L}} \hat{S}^\alpha_i \qquad \left[\hat{S}^\alpha, \hat{S}^\beta\right] = i\sum_\gamma \epsilon_{\alpha\beta\gamma}\hat{S}^\gamma$$

being $\mathcal{L}$ the lattice. Explicitly,

$$\hat{S}^z \equiv \frac{1}{2}\sum_{i\in\mathcal{L}} (\hat{n}_{i\uparrow} - \hat{n}_{i\downarrow}) \qquad \hat{S}^+ \equiv \frac{1}{2}\sum_{i\in\mathcal{L}} \hat{c}^\dagger_{i\uparrow}\hat{c}_{i\downarrow} \qquad \hat{S}^- \equiv \left(\hat{S}^+\right)^\dagger$$

The EHM hamiltonian and the number operator commutate with the global spin operator. For the kinetic part $\hat{H}_t$ and the local repulsion $\hat{H}_U$ this is a well known result [12, 27],

$$\left[\hat{H}_t + \hat{H}_U, \hat{\mathbf{S}}\right] = 0 \qquad \left[\hat{N}, \hat{\mathbf{S}}\right] = 0$$

For the non-local attraction, since it depends solely on total densities on neighbouring sites, spin-rotational invariance must hold. Anyway, let us give a short formal proof following this chain of results:

1. The non-local attraction can be written as:

$$\begin{aligned}\hat{H}_V &= V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'} \hat{n}_{i\sigma}\hat{n}_{j\sigma'} \\ &= V\sum_{\langle ij\rangle} \hat{\Psi}^\dagger_i\hat{\Psi}_i\hat{\Psi}^\dagger_j\hat{\Psi}_j\end{aligned} \tag{2.6}$$

   having we used $\hat{\Psi}^\dagger_i\hat{\Psi}_i = \hat{c}^\dagger_{i\uparrow}\hat{c}_{i\uparrow} + \hat{c}^\dagger_{i\downarrow}\hat{c}_{i\downarrow}$.

2. Similarly, the number operator is given by:

$$\begin{aligned}\hat{N} &= \sum_{i\in\mathcal{L}}\sum_\sigma \hat{n}_{i\sigma} \\ &= \sum_{i\in\mathcal{L}} \hat{\Psi}^\dagger_i\hat{\Psi}_i\end{aligned} \tag{2.7}$$

3. Let now $\hat{R}(\boldsymbol{\theta})$ be any global spin rotation,

$$\hat{R}(\boldsymbol{\theta}) \equiv \exp\left(-i\boldsymbol{\theta}\cdot\hat{\mathbf{S}}\right) \quad \text{where} \quad \boldsymbol{\theta} = (\theta_x, \theta_y, \theta_z)$$

   Spinors transform as:

$$\hat{R}(\boldsymbol{\theta})\hat{\Psi}_i\hat{R}(\boldsymbol{\theta})^\dagger = \mathcal{R}(\boldsymbol{\theta})\hat{\Psi}_i$$

   with $\mathcal{R}(\boldsymbol{\theta}) \in \mathrm{SU}(2)$ the unitary matrix associated to said rotation,

$$\mathcal{R}(\boldsymbol{\theta}) = \exp\left(-\frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\right) \quad \text{where} \quad \boldsymbol{\tau} = (\tau_x, \tau_y, \tau_z) \tag{2.8}$$

   Let us prove this. Let $\hat{A} = i\boldsymbol{\theta}\cdot\hat{\mathbf{S}}$ e $\hat{B} = \hat{\Psi}_i$. The following identity, a consequence of Baker-Campbell-Housdorff formula, holds:

$$e^{\hat{A}}\hat{B}e^{-\hat{A}} = \hat{B} + \left[\hat{A}, \hat{B}\right] + \frac{1}{2!}\left[\hat{A}, \left[\hat{A}, \hat{B}\right]\right] + \frac{1}{3!}\left[\hat{A}, \left[\hat{A}, \left[\hat{A}, \hat{B}\right]\right]\right] + \cdots$$

The first order of the expansion gives:

$$
\begin{aligned}
[i\boldsymbol{\theta}\cdot\hat{\mathbf{S}},\hat{\Psi}_i] &= i\sum_\alpha \theta_\alpha[\hat{S}^\alpha,\hat{\Psi}_i] \\
&= -\frac{i}{2}\sum_\alpha \theta_\alpha \tau^\alpha \hat{\Psi}_i \\
&= -\frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\hat{\Psi}_i
\end{aligned}
$$

We omitted some purely algebraic passages. The second order:

$$
\begin{aligned}
\Big[i\boldsymbol{\theta}\cdot\hat{\mathbf{S}},\Big[i\boldsymbol{\theta}\cdot\hat{\mathbf{S}},\hat{\Psi}_i\Big]\Big] &= \Big[i\boldsymbol{\theta}\cdot\hat{\mathbf{S}},-\frac{i}{2}(\boldsymbol{\theta}\cdot\boldsymbol{\tau})\hat{\Psi}_i\Big] \\
&= -\frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\Big[i\boldsymbol{\theta}\cdot\hat{\mathbf{S}},\Psi_i\Big] \\
&= \frac{1}{4}(\boldsymbol{\theta}\cdot\boldsymbol{\tau})^2\hat{\Psi}_i
\end{aligned}
$$

and so on. Collecting all terms,

$$
\begin{aligned}
\hat{R}(\boldsymbol{\theta})\hat{\Psi}_i\hat{R}(\boldsymbol{\theta})^\dagger &= \hat{\Psi}_i - \frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\hat{\Psi}_i + \frac{1}{2!\times 4}(\boldsymbol{\theta}\cdot\boldsymbol{\tau})^2\hat{\Psi}_i + \cdots \\
&= \sum_{k=1}^{\infty}\frac{1}{k!}\left(-\frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\right)^k\hat{\Psi}_i \\
&= \mathcal{R}(\boldsymbol{\theta})\hat{\Psi}_i
\end{aligned}
$$

In last passage, we identified the series as the Taylor expansion of the exponential, recognising the expression for $\mathcal{R}(\boldsymbol{\theta})$ given in Eq. (2.8).

4. Then, the group acts on the product $\hat{\Psi}_i^\dagger\hat{\Psi}_i$ as

$$
\begin{aligned}
\hat{R}\hat{\Psi}_i^\dagger\hat{\Psi}_i\hat{R}^\dagger &= \hat{R}\hat{\Psi}_i^\dagger\hat{R}^\dagger\hat{R}\hat{\Psi}_i\hat{R}^\dagger \\
&= \hat{\Psi}_i^\dagger\mathcal{R}^\dagger\mathcal{R}\hat{\Psi}_i \\
&= \hat{\Psi}_i^\dagger\hat{\Psi}_i
\end{aligned}
$$

being $\mathcal{R}\in\mathrm{SU}(2)$. Recalling Eqns. (2.6), (2.7), both $\hat{H}_V$ and $\hat{N}$ are SU(2) symmetric.

The EHM is symmetric under global spin rotations. Note that $\hat{H}_V$ is actually symmetric under local spin rotations, meaning that we could rotate the spin at each site by a different amount and get the same interaction. The same holds for $\hat{H}_U$. It is the kinetic operator that makes the EHM globally (and not locally) SU(2) symmetric.

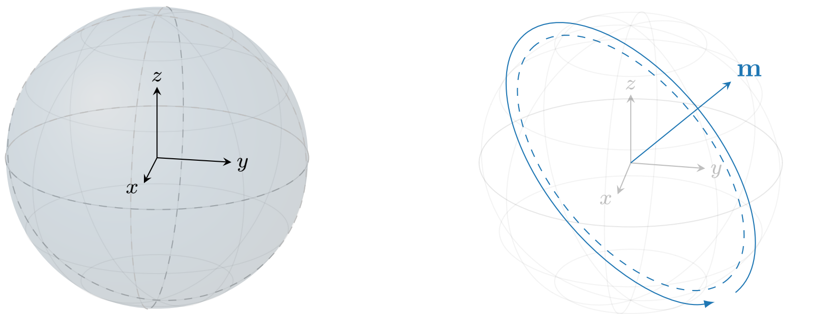
Note that the $U^c(1)$ gauge symmetry and the SU(2) spin rotations symmetry can be combined,

$$
\mathrm{U}^c(1)\otimes\mathrm{SU}(2)=\mathrm{U}(2)
$$

We will refer to U(2) symmetry to indicate full charge-spin conservation. Finally, note the group of 3D rotations contains the subgroup of 2D rotations around a given axis z,

$$
\mathrm{U}^z(1)\subset\mathrm{SU}(2)
$$

where the z superscript serves to distinguish from the gauge group. The z axis does not need to be identified with the "real" z axis, perpendicular with the lattice plane. Any given direction works. Inverting our logic, we can build the SU(2) generators, starting from $\hat{S}^z$, after we have chosen the direction z. In terms of commutation relations, a given operator can respect $\mathrm{U}^z(1)$ but not SU(2): in that case, it will commutate to zero with $\hat{S}^z$, and not the other spin operators. A graphic rendition of this SSB is given in Fig. 2.2, representing for a given vector $\mathbf{m}$ the passage from full SU(2) rotational symmetry around any direction to reduced U(1) rotational symmetry around just one direction.



a | SU(2) rotations domain before SSB.

b | U(1) rotations domain after SSB.

Figure 2.2 | Graphic representation of the concept of SU(2) $\rightarrow$ U^z(1) SSB. "Before" SSB, we have full symmetry under any possible rotation (solid color sphere, left panel). "After" SSB we have symmetry only for rotations around some vector $\mathbf{m}$ (dashed tilted circumference, right panel).

2.1.3 Particle-hole symmetry at half-filling

The half-filled EHM on the square lattice exhibits particle-hole symmetry (PHS). The presence of this additional symmetry is a consequence of the fact that the square lattice is bipartite. Let us prove this. Consider the generic unitary particle-hole (PH) transformation

$$\mathrm{P} \quad : \quad \hat{c}_{i\sigma} \quad \rightarrow \quad e^{i\eta_{i\sigma}}\hat{h}^{\dagger}_{i\sigma} \tag{2.9}$$

being in general η_i a phase dependent on the fermionic quantum state $(i\sigma)$. Inserting the above transformation in the EHM, we get

$$\mathrm{P}\{\hat{H}\} = -t\sum_{\langle ij\rangle}\sum_{\sigma} e^{i(\eta_{i\sigma}+\eta_{j\sigma})}\hat{h}_{i\sigma}\hat{h}^{\dagger}_{j\sigma} + U\sum_{i}\hat{h}_{i\uparrow}\hat{h}_{i\downarrow}\hat{h}^{\dagger}_{i\downarrow}\hat{h}^{\dagger}_{i\uparrow} - V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^{\dagger}_{j\sigma'}\hat{h}^{\dagger}_{i\sigma}$$

For the two interactions $\hat{H}_U$, $\hat{H}_V$ the phase factors cancelled out. We aim to find the conditions under which the above hamiltonian is mapped onto the EHM of (**??**),

$$\mathrm{P}\{\hat{H}\} \overset{?}{=} -t\sum_{\langle ij\rangle}\sum_{\sigma}\hat{h}^{\dagger}_{i\sigma}\hat{h}_{j\sigma} + U\sum_{i\in\mathcal{L}}\hat{h}^{\dagger}_{i\uparrow}\hat{h}^{\dagger}_{i\downarrow}\hat{h}_{i\downarrow}\hat{h}_{i\uparrow} - V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{h}^{\dagger}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma}$$

Before starting the derivation, which is an extension of the argument carried out for the conventional HM [12], let us discuss some properties of the hole operators. Due to the presence of the phase factors in the PHS operative definition, the Fermi anticommutation rules hold regardlessly of the phase factor:

$$\begin{aligned}\{\hat{h}_{i\sigma}, \hat{h}^{\dagger}_{j\sigma'}\} &= e^{i(\eta_{i\sigma}-\eta_{j\sigma'})}\{\hat{c}^{\dagger}_{i\sigma}, \hat{c}_{j\sigma'}\}\\ &= e^{i(\eta_{i\sigma}-\eta_{j\sigma'})}\delta_{ij}\delta_{\sigma\sigma'}\\ &= \delta_{ij}\delta_{\sigma\sigma'}\end{aligned}$$

The on-site density operator is given by:

$$\hat{h}_{i\sigma}\hat{h}^{\dagger}_{i\sigma} = \hat{n}_{i\sigma}$$

being $\hat{n}_{i\sigma}$ the fermionic number operator on site i with spin σ. It follows, for quartic interactions

$$\begin{aligned}
\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'}\hat{h}^\dagger_{i\sigma} &= -\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'} \\
&= \hat{h}_{i\sigma}\hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'} \\
&= -\hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} + \hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'} \\
&= \hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma} + \hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'} \\
&= -\hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{i\sigma}\hat{h}_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma} + \hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \left(\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma'}\right)\delta_{ij}\delta_{\sigma\sigma'} \\
&= \hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma} + (\hat{n}_{i\sigma} + \hat{n}_{j\sigma'}) - 1 - \left(\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma'}\right)\delta_{ij}\delta_{\sigma\sigma'}
\end{aligned} \tag{2.10}$$

The first term of the last passage is mirrored with respect to the starting term. Now, let us deal with the kinetic, local and non-local interactions separately.

Kinetic term. Consider the kinetic term,

$$\hat{H}_t = -t\sum_{\langle ij\rangle}\sum_{\sigma}\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}$$

Apply the PH transform of Eq. (2.9)

$$\mathrm{P}\{\hat{H}_t\} = -t\sum_{\langle ij\rangle}\sum_{\sigma}e^{i(\eta_{i\sigma}+\eta_{j\sigma})}\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma}$$

Anti-commutate the two fermionic operators, and exchange site labels $i \leftrightarrow j$. We get:

$$\mathrm{P}\{\hat{H}_t\} = -t\sum_{\langle ij\rangle}\sum_{\sigma}e^{i(\eta_{i\sigma}+\eta_{j\sigma}+\pi)}\hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma}$$

A necessary condition on the unitary operation P is the following,

$$\eta_{i\sigma} + \eta_{j\sigma} = (2k-1)\pi \quad \text{with} \quad k \in \mathbb{Z} \tag{2.11}$$

valid for NN sites.

Local repulsion. Substitute Eq. (2.10) in $\hat{H}_U$,

$$\begin{aligned}
\mathrm{P}\{\hat{H}_U\} &= U\sum_{i\in\mathcal{L}}\hat{h}_{i\uparrow}\hat{h}_{i\downarrow}\hat{h}^\dagger_{i\downarrow}\hat{h}^\dagger_{i\uparrow} \\
&= U\sum_{i\in\mathcal{L}}\hat{h}^\dagger_{i\uparrow}\hat{h}^\dagger_{i\downarrow}\hat{h}_{i\downarrow}\hat{h}_{i\uparrow} + U\sum_{i\in\mathcal{L}}(\hat{n}_{i\uparrow} + \hat{n}_{i\downarrow}) - U\sum_{i\in\mathcal{L}}(1) \\
&= U\sum_{i}\hat{h}^\dagger_{i\uparrow}\hat{h}^\dagger_{i\downarrow}\hat{h}_{i\downarrow}\hat{h}_{i\uparrow} + U(\hat{N} - L_xL_y)
\end{aligned}$$

For the half-filled state $\langle\hat{N}\rangle = L_xL_y$. The unitary map of Eq. (2.9), independently of the condition of Eq. (2.11), leaves the hamiltonian identical at half-filling.

Non-local attraction. Proceed identically for $\hat{H}_V$,

$$\begin{aligned}
\mathrm{P}\{\hat{H}_V\} &= -V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'}\hat{h}^\dagger_{i\sigma} \\
&= -V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma} - V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}(\hat{n}_{i\sigma} + \hat{n}_{j\sigma'}) + V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}(1) \\
&= -V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma} - 2zV(\hat{N} - L_xL_y)
\end{aligned}$$

Symmetry	Operations
$\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)}$	One-site discrete translations separately in both directions
C_4	Four-fold point group (real $\pi/2$ angle rotations)
$\mathrm{U}^{\mathrm{c}}(1)$	Charge-conserving global phase shifts
$\mathrm{SU}(2)$	Three-dimensional rotations in global spin space

Table 2.1 Model symmetries and relative group operations. Note that $\mathrm{U}^{\mathrm{c}}(1)$ represents the gauge symmetry linked to charge conservation.

with $z = 4$ the lattice coordination number. In the last passage, we used that on the square lattice

$$\begin{aligned}\sum_{\langle ij \rangle}(1) &= \sum_{i\in\mathcal{L}/2} z \\ &= \frac{L_x L_y}{2} \times z \end{aligned} \tag{2.12}$$

As before, the non-local attraction is particle-hole symmetric at half-filling. As for the local repulsion, PHS is given just the operation of Eq. (2.9). The condition of Eq. (2.11) is irrelevant.

Recollecting everything, if P is defined by the unitary operation

$$\mathrm{P} \quad : \quad \begin{cases} \hat{c}_{i\sigma} \to \hat{h}^{\dagger}_{i\sigma} & \text{for} \quad i \in \mathcal{S}_1 \\ \hat{c}_{i\sigma} \to -\hat{h}^{\dagger}_{i\sigma} & \text{for} \quad i \in \mathcal{S}_2 \end{cases}$$

and the system is half-filled, PHS is respected. Any other phase, as long as it alternates by π on the two sublattices, is good enough. As for the conventional HM [12], this result can be generalised to any bipartite lattice. In turn, if a lattice is not bipartite, PHS is not present. To summarise, we report in Tab. 2.1 all identified symmetries.

2.2 Symmetry breaking channels in real space

In many-body Physics, a phase transition is described by some order parameter rising from zero to a non-zero value. This is associated with some spontaneous symmetry breaking (SSB). In order to simulate different phases, we need a way to distinguish them based on which symmetries they break (or not). We also need a way to work computationally in a given symmetry sector. HF methods, by reducing the hamiltonian to a non-interacting approximation, offer a simple framework to these ends. Indeed, once we identified all symmetries of a given phase, we just impose them on our approximation of the full hamiltonian. The found ground-state approximation will entail these symmetries by construction.

Recall now the discussion of the HF method of Sec. 1.2. The ending point were Eqns. (1.16), (1.17), which identify the self-consistent single-particle fields which approximate the original quartic hamiltonian. In order to get there, we applied Wick's theorem to the quartic part of the hamiltonian and derived the most general expressions of $\mathbf{h}$. In next sections here apply Wick's theorem to the quartic operators of $\hat{H}_{\mathrm{EHM}}$ and discuss the symmetry properties of the resulting decomposition.

Before starting, let us define precisely the three phases we will investigate in terms of symmetries:

- Normal phase: no broken symmetry.
- Antiferromagnetic phase: simultaneous $\mathrm{SU}(2) \overset{\mathrm{SSB}}{\to} \mathrm{U}^z(1)$ and $\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)} \overset{\mathrm{SSB}}{\to} \mathrm{T}_2^{(x)} \otimes \mathrm{T}_2^{(y)}$. Among all possible realisations of antiferromagnetism, we realise a SDW with alternating distribution of spin densities;
- Superconducting phase: simultaneous $\mathrm{U}^c(1) \overset{\mathrm{SSB}}{\to} 0$ and $\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2$. We investigate anisotropic superconductivity, allowing for the particle number to be not conserved and for nematic effects.

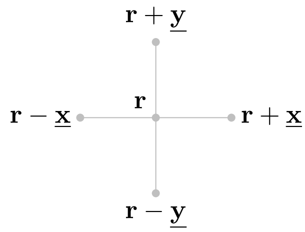


Figure 2.3 | Schematic representation of the four NNs of a given site i for a planar square lattice.

Nematic effects are observed in unconventional superconductors [22], and this is the reason of our interest in the possibility of rotational SSBs. We will theoretically discuss the nematic SSB $C_4 \to C_2$ also for normal and AF phases, as if discrete rotational symmetry was broken. As we will see, while it not trivial to exclude it *a priori*, it will actually turn out that the HF ground state preserves C_4 in both cases.

We now move to a more detailed discussion of each hamiltonian contribution. In next sections, we switch from the site index i to the vector notation $\mathbf{r}$, indicating the site 2D position on the lattice

$$i \quad \to \quad \mathbf{r} = (x, y)$$

with $x, y \in \mathbb{N}$. The lattice versors are:

$$\underline{\mathbf{x}} \equiv (1, 0) \qquad \underline{\mathbf{y}} \equiv (0, 1)$$

as in Fig. 2.3.

2.2.1 Local repulsion

The local repulsion $\hat{H}_U$ is the quartic operator of the conventional HM,

$$\begin{aligned} \hat{H}_U &= U \sum_{\mathbf{r}\in\mathcal{L}} \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}\downarrow} \\ &= U \sum_{\mathbf{r}\in\mathcal{L}} \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow} \end{aligned}$$

In order to obtain the self-consistent values of $h^{+-}_{\alpha\beta}$, $h^{++}_{\alpha\beta}$ we need to compute an expectation value of the full hamiltonian. Wick's theorem guarantees that, for the wavefunctions we are using, the expectation value of a quartic fermionic operator is the sum of all fully contracted terms. Recalling the nomenclature of Eq. (1.3),

$$\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle = \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}\rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\downarrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}\rangle\langle \hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Cooper}} \tag{2.13}$$

where the equal sign holds as long as we perform the averaging operation on the subspace of **gaussian states**. Now we discuss the above EVs separately, identifying if any can be neglected based on selection rules in a given phase.

Hartree term. The Hartree part of the above decomposition is a product of density EVs,

$$\langle \hat{n}_{\mathbf{r}\uparrow}\rangle\langle \hat{n}_{\mathbf{r}\downarrow}\rangle$$

Recalling the groups of Tab. 2.1, we see that the density operator preserves all symmetries. Single-site density operators commutate with spin operators and are gauge-invariant. Moreover, shifting or rotating the lattice leaves density operators unchanged. Thus no selection rule applies, and the above EVs are in principle non-zero.

Fock terms. For the Fock part, notice that

$$\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\downarrow} = \hat{S}^+_{\mathbf{r}}$$

being $S^+_{\mathbf{r}}$ the local raising operator. From SU(2) algebra we have:

$$\left[\hat{S}^+_{\mathbf{r}}, \hat{S}^\alpha\right] \neq 0 \qquad \alpha \in \{x, y, z\}$$

If the ground-state is invariant under global spin rotations, these EVs must be zero. Gauge symmetry gives no selection rules, due to Eq. (2.5).

Cooper terms. The Cooper part can be non-zero only if charge is not conserved. In terms of the gauge symmetry, it suffices to apply the map of Eq. (2.4) to $\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}$

$$\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow} \quad \rightarrow \quad e^{2i\eta}\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}$$

to conclude that this operator is in general not gauge invariant. It is instead SU(2)-symmetric. To prove this, it suffices to compute commutators of $\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}$ with $\hat{S}^z$, $\hat{S}^+$:

$$\begin{aligned}
\left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{S}^z\right] &= \left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \frac{1}{2}\sum_{\mathbf{r}'\in\mathcal{L}}(\hat{n}_{\mathbf{r}'\uparrow} - \hat{n}_{\mathbf{r}'\downarrow})\right] \\
&= \frac{1}{2}\left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}\right] \\
&= \frac{1}{2}\left[\hat{c}^\dagger_{\mathbf{r}\uparrow}, \hat{n}_{\mathbf{r}\uparrow}\right]\hat{c}^\dagger_{\mathbf{r}\downarrow} - \frac{1}{2}\hat{c}^\dagger_{\mathbf{r}\uparrow}\left[\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{n}_{\mathbf{r}\downarrow}\right]
\end{aligned}$$

Since:

$$\begin{aligned}
\left[\hat{c}^\dagger_{\mathbf{r}\sigma}, \hat{n}_{\mathbf{r}\sigma}\right] &= \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma} - \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\sigma} \\
&= -\hat{c}^\dagger_{\mathbf{r}\sigma}\left(1 - \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma}\right) \\
&= -\hat{c}^\dagger_{\mathbf{r}\sigma}
\end{aligned} \tag{2.14}$$

we have:

$$\left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{S}^z\right] = -\frac{1}{2}\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow} + \frac{1}{2}\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow} = 0$$

Moreover, considering the raising operator:

$$\begin{aligned}
\left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{S}^+\right] &= \left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \frac{1}{2}\sum_{\mathbf{r}'\in\mathcal{L}}\hat{c}^\dagger_{\mathbf{r}'\uparrow}\hat{c}_{\mathbf{r}'\downarrow}\right] \\
&= \left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\downarrow}\right] \\
&= \left[\hat{c}^\dagger_{\mathbf{r}\uparrow}, \hat{c}^\dagger_{\mathbf{r}\uparrow}\right]\hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow} - \left(\hat{c}^\dagger_{\mathbf{r}\uparrow}\right)^2\left[\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{c}_{\mathbf{r}\downarrow}\right] \\
&= 0
\end{aligned}$$

where we used the facts that every operator commutates with itself, and a squared creation operator is null. These commutation relations, together, prove that Cooper EVs are legitimately non-zero for a SU(2)-preserving phase.

In Tab. 2.2 we collect the selection rules for the Hartree, Fock and Cooper terms of Eq. (2.13), based on the symmetry specifications given for each phase in above paragraphs. Note that Fock-like EVs are prohibited for all phases.

Term	Normal phase	Antiferromagnet	Superconductor
Hartree			
Fock	Prohibited	Prohibited	Prohibited
Cooper	Prohibited	Prohibited	

Table 2.2 Summary of selection rules for the Wick's decomposition of Eq. (2.13) of the local repulsion $\hat{H}_U$. An empty entry indicates no selection rule.

2.2.2 Non-local attraction

In order to analyse similarly the non-local attraction $\hat{H}_V$, it is useful to decompose the operator in spin channels:

$$\begin{aligned}\hat{H}_V &\equiv \underbrace{\hat{H}_V^{\uparrow\uparrow} + \hat{H}_V^{\downarrow\downarrow}}_{\text{Same spin}} + \underbrace{\hat{H}_V^{\uparrow\downarrow} + \hat{H}_V^{\downarrow\uparrow}}_{\text{Opposite spin}} \\ &= \hat{H}^{(\text{s.s.})} + \hat{H}^{(\text{o.s.})}\end{aligned} \tag{2.15}$$

being

$$\hat{H}_V^{\sigma\sigma'} \equiv -V \sum_{\langle \mathbf{r},\mathbf{r}'\rangle} \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}'\sigma'}$$

Let us start from the opposite spin sector. On the square lattice (as for any bipartite lattice) NNs belong to different sublattices. This implies that any NNs sum can be rewritten as:

$$\sum_{\langle \mathbf{r}\mathbf{r}'\rangle} f(\mathbf{r},\mathbf{r}') = \sum_{\mathbf{r}\in\mathcal{L}/2}\ \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} f(\mathbf{r},\mathbf{r}+\boldsymbol{\delta}) \tag{2.16}$$

with f some function. We are using the notation of Fig. 2.3. The symbol "$\mathcal{L}/2$" indicates one of the two sublattices $\mathcal{S}_1$, $\mathcal{S}_2$ represented in Fig. 2.1. We can rewrite:

$$\hat{H}_V^{\sigma\sigma'} = -V \sum_{\mathbf{r}\in\mathcal{L}/2}\ \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left(\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma'} + \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\sigma'}\right)$$

Now, if we choose $\mathcal{L}/2 = \mathcal{S}_1$, this gives:

$$\begin{aligned}\hat{H}_V^{\uparrow\downarrow} &= -V \sum_{\mathbf{r}\in\mathcal{S}_1}\ \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left(\hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow} + \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\downarrow}\right) \\ \hat{H}_V^{\downarrow\uparrow} &= -V \sum_{\mathbf{r}\in\mathcal{S}_1}\ \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left(\hat{n}_{\mathbf{r}\downarrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\uparrow} + \hat{n}_{\mathbf{r}\downarrow}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\uparrow}\right)\end{aligned}$$

These two operators are "complementary": $\sigma =\uparrow$ densities act on $\mathcal{S}_1$ for $\hat{H}_V^{\uparrow\downarrow}$, on $\mathcal{S}_2$ for $\hat{H}_V^{\downarrow\uparrow}$; $\sigma =\downarrow$ densities, the opposite. Then their sum can be expressed as:

$$\begin{aligned}\hat{H}_V^{(\text{o.s.})} &\equiv \hat{H}_V^{\uparrow\downarrow} + \hat{H}_V^{\downarrow\uparrow} \\ &= -V \sum_{\mathbf{r}\in\mathcal{L}}\ \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left(\hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow} + \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\downarrow}\right)\end{aligned} \tag{2.17}$$

Wick's theorem applied to this quartic operator gives:

$$\begin{aligned}\langle \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\rangle &= \langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle \\ &\simeq \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\rangle\langle \hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Cooper}}\end{aligned} \tag{2.18}$$

For the same-spin sector we define

$$\begin{aligned}\hat{H}_V^{(\text{s.s.})} &\equiv \hat{H}_V^{\uparrow\uparrow} + \hat{H}_V^{\downarrow\downarrow} \\ &= -V \sum_{\mathbf{r}\in\mathcal{L}/2}\ \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}}\ \sum_{\sigma\in\{\uparrow,\downarrow\}} \left(\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma} + \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\sigma}\right)\end{aligned} \tag{2.19}$$

Sector	Term	Normal phase	Antiferromagnet	Superconductor
o.s.	Hartree			
	Fock	Prohibited	Prohibited	Prohibited
	Cooper	Prohibited	Prohibited	
s.s.	Hartree			
	Fock			
	Cooper	Prohibited	Prohibited	

Table 2.3 | Summary of selection rules for the Wick's decomposition of Eqns. (2.18), (2.20) of the non-local attraction $\hat{H}_V$. An empty entry indicates no selection rule.

Note that, with respect to the o.s. sector, we are summing on half the sites two times: one for spin $\uparrow$, one for spin $\downarrow$. Applying Wick's theorem:

$$\begin{aligned}\langle \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\rangle &= \langle \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\hat{c}_{\mathbf{r}\sigma}\rangle \\ &\simeq \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\hat{c}_{\mathbf{r}\sigma}\rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\rangle\langle \hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\hat{c}_{\mathbf{r}\sigma}\rangle}_{\text{Cooper}}\end{aligned} \tag{2.20}$$

As we did for $\hat{H}_U$, we analyse each separately.

Hartree terms. As it was for $\hat{H}_U$, Hartree EVs of Eqns. (2.17), (2.19) are not subject to selection rules for any symmetry of Tab. 2.1. Thus they are allowed in both sectors.

Fock terms. The Fock EVs from the two sectors behave differently. Let us compute the following generic commutation relations:

$$\begin{aligned}\left[\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \hat{S}^z\right] &= \left[\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \frac{1}{2}\sum_{\mathbf{r}'\in\mathcal{L}}(\hat{n}_{\mathbf{r}'\uparrow}-\hat{n}_{\mathbf{r}'\downarrow})\right] \\ &= \frac{(-1)^{\delta_{\sigma=\downarrow}}}{2}\left[\hat{c}^\dagger_{\mathbf{r}\sigma}, \hat{n}_{\mathbf{r}\sigma}\right]\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'} + \frac{(-1)^{\delta_{\sigma'=\downarrow}}}{2}\hat{c}^\dagger_{\mathbf{r}\sigma}\left[\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma}\right] \\ &= \left(-\frac{(-1)^{\delta_{\sigma=\downarrow}}}{2} + \frac{(-1)^{\delta_{\sigma'=\downarrow}}}{2}\right)\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}\end{aligned} \tag{2.21}$$

where in last passage we used Eq. (2.14) and its conjugate version,

$$[\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma}] = \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}$$

Now, if $\sigma = \sigma'$ the last passage of Eq. (2.21) is zero; otherwise is not. This proves that the o.s. Fock EVs are non-zero only for a phase that completely breaks SU(2); instead, in the s.s. sector we can have non-zero EVs for a phase that reduces SU(2) to $\mathrm{U}^z(1)$.

Cooper terms. For the Cooper terms, identical considerations as for $\hat{H}_U$ hold. Gauge symmetry must be broken to allow for non zero EVs, while SU(2) symmetry survives. The proof is analogous to the one given for $\hat{H}_U$.

In Tab. 2.3 we summarise these considerations, giving the selection rules on the EVs of Eqns. (2.18), (2.20), dividing in the two sectors "o.s." and "s.s.". By identifying these selection rules, we set grounds for the detailed derivation of each chapter and heavily simplify application of HF model to the EHM. We now move to the last part of this chapter: reciprocal space description of the EHM.

2.3 The EHM in reciprocal space

The EHM is symmetric under discrete lattice translations. We will investigate phases that do not (completely) break translational invariance. It is thus convenient to move to reciprocal space, by

Fourier-transforming the hamiltonian. We will use the following convention:

$$\hat{c}_{\mathbf{r}\sigma} \overset{\mathcal{F}}{=} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}\in \mathrm{BZ}} e^{-i\mathbf{k}\cdot\mathbf{r}} \hat{c}_{\mathbf{k}\sigma} \tag{2.22}$$

Throughout the text, $\overset{\mathcal{F}}{=}$ will indicate a passage where a Fourier transform is used. "BZ" is shorthand notation for the Brillouin Zone.

It is useful to recall that the following equation for Kronecker's delta holds:

$$\frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}} e^{i\mathbf{k}\cdot\mathbf{r}} = \delta_{\mathbf{k}=\mathbf{0}} \tag{2.23}$$

On one sublattice:

$$\frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}/2} e^{i\mathbf{k}\cdot\mathbf{r}} = \frac{1}{2}\delta_{\mathbf{k}=\mathbf{0}} \tag{2.24}$$

The notation $\mathcal{L}/2$ is shorthand to indicate $\mathcal{S}_1$ or $\mathcal{S}_2$.

Kinetic part. Let us transform $\hat{H}_t$:

$$\begin{aligned}
\hat{H}_t &= -t \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma} \left[\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}'\sigma} + \text{h.c.}\right] \\
&\overset{\mathcal{F}}{=} -t \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}\in\mathrm{BZ}} e^{i\mathbf{k}\cdot\mathbf{r}} \hat{c}^\dagger_{\mathbf{k}\sigma} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}'\in\mathrm{BZ}} e^{-i\mathbf{k}'\cdot\mathbf{r}'} \hat{c}_{\mathbf{k}'\sigma} + \text{h.c.} \\
&= -2t \sum_{\mathbf{k},\mathbf{k}'} \sum_{\sigma} \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}'\sigma} \times \frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}/2} e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}} \times \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} e^{i\mathbf{k}'\cdot\boldsymbol{\delta}} \\
&= -2t \sum_{\mathbf{k},\mathbf{k}'} \sum_{\sigma} \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} \left[\frac{e^{ik_x}+e^{-ik_x}}{2} + \frac{e^{ik_y}+e^{-ik_y}}{2}\right] &(2.25) \\
&= \sum_{\mathbf{k}\in\mathrm{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}} \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} &(2.26)
\end{aligned}$$

where we used Eqns. (2.16), (2.24). We defined the bare bands:

$$\epsilon_{\mathbf{k}} \equiv -2t\left(\cos k_x + \cos k_y\right) \tag{2.27}$$

Local repulsion. The local repulsion transforms as:

$$\begin{aligned}
\hat{H}_U &= U \sum_{\mathbf{r}\in\mathcal{L}} \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow} \\
&\overset{\mathcal{F}}{=} U \sum_{\mathbf{r}\in\mathcal{L}} \frac{1}{(L_x L_y)^2} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} e^{i\mathbf{k}_1\cdot\mathbf{r}}\hat{c}^\dagger_{\mathbf{k}_1\uparrow} e^{i\mathbf{k}_2\cdot\mathbf{r}}\hat{c}^\dagger_{\mathbf{k}_2\downarrow} e^{-i\mathbf{k}_3\cdot\mathbf{r}}\hat{c}_{\mathbf{k}_3\downarrow} e^{-i\mathbf{k}_4\cdot\mathbf{r}}\hat{c}_{\mathbf{k}_4\uparrow} \\
&= \frac{U}{L_x L_y} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} \hat{c}^\dagger_{\mathbf{k}_1\uparrow}\hat{c}^\dagger_{\mathbf{k}_2\downarrow}\hat{c}_{\mathbf{k}_3\downarrow}\hat{c}_{\mathbf{k}_4\uparrow} \times \frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} \\
&= \frac{U}{L_x L_y} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} \hat{c}^\dagger_{\mathbf{k}_1\uparrow}\hat{c}^\dagger_{\mathbf{k}_2\downarrow}\hat{c}_{\mathbf{k}_3\downarrow}\hat{c}_{\mathbf{k}_4\uparrow}\delta_{\mathbf{k}_1+\mathbf{k}_2=\mathbf{k}_3+\mathbf{k}_4}
\end{aligned}$$

where we used Eq. (2.23). Let $\mathbf{k}_1+\mathbf{k}_2=\mathbf{k}_3+\mathbf{k}_4=\mathbf{K}$, and define $\mathbf{k}$, $\mathbf{k}'$ such that

$$\mathbf{k}_1 \equiv \mathbf{K}+\mathbf{k} \qquad \mathbf{k}_2 \equiv \mathbf{K}-\mathbf{k} \qquad \mathbf{k}_3 \equiv \mathbf{K}-\mathbf{k}' \qquad \mathbf{k}_4 \equiv \mathbf{K}+\mathbf{k}'$$

Sums over these variables must be intended as over the Brillouin Zone (BZ). Hence:

$$\hat{H}_U \overset{\mathcal{F}}{=} \frac{U}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow}\hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \tag{2.28}$$

Non-local attraction. To transform the non-local attraction $\hat{H}_V$, it is useful to start by defining:

$$h_{\mathbf{r}\sigma\sigma'} \equiv -V \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left(\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma'} + \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\sigma'}\right)$$

with:

$$\hat{H}_V = \sum_{\mathbf{r}\in\mathcal{L}/2}\sum_{\sigma,\sigma'} h_{\mathbf{r}\sigma\sigma'}$$

The product of densities on neighbouring sites transforms as:

$$\begin{aligned}
\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'} &= \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'}\hat{c}_{\mathbf{r}\sigma} \\
&\stackrel{\mathcal{F}}{=} \frac{1}{(L_xL_y)^2}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} e^{\pm i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}} \hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma}
\end{aligned}$$

Then:

$$\begin{aligned}
h_{\mathbf{r}\sigma\sigma'} &\stackrel{\mathcal{F}}{=} -\frac{V}{(L_xL_y)^2}\sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} \\
&\qquad\qquad \times\left(e^{i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}} + e^{-i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}}\right)\hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma} \\
&= -\frac{2V}{(L_xL_y)^2}\sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}}\cos\left[(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}\right]\hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma}
\end{aligned}$$

The full non-local interaction is given by summing over all sites of $\mathcal{L}/2$ (which is, one sublattice). Applying Eq. (2.24) gives back momentum conservation

$$\frac{1}{L_xL_y}\sum_{\mathbf{r}\in\mathcal{L}/2} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} = \frac{1}{2}\delta_{\mathbf{k}_1+\mathbf{k}_2=\mathbf{k}_3+\mathbf{k}_4}$$

where we used Eq. (2.24). Define $\mathbf{k}$, $\mathbf{k}'$ and $\mathbf{K}$ as we did fo the local repulsion, and

$$\delta\mathbf{k} \equiv \mathbf{k}-\mathbf{k}'$$

Hence:

$$\begin{aligned}
\hat{H}_V &= \sum_{\mathbf{r}\in\mathcal{L}/2}\sum_{\sigma,\sigma'} h_{\mathbf{r}\sigma\sigma'} \\
&\stackrel{\mathcal{F}}{=} -\frac{V}{L_xL_y}\sum_{\sigma\sigma'}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\cos\left(\delta\mathbf{k}\cdot\boldsymbol{\delta}\right)\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma'}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \\
&= -\frac{V}{L_xL_y}\sum_{\sigma\sigma'}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma'}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}
\end{aligned} \tag{2.29}$$

Finally, we can separate in same spin and opposite spin channels. This gives:

$$\begin{aligned}
\hat{H}_V \stackrel{\mathcal{F}}{=} & \overbrace{-\frac{V}{L_xL_y}\sum_{\sigma}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}^{\text{s.s.}} \\
& \underbrace{-\frac{V}{L_xL_y}\sum_{\sigma}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}_{\text{o.s.}}
\end{aligned}$$

The o.s. part can be simplified: since

$$\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} = \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}$$

due to fermionic anti-commutation rules, and since the composite transform

$$\mathbf{k} \to -\mathbf{k} \qquad \mathbf{k}' \to -\mathbf{k}'$$

leaves the form factor $\cos(\delta k_x) + \cos(\delta k_y)$ invariant, we reduce the spin sum to two times the $\sigma = \uparrow$ part,

$$\hat{H}_V^{(\text{o.s.})} = -\frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} [\cos(\delta k_x) + \cos(\delta k_y)]\, \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \tag{2.30}$$

The $\mathcal{F}$-transform of the s.s. part is given explicitly by:

$$\begin{aligned}\hat{H}_V^{(\text{s.s.})} = -\frac{V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} &[\cos(\delta k_x) + \cos(\delta k_y)] \\ &\times \left[\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\uparrow} \hat{c}_{\mathbf{K}-\mathbf{k}'\uparrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} + \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\downarrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\downarrow}\right]\end{aligned} \tag{2.31}$$

All these expressions are going to be used in the analysis of each phase. In reciprocal space, for a quartic operator $\hat{c}^\dagger_{\mathbf{k}_1} \hat{c}^\dagger_{\mathbf{k}_2} \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4}$ we distinguish contractions as follows:

- Hartree contractions:
$$\hat{c}^\dagger_{\mathbf{k}_1} \hat{c}^\dagger_{\mathbf{k}_2} \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4} \qquad \hat{c}^\dagger_{\mathbf{k}_1} \hat{c}^\dagger_{\mathbf{k}_2} \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4}$$
- Fock contractions:
$$\hat{c}^\dagger_{\mathbf{k}_1} \hat{c}^\dagger_{\mathbf{k}_2} \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4} \qquad \hat{c}^\dagger_{\mathbf{k}_1} \hat{c}^\dagger_{\mathbf{k}_2} \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4}$$
- Cooper contractions:
$$\hat{c}^\dagger_{\mathbf{k}_1} \hat{c}^\dagger_{\mathbf{k}_2} \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4} \qquad \hat{c}^\dagger_{\mathbf{k}_1} \hat{c}^\dagger_{\mathbf{k}_2} \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4}$$

with appropriate sign factors. With the EHM expressed in reciprocal space, there is no general feature left to address. We can move to the detailed discussion of each phase.

2.4 Square lattice spatial symmetry channels

This short conclusive section is apparently rather uncorrelated with all we said in the previous parts. We here introduce a set of orthonormal functions that results particularly useful to represent quantities defined on NNs sites on the square lattice. Let us start from a generic function of two variables $f(\mathbf{r}, \mathbf{r}')$. Suppose that, for a given centre $\mathbf{r}$, the function is non-zero only on the site itself ($\mathbf{r}' = \mathbf{r}$) and its NNs $\mathbf{r}' = \mathbf{r} \pm \boldsymbol{\delta}$, with $\boldsymbol{\delta} \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\}$. In other words,

$$f(\mathbf{r}, \mathbf{r}') \neq 0 \qquad \Longleftrightarrow \qquad |\mathbf{r} - \mathbf{r}'| = 0, 1$$

in units of the lattice step. The following identity holds:

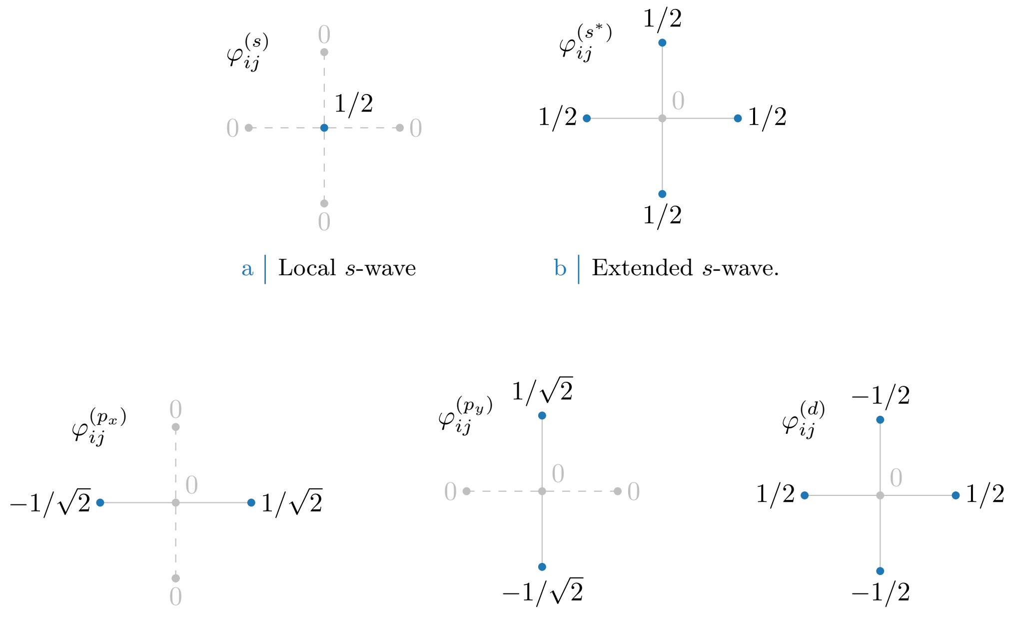
$$\begin{aligned} f(\mathbf{r}, \mathbf{r} + \underline{\mathbf{x}}) &= f(\mathbf{r}, \mathbf{r}')\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} \\ &= f(\mathbf{r}, \mathbf{r}') \left(\varphi^{(s^*)}_{\mathbf{r}\mathbf{r}'} + \varphi^{(d)}_{\mathbf{r}\mathbf{r}'} + \sqrt{2}\varphi^{(p_x)}_{\mathbf{r}\mathbf{r}'}\right) \end{aligned}$$

where we defined the set of orthonormal functions $\varphi^{(\gamma)}_{\mathbf{r}\mathbf{r}'}$ listed explicitly in Tab. 2.4 and sketched in Fig. 2.4. The orthonormality condition reads:

$$\frac{1}{L_x L_y} \sum_{\mathbf{r}\mathbf{r}'} \varphi^{(\gamma)}_{\mathbf{r}\mathbf{r}'} \varphi^{(\gamma')*}_{\mathbf{r}\mathbf{r}'} = \delta_{\gamma\gamma'}$$

Similarly, also the following relations hold:

$$\begin{aligned} f(\mathbf{r}, \mathbf{r}) &= f(\mathbf{r}, \mathbf{r}')\varphi^{(s)}_{\mathbf{r}\mathbf{r}'} \\ f(\mathbf{r}, \mathbf{r} - \underline{\mathbf{x}}) &= f(\mathbf{r}, \mathbf{r}') \left(\varphi^{(s^*)}_{\mathbf{r}\mathbf{r}'} + \varphi^{(d)}_{\mathbf{r}\mathbf{r}'} - \sqrt{2}\varphi^{(p_x)}_{\mathbf{r}\mathbf{r}'}\right) \\ f(\mathbf{r}, \mathbf{r} + \underline{\mathbf{y}}) &= f(\mathbf{r}, \mathbf{r}') \left(\varphi^{(s^*)}_{\mathbf{r}\mathbf{r}'} - \varphi^{(d)}_{\mathbf{r}\mathbf{r}'} + \sqrt{2}\varphi^{(p_y)}_{\mathbf{r}\mathbf{r}'}\right) \\ f(\mathbf{r}, \mathbf{r} - \underline{\mathbf{y}}) &= f(\mathbf{r}, \mathbf{r}') \left(\varphi^{(s^*)}_{\mathbf{r}\mathbf{r}'} - \varphi^{(d)}_{\mathbf{r}\mathbf{r}'} - \sqrt{2}\varphi^{(p_y)}_{\mathbf{r}\mathbf{r}'}\right) \end{aligned}$$



a | Local s-wave

b | Extended s-wave.

c | p_x-wave.

d | p_y-wave.

e | $d_{x^2-y^2}$-wave.

Figure 2.4 | Form factors at different topologies, as listed in Tab. 2.4. In figures five sites are represented: the hub site i and its four NN. Solid lines represent non-zero values for $\varphi_{\boldsymbol{\delta}}$, while dashed lines represent vanishing factors.

Structure	Form factor	Graph
s-wave	$\varphi^{(s)}_{\mathbf{rr}'} = \delta_{\mathbf{rr}'}$	Fig. 2.4a
Extended s-wave	$\varphi^{(s^*)}_{\mathbf{rr}'} = \frac{1}{2}\left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}}\right]$	Fig. 2.4b
p_x-wave	$\varphi^{(p_x)}_{\mathbf{rr}'} = \frac{1}{\sqrt{2}}\left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} - \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}}\right]$	Fig. 2.4c
p_y-wave	$\varphi^{(p_y)}_{\mathbf{rr}'} = \frac{1}{\sqrt{2}}\left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} - \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}}\right]$	Fig. 2.4d
$d_{x^2-y^2}$-wave	$\varphi^{(d)}_{\mathbf{rr}'} = \frac{1}{2}\left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} - \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} - \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}}\right]$	Fig. 2.4e

Table 2.4 | Form factors for a function defined on a square lattice, with a prefactor included for normalization. We use lattice vector notation here. The last column indicates the graph representation of each contribution given in Fig. 2.4. Subscript $x^2 - y^2$ is omitted for notational clarity.

Thus, defining:

$$\begin{aligned}
f^{(s)} &\equiv f(\mathbf{r},\mathbf{r}) \\
f^{(s^*)} &\equiv f(\mathbf{r},\mathbf{r}+\underline{\mathbf{x}}) + f(\mathbf{r},\mathbf{r}-\underline{\mathbf{x}}) + f(\mathbf{r},\mathbf{r}+\underline{\mathbf{y}}) + f(\mathbf{r},\mathbf{r}-\underline{\mathbf{y}}) \\
f^{(p_x)} &\equiv f(\mathbf{r},\mathbf{r}+\underline{\mathbf{x}}) - f(\mathbf{r},\mathbf{r}-\underline{\mathbf{x}}) \\
f^{(p_y)} &\equiv f(\mathbf{r},\mathbf{r}+\underline{\mathbf{y}}) - f(\mathbf{r},\mathbf{r}-\underline{\mathbf{y}}) \\
f^{(d)} &\equiv f(\mathbf{r},\mathbf{r}+\underline{\mathbf{x}}) + f(\mathbf{r},\mathbf{r}-\underline{\mathbf{x}}) - f(\mathbf{r},\mathbf{r}+\underline{\mathbf{y}}) - f(\mathbf{r},\mathbf{r}-\underline{\mathbf{y}})
\end{aligned}$$

we get the decomposition:

$$f(\mathbf{r},\mathbf{r}') = \sum_{\gamma} f^{(\gamma)} \varphi^{(\gamma)}_{\mathbf{rr}'}$$

Structure	Structure factor	Graph
s-wave	$\varphi_{\mathbf{k}}^{(s)} = 1$	Fig. 2.4a
Extended s-wave	$\varphi_{\mathbf{k}}^{(s^*)} = \cos k_x + \cos k_y$	Fig. 2.4b
p_x-wave	$\varphi_{\mathbf{k}}^{(p_x)} = i\sqrt{2}\sin k_x$	Fig. 2.4c
p_y-wave	$\varphi_{\mathbf{k}}^{(p_y)} = i\sqrt{2}\sin k_y$	Fig. 2.4d
$d_{x^2-y^2}$-wave	$\varphi_{\mathbf{k}}^{(d)} = \cos k_x - \cos k_y$	Fig. 2.4e

Table 2.5 Structure factors derived from the correlation structures of Tab. 2.4. The functions hereby defined are orthonormal, and are plotted in Fig. 2.5.

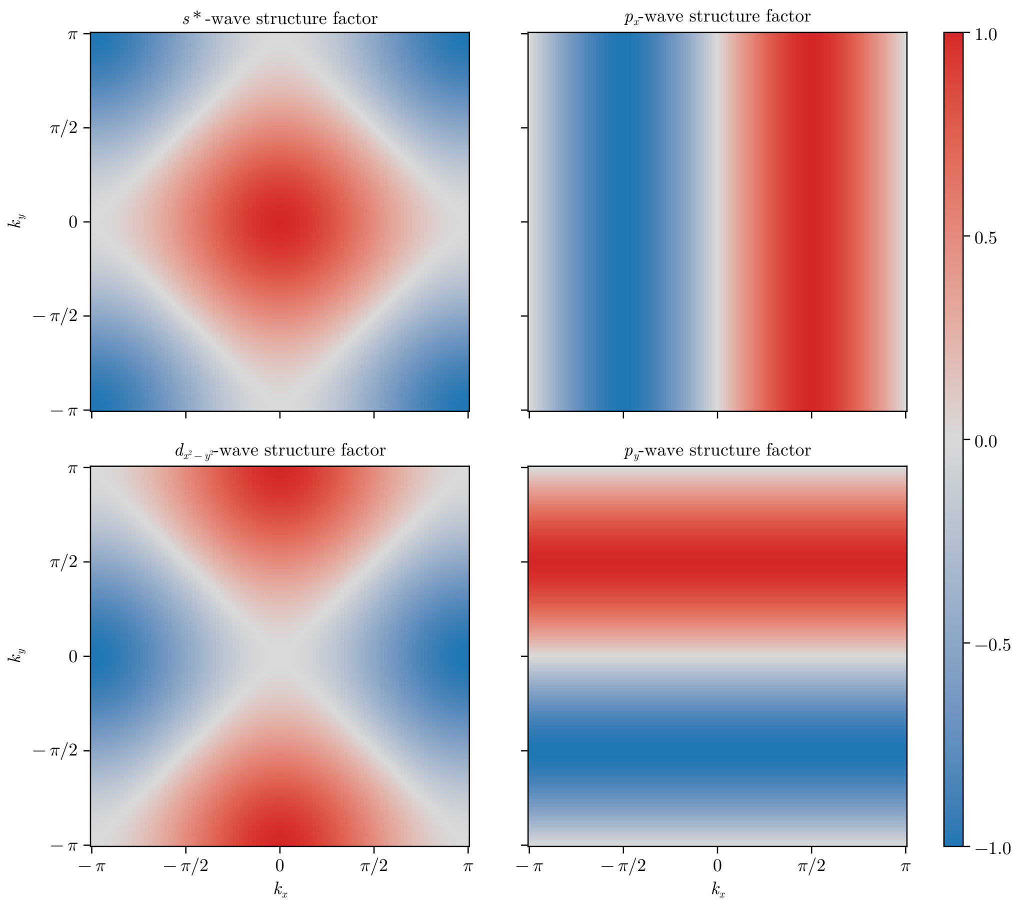


Figure 2.5 Representation of the various structure factors listed in Tab. 2.5 on the BZ. For the s^* and $d_{x^2-y^2}$-wave factors, the real part is plotted; for the p_ℓ factors, the imaginary part is plotted.

We decomposed the original function in its harmonics. Moving to reciprocal space, we get the $\mathcal{F}$-transformed functions of Tab. 2.5, also satisfying the orthonormality condition:

$$\frac{1}{L_x L_y} \sum_{\mathbf{k}} \varphi_{\mathbf{k}}^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma')*} = \delta_{\gamma\gamma'}$$

The functions of Tab. 2.5 are plotted in the BZ in Fig. 2.5.

We will refer to these harmonics as "symmetry channels", indicating the various spatial components of the order parameter associated to a given phase transition. As we will see, indeed, we consider order parameters that have specific transformation properties under planar rotations.

To distinguish the various spatial harmonics of the order parameter is useful in order to apply symmetry-based selection rules.

Finally, a very useful mathematical identity is given by the following argument. Consider Eq. (2.29), where is present the factor $\cos(\delta k_x) + \cos(\delta k_y)$ (with $\delta k_x = k_x - k'_x$ and $\delta k_y = k_y - k'_y$). Expanding,

$$\cos(\delta k_x) + \cos(\delta k_y) = c_x c'_x + c_y c'_y + s_x s'_x + s_y s'_y$$

In last passage, for the sake of readability, we used the notations

$$c_\ell \equiv \cos k_\ell \qquad s_\ell \equiv \sin k_\ell \qquad c'_\ell \equiv \cos k'_\ell \qquad s'_\ell \equiv \sin k'_\ell$$

Group the four terms,

$$\underbrace{\left(c_x c'_x + c_y c'_y\right)}_{\text{Symmetric}} + \underbrace{\left(s_x s'_x + s_y s'_y\right)}_{\text{Anti-symmetric}} \tag{2.32}$$

All four terms are invariant under simultaneous inversion of both momenta, $(\mathbf{k}, \mathbf{k}') \to (-\mathbf{k}, -\mathbf{k}')$. Instead, if we invert just one momentum of the pair, the first two are symmetric for both arguments $\mathbf{k}$, $\mathbf{k}'$; the second two are anti-symmetric. Decoupling the symmetric part, we obtain

$$c_x c'_x + c_y c'_y = \frac{1}{2}(c_x + c_y)(c'_x + c'_y) + \frac{1}{2}(c_x - c_y)(c'_x - c'_y)$$

which finally gives:

$$\begin{aligned} \cos(\delta k_x) + \cos(\delta k_y) = {} & \frac{1}{2}(c_x + c_y)(c'_x + c'_y) && (s^*\text{-wave}) \\ & + s_x s'_x && (p_x\text{-wave}) \\ & + s_y s'_y && (p_y\text{-wave}) \\ & + \frac{1}{2}(c_x - c_y)(c'_x - c'_y) && (d_{x^2-y^2}\text{-wave}) \end{aligned}$$

or, in a more compact form

$$\cos(\delta k_x) + \cos(\delta k_y) = \frac{1}{2}\sum_\gamma \varphi_{\mathbf{k}}^{(\gamma)} \varphi_{\mathbf{k}'}^{(\gamma)*} \quad \text{for } \gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\} \tag{2.33}$$

$$= \frac{1}{2}\sum_\gamma \varphi_{\mathbf{k}}^{(\gamma)*} \varphi_{\mathbf{k}'}^{(\gamma)} \tag{2.34}$$

This decomposition will result particularly handy in later calculations.

Chapter 3

The normal phase

ToDo

- Clarify the Mean-Field decomposition of quartic operators OR use the self-consistent fields

In this chapter we apply the HF methods sketched in Sec. 1.2 to the “normal phase”. We are studying the model in a state were no symmetry of those enumerated in Tab. 2.1 is broken. Even though this phase may appear not so interesting, it actually shows a significant physical difference with respect to the conventional perfectly degenerate Fermi gas.

HF methods applied in this context will lead us to a self-consistent hopping renormalisation effect. For certain values of the non-local attraction V and the doping level, a complete Mott localisation effect takes place. The resulting bands shrinking and flattening is the main net effect of the presence of $\hat{H}_V$ in the EHM.

This chapter is divided in three main parts. First, we discuss the symmetries of the normal phase, following the general discussion of Chap. 2, and apply HF methods to derive the self-consistent approximated hamiltonian. Second, we impose some theoretical constraints on the self-consistent solution. Lastly, we present numeric results.

3.1 The “normal” phase and its symmetries

In the normal phase no symmetry is broken. Thus, the HF wavefunction approximating the ground state must exhibit the full $\mathrm{U}(2) \otimes \mathrm{C}_4 \otimes \mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)}$ symmetry discussed in Sec. 2.1. We take a slight detour from this, and instead allow for

$$\mathrm{C}_4 \overset{\mathrm{SSB}}{\rightarrow} \mathrm{C}_2$$

This is the only potential SSB we consider. It will turn out in Sec. 3.5.2 that even allowing for breaking of rotational symmetry, the ground-state actually does not break C_4. This can be seen as a convolute way to confirm that there is not a normal-nematic phase. While this may seem as an useless complication, the derivation will set grounds for actual nematic effects found in the superconducting phase.

Our HF method relies on the application of Wick’s theorem on the quartic part of the interacting hamiltonian, expressed in Eq. (1.3). [It’s not that immediate how Wick’s theorem gives HF decomposition.]

Recalling Tabs. 2.2 and 2.3, for the normal phase we must consider the following contributes to the EVs of Eqns. (2.13), (2.18), (2.20):

$\hat{H}_U$	Hartree contraction
$\hat{H}_V$ (o.s. sector)	Hartree contraction
$\hat{H}_V$ (s.s. sector)	Hartree and Fock contractions

As we will see in detail and is summarised in Tab. 3.1, Hartree contractions give a chemical potential shift, while the unique Fock contraction effectively renormalises the bare bands.

Operator	Sector	RWCs	Net effect
$\hat{H}_U$		Hartree	μ shift by $-nU$
$\hat{H}_V$	o.s.	Hartree	μ shift by nzV
	s.s.	Hartree	μ shift by nzV
		Fock	Bands renormalization and resymmetrization

Table 3.1 | Relevant Wick's contractions and net effect in the normal phase.

In the normal phase, spin is a pure physical degeneracy, being the $\uparrow$ and $\downarrow$ sectors separated; moreover, $\hat{H}$ is $\mathbb{Z}_2$ invariant with respect to spin inversion, $\sigma \leftrightarrow \overline{\sigma}$. This implies that all physical quantities, as is for the normal phase, should not depend on the spin index. We will perform all computations leaving σ explicit and using this consideration at the end.

3.2 Hartree effects: chemical potential renormalization

Let us start by the analysis of the Hartree channel. The Hartree channel is defined as operator resulting from summing Hartree-like terms originating from Wick's decomposition of the quartic parts of $\hat{H}$. Each quartic term decomposes as a sum of fully contracted pairs, as in Eq. (1.3).

In the final hamiltonian, the net effect originating from Hartree terms will be a renormalisation of μ. To derive analytically this effect is only important to the ends of computing correctly the free energy density of the approximated ground state. As is discussed in Sec. 1.3.1, in the context of numerical simulations at fixed density, the chemical potential is determined self-consistently at each step; thus any net effect that shifts μ is effectively ignored within (and limitedly to) the iterative algorithm.

3.2.1 Hartree channel in the normal phase: analytic derivation

The quartic part of $\hat{H}$ is given by the the local repulsion $\hat{H}_U$ and the non-local attraction $\hat{H}_V$. [To be continued...]

Local repulsion. In the normal phase, the local repulsion $\hat{H}_U$ decomposes in Hartree channel as

$$\hat{H}_U \simeq U \sum_i \left[\langle \hat{n}_{i\uparrow}\rangle \hat{n}_{i\downarrow} + \hat{n}_{i\uparrow}\langle \hat{n}_{i\downarrow}\rangle - \langle \hat{n}_{i\downarrow}\rangle\langle \hat{n}_{i\uparrow}\rangle\right]$$

For a translationally invariant phase such as the normal phase, it holds

$$\langle \hat{n}_{i\sigma}\rangle = n \tag{3.1}$$

which gives

$$\begin{aligned}\hat{H}_U &\simeq nU \sum_i \left[\hat{n}_{i\uparrow} + \hat{n}_{i\downarrow}\right] - U \sum_i n^2 \\ &= nU \times \hat{N} - E^{(\mathrm{N})}_{\mathrm{H}/U}\end{aligned}$$

being $\hat{N}$ the total number operator and $E^{(\mathrm{N})}_{\mathrm{H}/U}$ the "contraction energy" (also known as double counting term). In Sec. 3.2.2 we deal with it. The chemical potential is corrected by

$$\mu \to \mu - nU \tag{3.2}$$

The non-local attraction further corrects μ.

Non-local attraction. The non-local repulsion is divided in two sectors, as in Eq. (2.15). Its Hartree-like terms are:

$$\begin{aligned}\hat{H}_V \simeq &\overbrace{-V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{n}_{i\sigma}\rangle\hat{n}_{j\sigma}+\hat{n}_{i\sigma}\langle\hat{n}_{j\sigma}\rangle-\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\sigma}\rangle\right]}^{\text{s.s.}}\\ &\underbrace{-V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{n}_{i\sigma}\rangle\hat{n}_{j\bar{\sigma}}+\hat{n}_{i\sigma}\langle\hat{n}_{j\bar{\sigma}}\rangle-\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\bar{\sigma}}\rangle\right]}_{\text{o.s.}}+(\text{all the rest})\end{aligned} \tag{3.3}$$

where “all the rest” collects all non-Hartree terms. Recalling the Ansatz of Eq. (3.1), we get

$$\hat{H}_V \simeq \underbrace{-nV\sum_{\langle ij\rangle}\sum_{\sigma}\left[\hat{n}_{j\sigma}+\hat{n}_{i\sigma}\right]}_{\text{s.s.}}\underbrace{-nV\sum_{\langle ij\rangle}\sum_{\sigma}\left[\hat{n}_{j\bar{\sigma}}+\hat{n}_{i\sigma}\right]}_{\text{o.s.}}-E^{(\mathrm{N})}_{\mathrm{H}/V}+(\text{all the rest})$$

with $E^{(\mathrm{N})}_{\mathrm{H}/V}$ the normal state shift to energy brought by $\hat{H}_V$. Both sums above give a number operator. Note that we sum over NNs: each site is connected to $z = 4$ sites. Including this combinatorial factor, we have

$$\hat{H}_V \simeq -2nzV\times\hat{N}-E^{(\mathrm{N})}_{\mathrm{H}/V}+(\text{all the rest})$$

This accounts for the final Hartree shift of μ,

$$\mu \to \mu + 2znV \tag{3.4}$$

Considering the net shift to μ due to both interactions by combining Eqns. (3.2), (3.4), we get the final RMP anticipated in Tab. 3.1,

$$\tilde{\mu} \equiv \mu + n(2zV - U) \tag{3.5}$$

This result will remain valid throughout all phases discussed in this text. For the AF phase, Sec. **??**, the SDW character leaves this shift untouched. The SC phase, Sec. **??**, we will discuss is inherently translational invariant, thus the computation is the same as here.

3.2.2 Hartree contraction energy

The double counting terms accounting for energy shifts in the Hartree channel are, as anticipated:

$$\begin{aligned}E^{(\mathrm{N})}_{\mathrm{H}/U} &= \sum_i\langle\hat{n}_{i\downarrow}\rangle\langle\hat{n}_{i\uparrow}\rangle\\ &= L_xL_y\times n^2U\end{aligned} \tag{3.6}$$

for the local repulsion and

$$\begin{aligned}E^{(\mathrm{N})}_{\mathrm{H}/V} &= -V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\sigma}\rangle+\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\bar{\sigma}}\rangle\right]\\ &= -2V\sum_{\langle ij\rangle}\sum_{\sigma}n^2\\ &= -L_xL_y\times 2zn^2V\end{aligned} \tag{3.7}$$

for the non-local attraction. In the last passage, we used Eq. (2.12) and spin degeneracy. The contraction energy is an extensive quantity. When computing total free energy density, we will use the contraction energy per site.

3.3 Fock effects: bare bands renormalization

Analysis of Fock terms will lead us to the effect that makes the normal phase interesting, when compared to the free Fermi gas on the square lattice. From Wick's decomposition of $\hat{H}_V$, the unique allowed Fock term comes from the same-spin sector. This selection rules is implied by the SU(2) symmetry of the normal phase. Then:

$$\hat{H}_V \simeq (\text{Hartree channel}) + V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle \right] \tag{3.8}$$

Note the + sign in front of the displayed term: it originates from the − sign in Fock term of Eq. (1.3). A bond-wise hopping amplitude can be defined,

$$\tilde{t}_{ij\sigma} \equiv t - V \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle$$

The effective kinetic hamiltonian is given by

$$\begin{aligned} \hat{H}_{\tilde{t}} &= \hat{H}_t + V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} + \text{h.c.} \right] \\ &= - \sum_{\langle ij \rangle} \sum_{\sigma} \left[\tilde{t}_{ij\sigma} \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} + \text{h.c.} \right] \end{aligned}$$

To proceed, it is useful to move to reciprocal space.

3.3.1 Fock channel in the normal phase: analytic derivation

In reciprocal space, the effective hopping must be transformed as well. Consider the Fourier Transform $\mathcal{F}$ given in Eq. (2.31), applied to the displayed part of Eq. (3.8),

$$\begin{aligned} &V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle \right] \\ &\qquad \stackrel{\mathcal{F}}{=} \frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \sum_{\sigma} \left[\cos(\delta k_x) + \cos(\delta k_y) \right] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \end{aligned} \tag{3.9}$$

Here, the 2 prefactor comes from recognizing that the first two operators in square brackets in the first line generate identical contributions to the full sum (as is easy to see by performing trivial variables changes); the double counting energy shift is given by

$$E^{(\mathrm{N})}_{\mathrm{F}/V} \equiv \frac{V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \sum_{\sigma} \left[\cos(\delta k_x) + \cos(\delta k_y) \right] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \rangle \langle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \rangle \tag{3.10}$$

Up to this point, this derivation is general and will be re-used in next chapters. Here the specificity of the physical phase settles in: if we assume the normal state to be a free degenerate Fermi gas with renormalized bands $\tilde{\epsilon}_{\mathbf{k}}$, we get

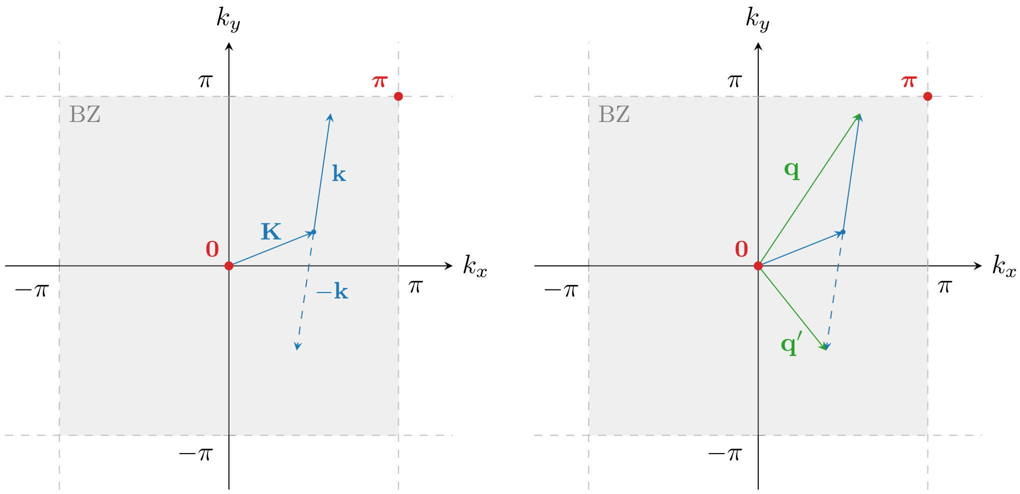
$$\langle \hat{c}^\dagger_{\mathbf{k}_1\sigma_1} \hat{c}_{\mathbf{k}_2\sigma_2} \rangle = \delta_{\mathbf{k}_1=\mathbf{k}_2} \delta_{\sigma_1=\sigma_2} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \tag{3.11}$$

being f the Fermi-Dirac distribution,

$$f(\epsilon; \beta, \mu) \equiv \frac{1}{e^{\beta(\epsilon-\mu)} + 1}$$

It follows that Eq. (3.9) becomes

$$\begin{aligned} &V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle \right] \\ &\qquad \stackrel{\mathcal{F}}{=} \frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_{\sigma} \left[\cos(2k_x) + \cos(2k_y) \right] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \end{aligned} \tag{3.12}$$



a | Generic vectors. b | Change of variables.

Figure 3.1 | Representation of the variable change employed to pass from Eq. (3.12) to (3.13). In Fig. 3.1a generic vectors are considered, cycling over all values of $\mathbf{K}, \mathbf{k} \in \mathrm{BZ}$. In Fig. 3.1b is depicted the variables change to the new vectors $\mathbf{q}, \mathbf{q}'$.

since the only contribution comes from $\mathbf{k}' = -\mathbf{k}$.

Let now $\mathbf{q} \equiv \mathbf{K} + \mathbf{k}$, $\mathbf{q}' \equiv \mathbf{K} - \mathbf{k}$. A representation of this variables change is given in Fig. 3.1. Being for $\ell = x, y$

$$\begin{aligned}\delta q_\ell &\equiv q_\ell - q'_\ell \\ &= (K_\ell + k_\ell) - (K_\ell - k_\ell) \\ &= 2k_\ell\end{aligned}$$

we reduce Eq. (3.9) to

$$V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle \right]$$

$$\stackrel{\mathcal{F}}{=} \frac{2V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_{\sigma} \left[\cos(\delta q_x) + \cos(\delta q_y) \right] \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \tag{3.13}$$

Recall now the result of the symmetry channels decomposition of Eq. (2.33),

$$\cos(\delta q_x) + \cos(\delta q_y) = \frac{1}{2} \sum_{\gamma} \varphi^{(\gamma)}_{\mathbf{q}} \varphi^{(\gamma)*}_{\mathbf{q}'} \quad \text{for } \gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\}$$

Thanks to this handy property, we reduce Eq. (3.13) to

$$V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle \right]$$

$$\stackrel{\mathcal{F}}{=} \sum_{\gamma,\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q} \in \mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} u_{\mathbf{q}\sigma} \right] \times V \sum_{\mathbf{q}' \in \mathrm{BZ}} \varphi^{(\gamma)*}_{\mathbf{q}'} \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \tag{3.14}$$

where we defined the EV:

$$u_{\mathbf{q}\sigma} \equiv \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle \qquad u_{\mathbf{q}\sigma} = u^*_{\mathbf{q}\sigma} \tag{3.15}$$

The numeric value of $u_{\mathbf{q}\sigma}$ is given by the Fermi-Dirac distribution, but leaving it implicit we can re-use this computation later. Let $u^{(\gamma)}_\sigma$ be its γ-wave component,

$$u^{(\gamma)}_\sigma = \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle$$

Symmetry	Self-consistency equation	Graph
s^*-wave	$u^{(s^*)} = \dfrac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$	Fig. 2.4b
$d_{x^2-y^2}$-wave	$u^{(d)} = \dfrac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x - \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$	Fig. 2.4e

Table 3.2 | Symmetry resolved self-consistency equations for the harmonics of $u_{\mathbf{k}}$.

The bare bands $\epsilon_{\mathbf{k}}$ are s^*-wave symmetric,

$$\epsilon_{\mathbf{k}} = -2t(\cos k_x + \cos k_y) = -4t\varphi_{\mathbf{k}}^{(s^*)}$$

We used one function of those listed in Tab. 2.5. The renormalised bands $\tilde{\epsilon}_{\mathbf{k}}$ can exhibit a more complex symmetry. In the normal phase inversion symmetry is unbroken, thus

$$u_{\mathbf{q}\sigma} = u_{-\mathbf{q}\sigma}$$

It follows that antisymmetric harmonics are zero,

$$u_\sigma^{(p_\ell)} = 0 \quad \text{for} \quad \ell \in \{x, y\}$$

The remaining s^* and d symmetries give legitimate HFPs with the respective self-consistency equations, reported in Tab. 3.2.Spin index is omitted due to spin degeneracy.

The two HFPs $u_\sigma^{(s^*)}$ and $u_\sigma^{(d)}$ are in general non-zero, since we are allowing for $\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2$. Note that when imposing C_4 invariance the $u_\sigma^{(d)}$ HFP vanishes, while in general $u_\sigma^{(s^*)}$ is non-zero. From these considerations Eq. (3.14) becomes

$$\begin{aligned} V \sum_{\langle ij\rangle} \sum_{\sigma} \Big[\langle \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} \rangle \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma} + \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} \langle \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma} \rangle - \langle \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} \rangle \langle \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma} \rangle \Big] \\ \overset{\mathcal{F}}{=} V \sum_{\mathbf{q}\in\mathrm{BZ}} \sum_{\sigma} \Big[u_\sigma^{(s^*)} \varphi_{\mathbf{q}}^{(s^*)*} + u_\sigma^{(d)} \varphi_{\mathbf{q}}^{(d)*} \Big] \hat{c}_{\mathbf{q}\sigma}^\dagger \hat{c}_{\mathbf{q}\sigma} - E_{\mathrm{F}/V}^{(\mathrm{N})} \end{aligned} \tag{3.16}$$

Let

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv \epsilon_{\mathbf{k}\sigma} + u_\sigma^{(s^*)} V(\cos k_x + \cos k_y) + u_\sigma^{(d)} V(\cos k_x - \cos k_y) \tag{3.17}$$

This equation defines the bands renormalisation anticipated in Tab. 3.1, and is actually spin-independent. The fact that we have both s^* and a d-wave components introduces an anisotropy that is discussed in next section. However, we anticipate here that, as will result from Sec. 3.5.2, the d-wave harmonic will be suppressed. For now we keep this component ignoring these anticipations.

3.3.2 Fock contraction energy

Finally, let us derive the Fock contraction energy by expanding Eq. (3.10):

$$\begin{aligned} E_{\mathrm{F}/V}^{(\mathrm{N})} &= \frac{V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_{\sigma} \left[\cos(\delta q_x) + \cos(\delta q_y)\right] \langle \hat{c}_{\mathbf{q}\sigma}^\dagger \hat{c}_{\mathbf{q}'\sigma} \rangle \langle \hat{c}_{\mathbf{q}\sigma}^\dagger \hat{c}_{\mathbf{q}'\sigma} \rangle \\ &= L_x L_y \times \frac{V}{2} \sum_{\gamma,\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} u_{\mathbf{q}\sigma} \right] \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}'\in\mathrm{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} u_{\mathbf{q}'\sigma}^* \right] \\ &= L_x L_y \times \frac{V}{2} \sum_{\sigma} \left[\left| u_\sigma^{(s^*)} \right|^2 + \left| u_\sigma^{(d)} \right|^2 \right] \end{aligned} \tag{3.18}$$

This expression gives the Fock contraction energy coming from the $\hat{H}_V$ operator. The squared spectral amplitude of $u_{\mathbf{k}\sigma}$ here is summed, and contributes to contraction energy linearly in V. This suggests that a mixed-symmetry band could be energetically preferred.

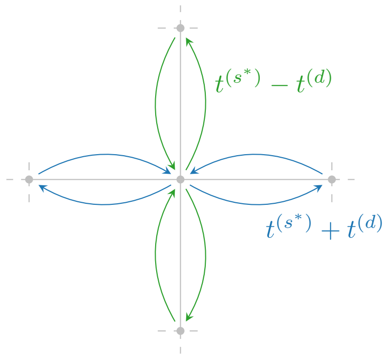


Figure 3.2 | Graphic rendition of anisotropic hopping, with a s^*-wave component (symmetric by $\pi/$ rotations) and a d-wave component (antisymmetric).

3.3.3 Nematicity: extending hopping to d-wave bands

For the normal phase, there is no reason to expect nematic effects, nor to exclude them. They are observed in superconducting phase [22]. Thus, we allow the possibility.

From Tabs. 2.4 and 2.5 we can inverse-Fourier transform the form factors to get the full effective hopping,

$$\tilde{t}_{ij\sigma} = \sum_{\gamma \in \{s^*, d\}} t_\sigma^{(\gamma)} \varphi_{ij}^{(\gamma)}$$

with

$$t_\sigma^{(s^*)} = t - \frac{u_\sigma^{(s^*)} V}{2} \qquad t_\sigma^{(d)} = -\frac{u_\sigma^{(d)} V}{2}$$

This is what we intend for "nematic effects". The possibility that planar rotational symmetry is broken in such a way that the kinetic operators in x and y directions differ. A schematics of anisotropic hopping is given in Fig. 3.2: for each bond linking two sites, the hopping parameter can be a positive or negative combination of the two components, whether the link direction is vertical or horizontal. The solution needs to be invariant under exchange $x \leftrightarrow y$, thus the actual SSB is $\mathrm{C}_4 \overset{\text{SSB}}{\to} \mathrm{C}_2/\mathbb{Z}_2$.

The fact that hopping is here anisotropic does not break full translational invariance $\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)}$. Imposing a free degenerate Fermi gas ground state in Eq. (3.11) leads to a uniform density. This is proven by Fourier-transforming the site number operator according to the convention of Eq. (2.22),

$$\hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}\sigma} \overset{\mathcal{F}}{=} \frac{1}{L_x L_y} \sum_{\mathbf{k},\mathbf{k}'} e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}} \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}'\sigma}$$

Then the on-site density EV is

$$\langle \hat{n}_{\mathbf{r}\sigma} \rangle = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

A perfectly degenerate Fermi gas is always translational invariant and thus cannot exhibit non-uniform symmetry. The fact that hopping is allowed to be anisotropic does not interfere with that.

Note that for null HFPs,

$$u_\sigma^{(s^*)} = 0 \qquad u_\sigma^{(d)} = 0$$

we have, recalling Tab. 2.4:

$$\tilde{t}_{\mathbf{r}\mathbf{r}'\sigma} = \frac{t}{2} \left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}} \right]$$

The fact that $t/2$ appears here instead of t is a consequence of the equivalence:

$$-t \sum_{\langle ij \rangle} \sum_\sigma \left[\hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + \text{h.c.} \right] = -\frac{t}{2} \sum_{i,j \in \mathcal{S}} \sum_\sigma \left[\hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + \text{h.c.} \right]$$

where we including a prefactor that corrects double-counting. The RMP $\tilde{t}_{ij\sigma}$ must be interpreted like the right-hand side of the above equation, with the kinetic operator becoming:

$$\hat{H}_{\tilde{t}} \equiv -\frac{1}{2}\sum_{i,j\in\mathcal{S}}\sum_{\sigma}\left[\tilde{t}_{ij\sigma}\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma} + \text{h.c.}\right]$$

To limit the sum to NNs is a matter of the symmetry structure of the function $\tilde{t}_{ij\sigma}$.

We can define the direction-dependent hopping as

$$\tilde{t}^{(x)} \equiv t - \frac{u^{(s^*)} - u^{(d)}}{2}V \tag{3.19}$$

$$\tilde{t}^{(y)} \equiv t - \frac{u^{(s^*)} + u^{(d)}}{2}V \tag{3.20}$$

Both components depend linearly on V. The HFP $u^{(s^*)}$ is a measure of the direction-averaged hopping shift with respect to V, while $u^{(d)}$ measures the anisotropy,

$$u^{(s^*)} = \frac{\left(t - \tilde{t}^{(x)}\right) + \left(t - \tilde{t}^{(y)}\right)}{V} \qquad u^{(d)} = \frac{\tilde{t}^{(y)} - \tilde{t}^{(x)}}{V}$$

This discussion is general, and not specific to the normal phase. In next chapters we will interpret phase renormalisation as in this section.

3.4 Normal free energy density

In the end, we want to compare phases. Thus we need a way to extract the free energy density of the approximated ground-state for each phase. The procedure sketched here will be re-used in next chapters. Let

$$f^{(\mathrm{N})}_{\mathrm{MFT}} \equiv \frac{F^{(\mathrm{N})}_{\mathrm{MFT}}}{L_x L_y}$$

being F_{MFT} the total free energy. Consider the complete hamiltonian

$$\hat{H} - \mu\hat{N} \simeq \sum_{\mathbf{k},\sigma}\tilde{\epsilon}_{\mathbf{k}\sigma}\hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} + E^{(\mathrm{N})}_{\mathrm{c}} - \tilde{\mu}\hat{N}$$

where $E^{(\mathrm{N})}_{\mathrm{c}}$ is the (negative) total contraction energy for the normal phase,

$$E^{(\mathrm{N})}_{\mathrm{c}} \equiv -\left[E^{(\mathrm{N})}_{\mathrm{H}/U} + E^{(\mathrm{N})}_{\mathrm{H}/V} + E^{(\mathrm{N})}_{\mathrm{F}/V}\right]$$

having included the various contraction energies from Eqns. (3.6), (3.7) and (3.18).

Let us now take together all terms. From now on, we will omit spin index σ, making use of the always present $\mathbb{Z}_2$ spin inversion symmetry discussed in Sec. 3.1. For a free Fermi gas, the free energy density is made of four contributions:

1. The free particles ground state energy density,

$$e^{(\mathrm{N})}_{\mathrm{g}} = \frac{2}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}(\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu})f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})$$

2. The contraction energy density,

$$\begin{aligned} e^{(\mathrm{N})}_{\mathrm{c}} &\equiv \frac{E^{(\mathrm{N})}_{\mathrm{c}}}{L_x L_y} \\ &= -n^2 U + V\left[2zn^2 - \left|u^{(s^*)}\right|^2 - \left|u^{(d)}\right|^2\right] \end{aligned}$$

3. The entropic contribution to free energy

$$e_s^{(\mathrm{N})} \equiv -Ts$$

with $T = 1/k_B\beta$ and s the free Fermi gas entropy density for a single spin sector [1],

$$s = -\frac{k_B}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\left[(1-f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}))\ln(1-f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) + f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\ln f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\right] \tag{3.21}$$

We use the identity:

$$\ln\left(1-\frac{1}{e^x+1}\right) = x + \ln\left(\frac{1}{e^x+1}\right)$$

and, substituting $x = \beta(\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu})$, reduce the above expression to:

$$\begin{aligned} s = -\frac{k_B}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\Big[&\ln(1-f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) + f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\ln f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) \\ &- \beta(\tilde{\epsilon}_{\mathbf{k}}-\tilde{\mu})f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\ln f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\Big]\end{aligned}$$

Second and fourth terms cancel out, and multiplying by $-2T$ (taking care of spin degeneracy) we have

$$e_s^{(\mathrm{N})} = \frac{2}{\beta L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\ln(1-f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) - \frac{2}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}(\tilde{\epsilon}_{\mathbf{k}}-\tilde{\mu})f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})$$

4. The chemical potential correction

$$e_n^{(\mathrm{N})} = \mu n$$

Note here that we are using the "physical" value of μ, not its RMP counterpart $\tilde{\mu}$. The reason for this is discussed in Sec. 1.3.1.

Thus, composing everything

$$\begin{aligned} f_{\mathrm{MFT}}^{(\mathrm{N})} &\equiv e_{\mathrm{g}}^{(\mathrm{N})} + e_{\mathrm{c}}^{(\mathrm{N})} + e_s^{(\mathrm{N})} + e_n^{(\mathrm{N})} \\ &= \frac{2}{\beta L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\ln(1-f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) - V\left[\left|u^{(s^*)}\right|^2 + \left|u^{(d)}\right|^2\right] + [\mu + n(2zV-U)]\,n\end{aligned}$$

In Sec. 1.3.1 we argued that our iterative scheme computes $\tilde{\mu}$, not μ. Using Eq. (3.5), we see

$$\mu + n(2zV - U) = \tilde{\mu}$$

Then, the final expression is:

$$f_{\mathrm{MFT}}^{(\mathrm{N})} = \frac{2}{\beta L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\ln(1-f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) - V\left[\left|u^{(s^*)}\right|^2 + \left|u^{(d)}\right|^2\right] + \tilde{\mu}n \tag{3.22}$$

including all effects.

3.5 Theoretical constraints on the normal phase HFPs

Within this section we discuss some theoretical constraints on the two HFPs of the normal phase. The discussion is divided in two parts: first we consider the simpler case of unviolated C_4 symmetry, then we move to anisotropic hopping and $\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2$.

Before starting, it is useful to spend a few words on the spatial structure of a mixed $s^*\otimes d$-wave band. Let in general for notational simplicity

$$\tilde{\epsilon}_{\mathbf{k}} = \epsilon^{(s^*)}(\cos k_x + \cos k_y) + \epsilon^{(d)}(\cos k_x - \cos k_y)$$

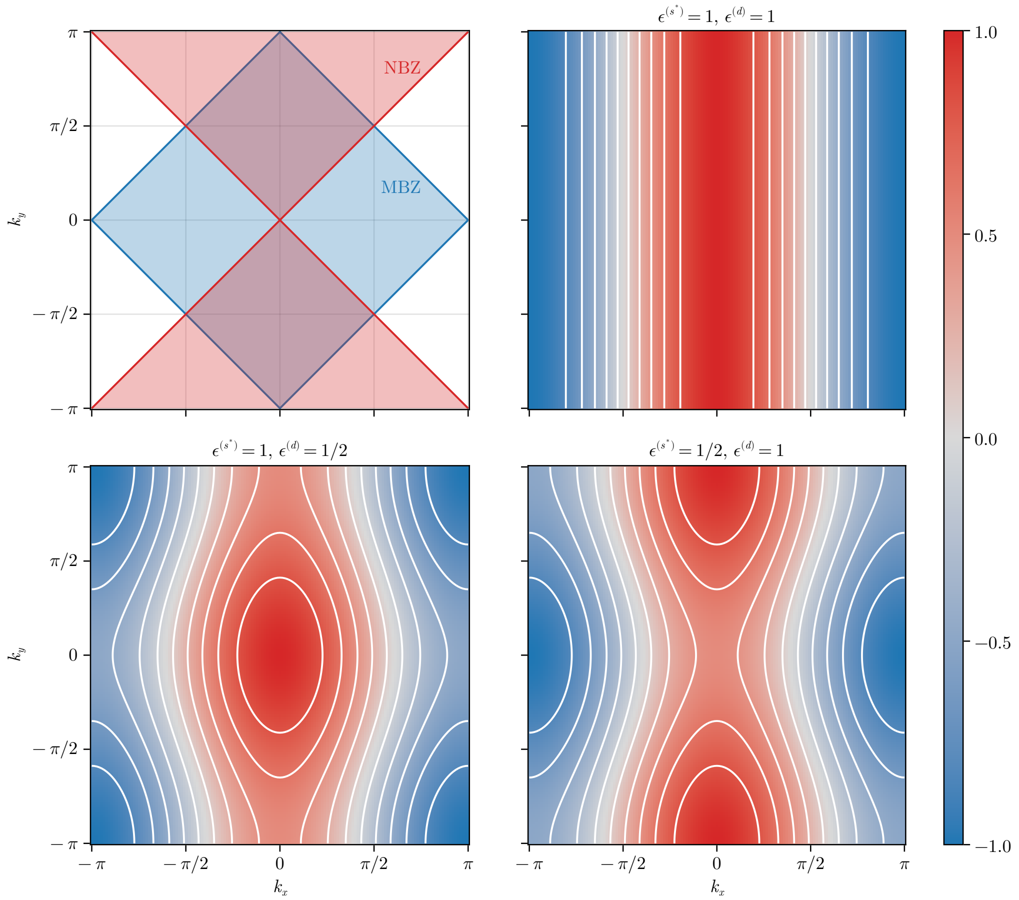


Figure 3.3 | Theoretical representation of mixed-symmetry nematic bands for the normal phase. The white contour represent level lines. In top-left panel, the MBZ and NBZ are depicted. While these plots only account for positive-defined components, in text is described how to map each one on its counterpart with different sign.

It is the balance between the two components $\epsilon^{(s^*)}$ and $\epsilon^{(d)}$ that gives the main features of the renormalised band. Consider the s^*-wave and d-wave structure factors already presented in Fig. 2.5, and define the Magnetic Brillouin Zone (MBZ) and Nematic Brillouin Zone (NBZ) as follows:

$$\text{MBZ} \equiv \left\{ \mathbf{k} \in \text{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \geq 0 \right\}$$
$$\text{NBZ} \equiv \left\{ \mathbf{k} \in \text{BZ} \mid \varphi_{\mathbf{k}}^{(d)} \geq 0 \right\}$$

represented in the top-left panel of Fig. 3.3. The reason for the name of the MBZ will be clear in Chap. **??**. In the rest of the panels of Fig. 3.3 are reported examples of mixed symmetry bands, for positive components. The respective opposite sign plots are obtained by the following rules:

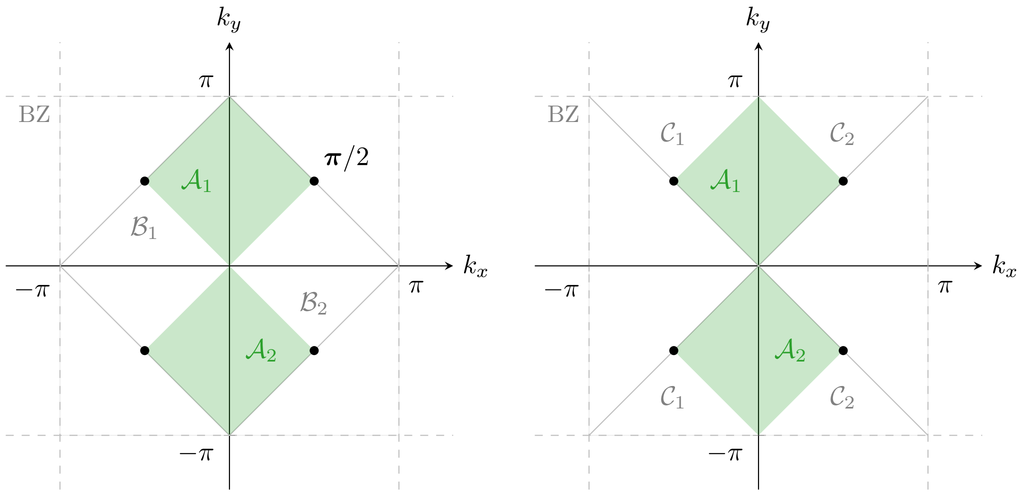
$$\begin{aligned}
\epsilon^{(s^*)} < 0 \quad &\text{and} \quad \epsilon^{(d)} < 0 \quad \Longrightarrow \quad \text{Invert colors in Fig. 3.3} \\
\epsilon^{(s^*)} > 0 \quad &\text{and} \quad \epsilon^{(d)} < 0 \quad \Longrightarrow \quad \text{Rotate plots by } \pi/2 \text{ in Fig. 3.3} \\
\epsilon^{(s^*)} < 0 \quad &\text{and} \quad \epsilon^{(d)} > 0 \quad \Longrightarrow \quad \text{Rotate plots by } \pi/2 \text{ and invert colors in Fig. 3.3}
\end{aligned}$$

Let us focus on the MBZ and the NBZ. Starting from the intersections of the two zones, both can be factored in two sectors of interest,

$$\text{MBZ} = \mathcal{A} \cup \mathcal{B} \qquad \text{NBZ} = \mathcal{A} \cup \mathcal{C}$$

being, accordingly to Fig. 3.4,

$$\mathcal{A} \equiv \mathcal{A}_1 \cup \mathcal{A}_2 \qquad \mathcal{B} \equiv \mathcal{B}_1 \cup \mathcal{B}_2 \qquad \mathcal{C} \equiv \mathcal{C}_1 \cup \mathcal{C}_2$$



a | MBZ sectors. b | NBZ sectors.

Figure 3.4 | Depiction of the sectors composing the MBZ (Fig. 3.4a) and the NBZ (Fig. 3.4b), based on the divisions shown in the top-left panel of Fig. 3.3. The full green zone composes $\mathcal{A}$, the region where both the s^*-wave and d-wave structure factors are positive defined; its MBZ complementary white counterpart in the left panel gives $\mathcal{B}$, for which the s^*-wave structure factors is positive-definite while the d-wave is negative. Similarly the NBZ complementary white counterpart in the right panel gives $\mathcal{C}$, for which signs are exchanged. The black dots, at the contours intersection, are band nodes.

These domains satisfy:

$$\begin{aligned}
\mathcal{A} &\equiv \left\{ \mathbf{k} \in \mathrm{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \geq 0 \quad \text{and} \quad \varphi_{\mathbf{k}}^{(d)} \geq 0 \right\} \\
&= \mathrm{MBZ} \cap \mathrm{NBZ} \\
\mathcal{B} &\equiv \left\{ \mathbf{k} \in \mathrm{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \geq 0 \quad \text{and} \quad \varphi_{\mathbf{k}}^{(d)} \leq 0 \right\} \\
&= \mathrm{MBZ} \setminus \mathrm{NBZ} \\
\mathcal{C} &\equiv \left\{ \mathbf{k} \in \mathrm{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \leq 0 \quad \text{and} \quad \varphi_{\mathbf{k}}^{(d)} \geq 0 \right\} \\
&= \mathrm{NBZ} \setminus \mathrm{MBZ}
\end{aligned}$$

and will be used later on. Whatever the shape of the mixed band, the following equation always holds

$$|k_x| = \frac{\pi}{2} \quad \text{and} \quad |k_y| = \frac{\pi}{2} \qquad \Longrightarrow \qquad \tilde{\epsilon}_{(k_x,k_y)} = 0$$

since only cosines are involved. This identifies four node points, represented as black dots in Fig. 3.4.

3.5.1 Constraints in presence of a robust C_4 symmetry and bands shrinking

Consider the simpler situation where C_4 symmetry is imposed. This implies that the renormalised bands are

$$\tilde{\epsilon}_{\mathbf{k}} \equiv (u^{(s^*)}V - 2t)(\cos k_x + \cos k_y)$$

with a single self-consistency equation

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

The following chain of deductions leads us to a constraint on the possible values of $u^{(s^*)}$:

1. For $t > 0$, there is no self-consistent solution admitting $u^{(s^*)} \leq 0$. Let us prove this: suppose instead $u^{(s^*)} \leq 0$. For any function $g_{\mathbf{k}}$ defined on the BZ:

$$\sum_{\mathbf{k}\in\mathrm{BZ}} g_{\mathbf{k}} = \sum_{\mathbf{k}\in\mathrm{MBZ}} [g_{\mathbf{k}} + g_{\mathbf{k}+\boldsymbol{\pi}}]$$

where we defined the vector $\boldsymbol{\pi} \equiv (\pi, \pi)$. The self-consistency equation thus reads

$$\begin{aligned} u^{(s^*)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) + f(\tilde{\epsilon}_{\mathbf{k}+\boldsymbol{\pi}}; \beta, \tilde{\mu}) \right] \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right] \end{aligned} \tag{3.23}$$

having we used the antisymmetry of s^*-wave functions by $\boldsymbol{\pi}$ shifts. Now, if $u^{(s^*)} \leq 0$ the renormalised band $\tilde{\epsilon}_{\mathbf{k}}$ is negative inside the MBZ and positive outside, implying

$$\forall \mathbf{k} \in \mathrm{MBZ} \quad : \quad f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \geq f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

the equality only being satisfied on the MBZ contour. Then, since inside the MBZ it holds by definition $\cos k_x + \cos k_y \geq 0$, the above equations imply

$$u^{(s^*)} \leq 0 \quad \Longrightarrow \quad u^{(s^*)} \geq 0$$

To exclude the possibility of $u^{(s^*)} = 0$, it suffices to notice that from the argument above such a situation only arises when the MBZ contour degenerates to the entirety of the MBZ itself – which is, when the band is flat. But that cannot be the case for $u^{(s^*)} = 0$, as long as $t > 0$. Thus we have proven that

$$u^{(s^*)} > 0$$

2. For $t > 0$, there is no self-consistent solution admitting $u^{(s^*)} \geq 2t/V$. We proceed exactly as did for the previous point, up to the passage:

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right]$$

Now, if $u^{(s^*)} \geq 2t/V$ this means the prefactor in front of the renormalised bands is positive. Then $\tilde{\epsilon}_{\mathbf{k}}$ is positive inside the MBZ and negative outside, implying

$$\forall \mathbf{k} \in \mathrm{MBZ} \quad : \quad f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \leq f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

thus we have

$$u^{(s^*)} \geq 2t/V \quad \Longrightarrow \quad u^{(s^*)} \leq 0$$

which is absurd. Thus

$$u^{(s^*)} < 2t/V$$

Collecting the two points above, we have the constraint on $u^{(s^*)}$

$$\mathrm{C}_4 \quad \Longrightarrow \quad 0 < u^{(s^*)} < 2t/V \tag{3.24}$$

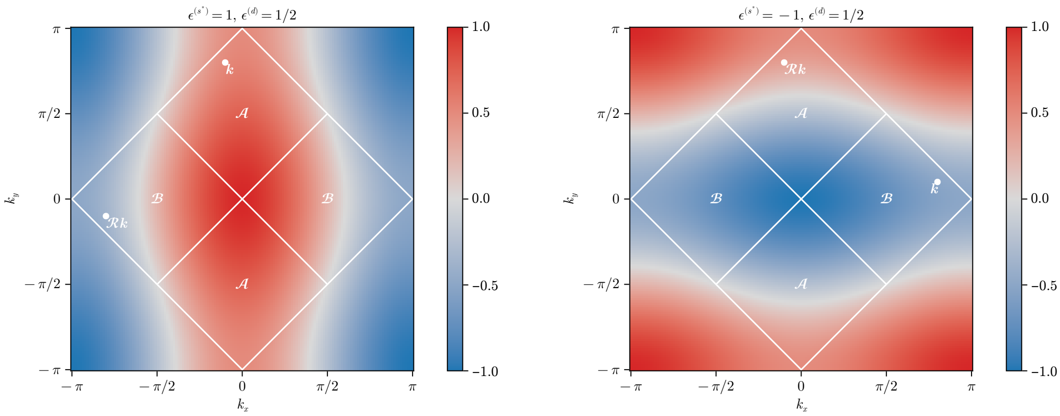
The effective hopping is smaller than the original hopping. We can denominate this net effect "bands shrinking".

3.5.2 Natural suppression of nematicity within $\mathrm{C}_2/\mathbb{Z}_2$ symmetry

Let us extend the arguments above to a weaker symmetry, allowing for

$$\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2/\mathbb{Z}_2$$

We work with the full, anisotropic band of Eq. (3.17). From the following chain of deductions we conclude that C_4 is not broken:



a | Same-sign mixed bands.

b | Opposite-sign mixed bands.

Figure 3.5 | Theoretical example of bands with $u^{(s^*)} > 2t/V$, $u^{(d)} > 0$ (top-left) and $u^{(s^*)} < 0$, $u^{(d)} > 0$ (top right). Two example points are reported, connected by a $\pi/2$ rotation as given in Eq. (3.25).

1. A pure d-wave band is impossible to realise at any finite $t > 0$: [ERROR IN THIS PROOF] if the band is purely d-wave symmetric, then it must be $u^{(s^*)} = 2t/V$. It follows that Fermi-Dirac distribution cannot admit s^*-wave components,

$$\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) = 0$$

thus it must be $u^{(s^*)} = 0$. We have

$$u^{(s^*)} = 2t/V \quad \Longrightarrow \quad u^{(s^*)} = 0$$

which is absurd at finite values. A purely d-wave band is impossible.

2. Let for instance

$$u^{(s^*)} > 2t/V > 0 \qquad u^{(d)} > 0$$

The resulting band is analogous to the ones of Fig. 3.3. Within this structure and using the notation of Fig. 3.4a, it always holds

$$\forall \mathbf{k} \in \mathcal{A} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} \geq 0$$

Recollecting Eq. (3.23) which is identically written as

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathcal{A}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right] + \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathcal{B}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right]$$

The sum with $\mathcal{B}$ requires a little care: in fact, as is evident from the bottom-left panel of Fig. 3.3, such a region contains nodal curves across which bands change sign. Let $\mathcal{R}$ be the clockwise $\pi/2$ rotation: then, any point in $\mathcal{B}$ can be seen as the rotation of a point in $\mathcal{A}$

$$\forall \mathbf{k}' \in \mathcal{B} \quad \exists \quad \mathbf{k} \in \mathcal{A} \mid \mathbf{k}' = \mathcal{R}\mathbf{k} \tag{3.25}$$

A graphical rendition of this idea is given in Fig. 3.5a.

Now: both $\epsilon^{(s^*)}$ and $\epsilon^{(d)}$ are positive by construction, then it holds

$$\forall \mathbf{k} \in \mathcal{A} \quad : \quad |\tilde{\epsilon}_{\mathbf{k}}| \geq |\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}|$$

which a trivial consequence of the fact that $|a-b| \leq |a+b|$ for positive a, b and

$$\forall \mathbf{k} \in \mathcal{A} \quad : \quad \begin{aligned} \tilde{\epsilon}_{\mathbf{k}} &= \overbrace{\epsilon^{(s^*)} \left(\cos k_x + \cos k_y\right)}^{\geq 0} + \overbrace{\epsilon^{(d)} \left(\cos k_x - \cos k_y\right)}^{\geq 0} \\ \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} &= \underbrace{\epsilon^{(s^*)} \left(\cos k_x + \cos k_y\right)}_{\geq 0} - \underbrace{\epsilon^{(d)} \left(\cos k_x - \cos k_y\right)}_{\geq 0} \end{aligned}$$

Now, the Fermi-Dirac distribution is monotonically descendent: this means that

$$\begin{aligned} -\tilde{\epsilon}_{\mathbf{k}} \leq \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} \quad &\Longrightarrow \quad f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \geq f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \\ -\tilde{\epsilon}_{\mathcal{R}\mathbf{k}} \leq \tilde{\epsilon}_{\mathbf{k}} \quad &\Longrightarrow \quad f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \geq f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \end{aligned}$$

thus we have

$$\begin{aligned} u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathcal{A}} &(\cos k_x + \cos k_y) \\ &\times \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) + f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \right] \end{aligned}$$

which leads to an absurd,

$$u^{(s^*)} > 2t/V \quad \Longrightarrow \quad u^{(s^*)} \leq 0$$

3. An identical argument, exchanging the roles of $\mathcal{A}$ and $\mathcal{B}$, can be carried out to exclude the possibility of
$$u^{(s^*)} > 2t/V > 0 \qquad u^{(d)} < 0$$
thus we can self-consistently claim $u^{(s^*)} < 2t/V$.

4. Let now
$$u^{(s^*)} < 0 \qquad u^{(d)} > 0$$
Within this structure, it always holds
$$\forall \mathbf{k} \in \mathcal{B} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} \geq 0$$
Proceed exactly as before, with identical meaning of the notation and inverted roles for $\mathcal{A}$ and $\mathcal{B}$, as drawn in Fig. 3.5b. First, we have
$$\forall \mathbf{k} \in \mathcal{B} \quad : \quad |\tilde{\epsilon}_{\mathbf{k}}| \leq |\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}|$$
which a trivial consequence of the fact that $|a-b| \geq |a+b|$ for $a \leq 0$ and $b \geq 0$, and
$$\forall \mathbf{k} \in \mathcal{B} \quad : \quad \begin{aligned} \tilde{\epsilon}_{\mathbf{k}} &= \overbrace{\epsilon^{(s^*)} \left(\cos k_x + \cos k_y\right)}^{\geq 0} + \overbrace{\epsilon^{(d)} \left(\cos k_x - \cos k_y\right)}^{\leq 0} \\ \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} &= \underbrace{\epsilon^{(s^*)} \left(\cos k_x + \cos k_y\right)}_{\geq 0} - \underbrace{\epsilon^{(d)} \left(\cos k_x - \cos k_y\right)}_{\leq 0} \end{aligned}$$
By repeating identical steps as before we end up with the implication
$$u^{(s^*)} < 0 \quad \Longrightarrow \quad u^{(s^*)} \geq 0$$
which is absurd.

5. An identical argument, exchanging the roles of $\mathcal{A}$ and $\mathcal{B}$, can be carried out to exclude the possibility of
$$u^{(s^*)} < 0 \qquad u^{(d)} < 0$$

From the chain of derivations above, we can conclude self-consistently that also in the nematic structure the constraint of Eq. (3.24) needs to be fulfilled. In the s^*-wave channel, partial breaking of rotational symmetry is not enough to modify the main effect identified under C_4 symmetry, "bands shrinking". What is missing is the behaviour of the d-wave channel when said effect takes place. Continuing the enumeration above,

6. Let $0 < u^{(s^*)} < 2t/V$ and $u^{(d)} > 0$. The self-consistency equation for $u^{(d)}$ reads

$$\begin{aligned} u^{(d)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} (\cos k_x - \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{NBZ}} (\cos k_x - \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \right] \end{aligned}$$

where we recognized, as is shown in Fig. 3.4b, that $\mathrm{BZ} \setminus \mathrm{NBZ} = \mathcal{R}\mathrm{NBZ}$. It holds:

$$\forall \mathbf{k} \in \mathrm{NBZ} \quad : \quad \begin{aligned} \tilde{\epsilon}_{\mathbf{k}} &= \epsilon^{(s^*)} (\cos k_x + \cos k_y) + \overbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}^{\geq 0} \\ \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} &= \epsilon^{(s^*)} (\cos k_x + \cos k_y) - \underbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}_{\geq 0} \end{aligned}$$

thus $\tilde{\epsilon}_{\mathcal{R}\mathbf{k}} < \tilde{\epsilon}_{\mathbf{k}}$, and inserting this fact in the self-consistency equation above we get

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} > 0 \quad \Longrightarrow \quad u^{(d)} < 0$$

which, as always, is absurd.

7. An identical argument, with inverted inequalities, can be carried out to exclude the possibility of

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} < 0 \quad \Longrightarrow \quad u^{(d)} > 0$$

These last two facts were the missing informations. Importantly, self-consistency excludes the possibility of a $C_4 \to C_2$ SSB in the normal phase, giving the constraint

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} = 0 \tag{3.26}$$

A consequence of this is that there is no need at all to bother using $u^{(d)}$ as an HFP. This is physically expected but non trivial.

At this point, bands shrinking is a guaranteed effect, and the point to be asked is where the converged value of $u^{(s^*)}$ will be positioned in its admissibility range, based on the model parameter. This question is the subject of next sections.

3.5.3 Expected dependence on δ at low temperature

By looking once again at the self-consistency equation, we can infer roughly the physical expectation for $u^{(s^*)}$ when the system is subject to electronic doping. Consider first the low-temperature limit, namely $\beta \gg 1$. Consider the self-consistency equation of Tab. 3.2 seen as a functional of temperature and chemical potential,

$$u^{(s^*)}[\beta, \tilde{\mu}] = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

in the zero-temperature limit, as long as the shrunk band is not flat, it holds

$$\lim_{\beta \to \infty} u^{(s^*)}[\beta, \tilde{\mu}] = \frac{1}{L_x L_y} \sum_{\epsilon_{\mathbf{k}} \leq \tilde{\mu}} (\cos k_x + \cos k_y)$$

Now, at low temperature the Fermi-Dirac slope becomes less steep, a fact that makes the above equation only approximately verified as long as the convergence value of $u^{(s^*)}$ results in a "not-so-strong" bands shrinking.

Let us assume the latter, which means we assume it holds

$$u^{(s^*)}[\beta,\tilde{\mu}]\big|_{\beta\gg 1} \simeq \frac{1}{L_x L_y}\sum_{\epsilon_{\mathbf{k}}\leq\tilde{\mu}}(\cos k_x + \cos k_y)$$

which, at this point, will be true only in an unspecified region of parameters space. Of course, the "not-so-strong" shrinking assumption implies the fact that a FS is well-defined, being the band non-flat. At half filling, because of PHS,

$$\delta = 0 \qquad \Longrightarrow \qquad \mathrm{FS} = \partial\mathrm{MBZ}$$

With $\partial\Gamma$ we indicate the contour of the simply connected domain Γ. At $\delta > 0$ the FS expands over the MBZ boundary, where the form factor $\cos k_x + \cos k_y$ becomes negative. This gives the self-consistency equation

$$u^{(s^*)}[\beta,\tilde{\mu}]\big|_{\beta\gg 1} \simeq \frac{1}{L_x L_y}\sum_{\epsilon_{\mathbf{k}}\leq 0}|\cos k_x + \cos k_y| - \frac{1}{L_x L_y}\sum_{0<\epsilon_{\mathbf{k}}\leq\tilde{\mu}}|\cos k_x + \cos k_y|$$

which, being the sum of a positive and a negative term, is maximized whenever $\tilde{\mu} = 0$, at half filling.

This holds as long as the effective hopping is non-zero, a condition implying together

$$\delta_1 > \delta_2 \qquad \Longrightarrow \qquad \mathrm{FS}_1 \text{ encircles } \mathrm{FS}_2$$

with obvious meaning of the notation. Using this property we conclude

$$\tilde{t} > 0 \quad \text{and} \quad \delta_1 > \delta_2 \qquad \Longrightarrow \qquad \lim_{\beta\to\infty} u^{(s^*)}[\beta,\tilde{\mu}]\Big|_{\delta_1} < \lim_{\beta\to\infty} u^{(s^*)}[\beta,\tilde{\mu}]\Big|_{\delta_2}$$

This is the first expected behaviour of the self-consistent functional $u^{(s^*)}$: if there exists a region of parametric space where the shrinking effect is "not-so-strong" to rise the doping level must lower $u^{(s^*)}$. Such a region must exist: at least all the segment $(V = 0, \delta)$ is included, since on said region of parametric space the model is just the common free electronic model.

Following this line of reasoning, we can extract easily the two limits of $u^{(s^*)}$ at half-filling and unitary filling.

1. At half-filling, the self-consistency equation reduces to

$$u^{(s^*)}[\beta,0]\big|_{\beta\gg 1} \simeq \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}(\cos k_x + \cos k_y)$$

   In thermodynamic limit, performing integration on the MBZ shown in the top-left panel of Fig. 3.3:

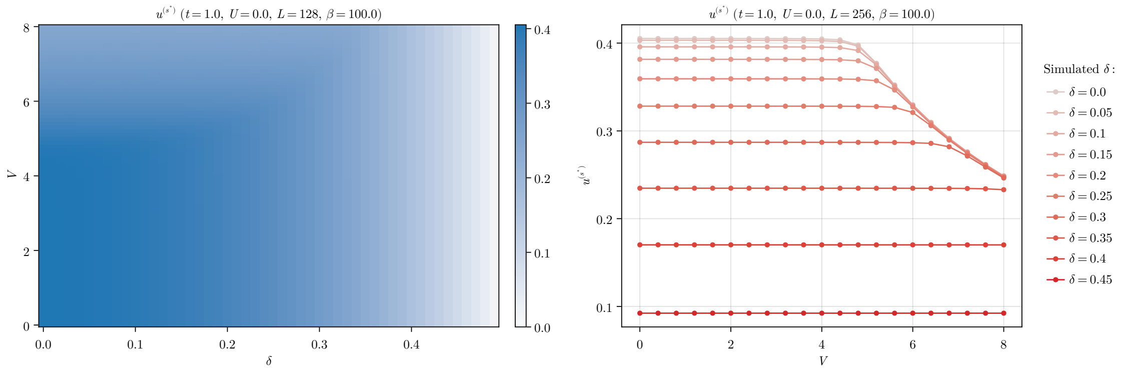
$$\begin{aligned}
u^{(s^*)}[\beta,0]\big|_{\beta\gg 1} &\simeq \frac{1}{(2\pi)^2}\int_{-\pi}^{\pi}\mathrm{d}k_y\int_{-\pi+|k_y|}^{\pi-|k_y|}\mathrm{d}k_x\cos k_x + \frac{1}{(2\pi)^2}\int_{-\pi}^{\pi}\mathrm{d}k_x\int_{-\pi+|k_x|}^{\pi-|k_x|}\mathrm{d}k_y\cos k_y \\
&= \frac{1}{2\pi^2}\int_{-\pi}^{0}\mathrm{d}k_y\int_{-\pi-k_y}^{\pi+k_y}\mathrm{d}k_x\cos k_x + \frac{1}{2\pi^2}\int_{0}^{\pi}\mathrm{d}k_y\int_{-\pi+k_y}^{\pi-k_y}\mathrm{d}k_x\cos k_x \\
&= \frac{1}{\pi^2}\int_{-\pi}^{0}\mathrm{d}k_y\sin(\pi+k_y) + \frac{1}{\pi^2}\int_{0}^{\pi}\mathrm{d}k_y\sin(\pi-k_y) \\
&= \left(\frac{2}{\pi}\right)^2 \approx 0.405
\end{aligned} \tag{3.27}$$

2. At unitary filling we have

$$u^{(s^*)}\left[\beta,2\tilde{t}^{(s^*)}\right]\Big|_{\beta\gg 1} = \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}(\cos k_x + \cos k_y) = 0$$

   which vanishes due to the structure factor periodicity by $\boldsymbol{\pi}$.

To summarize: around $V = 0$, $u^{(s^*)}$ is expected to descend monotonically from $4/\pi^2$ to 0 at increasing doping.



a | Heatmap of $u^{(s^*)}$ in δV plane. b | V-dependency of $u^{(s^*)}$ varying δ.

Figure 3.6 | Plot of the self-consistent convergence values for the HFP $u^{(s^*)}$ in the normal phase. To extract the data on the right side coarse-graining precision of the left-hand side data has been exchanged with higher lattice dimensions. As is evident, based on doping level $u^{(s^*)}$ exhibits non-trivial regions of V-independence.

3.6 Discussion of numeric results

In this section we present and discuss the numeric results, the physical relevance of the found phenomena and the boundaries of the algorithm. Let us start by sketching the algorithm used to determine self-consistently the mean-field structure of the normal phase in parameter space. For the normal phase, the HFPs "vector" defined in Sec. 1.3.2 has just one component:

$$\mathbf{v} = \left[u^{(s^*)}\right]$$

As before, we treat spin as a pure degeneracy. The only HFP is characterized by the first self-consistency equation of Tab. 3.2. The resulting model has two RMPs:

$$\tilde{t}_{ij} \equiv \left(t - \frac{u^{(s^*)}V}{2}\right)\varphi_{ij}^{(s^*)}$$
$$\tilde{\mu} \equiv \mu + n(2zV - U)$$

the second one being only dependent on the model parameters and not on HFPs. Finally, the phase free energy is given by Eq. (3.22). Evidently the normal phase is completely independent of U, since the local repulsion cannot develop any RWC other than those renormalizing the chemical potential. Because of this we present the results varying the other parameters, keeping everywhere $U = 0$. We ran simulations in the (V, δ) plane, setting up the following general parameters:

$$U = 0 \qquad \Delta n = 10^{-2} \qquad \Delta\mathbf{v} = \left[u^{(s^*)} : 5 \times 10^{-4}\right]$$

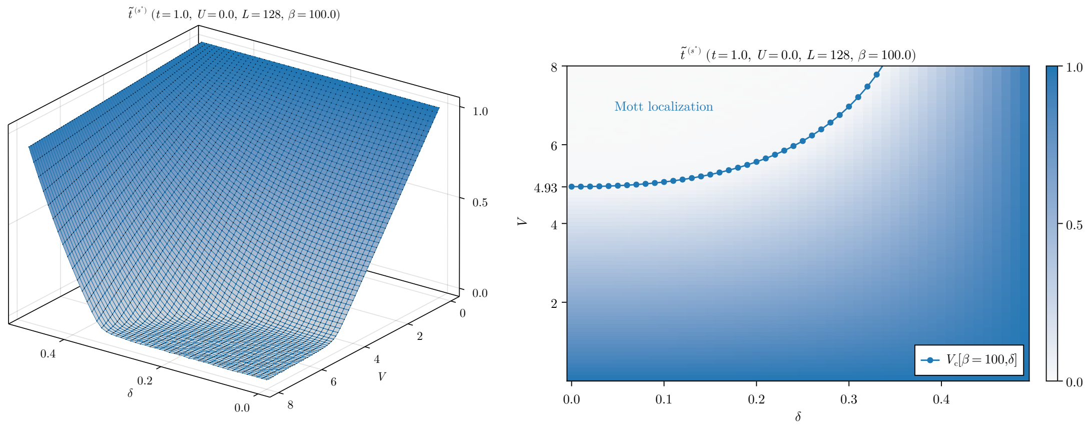
with $\Delta\mathbf{v}$ the tolerance vector defined as in Sec. 1.3.2.

In Fig. 3.6a are reported the results obtained for the HFP $u^{(s^*)}$ on a 128×128 lattice and "low" temperature $\beta = 100$, varying the non-local attraction and the doping level

$$V = 0,\ 0.1,\ 0.2,\ \ldots,\ 4.0 \qquad \delta = 0,\ 0.01,\ 0.02,\ \ldots,\ 0.49$$

The expectations of Sec. 3.5.3 are satisfied: at low V (indicatively $V \lesssim 5$) the trend is monotonically descendent at increasing doping. As V gets bigger, however, $u^{(s^*)}$ starts decreasing in the top-left corner of Fig. 3.6a. This effect is more visible in Fig. 3.6b, where we doubled L and ran simulations with a smaller δ resolution: the convergence value of $u^{(s^*)}$ is approximately independent of V up to some point, where this behaviour is broken. Note that, as was expected from the discussion of Sec. 3.5.3, the half-filled convergence value of $u^{(s^*)}$ in the V-independent region is approximately given by Eq. (3.27).

This effect can only be interpreted in one way, as we discuss in next section: $u^{(s^*)}$ maintains a constant value at fixed doping, as is described in Sec. 3.5.3, as long as $\tilde{t}^{(s^*)} > 0$. When V gets large enough, $u^{(s^*)}$ starts decreasing because of its constraints. The net effect is the nullification of $\tilde{t}^{(s^*)}$, which is, a total localization effect.



a | Simulated values of $\tilde{t}^{(s^*)}$

b | Critical Mott boundary.

Figure 3.7 | Plot of the self-consistent convergence values for the HFP $u^{(s^*)}$ in the normal phase and the Mott-boundary V_c, computed as is described in text. In the right panel, the tick $V = 4.93$ is specified explicitly as a graphic confirm of the computation of Eq. (3.28). For graphic reasons the points at $V_c > 8$ have been omitted.

3.6.1 Mott localization effect

As is evident from Fig. 3.6b, at low V the convergence value of $u^{(s^*)}$ is approximately V-independent at sufficiently low temperatures. Let $V_c[\beta, \delta]$ be the critical value such that

$$\forall V \mid 0 \leq V < V_c[\beta, \delta] \quad \implies \quad u^{(s^*)} \text{ is } V \text{ independent.}$$

Then within the entire segment $[0, V_c]$ the convergence value is the same obtained for $V = 0$, hence we can define in first instance the critical V as the value such that the hopping correction is maximized,

$$V_c[\beta, \delta] \equiv \frac{2t}{u^{(s^*)}[\beta, \tilde{\mu}(\delta)]\big|_{V=0}}$$

When V becomes larger than this critical value, the resulting RMP $\tilde{t}^{(s^*)}$ drops to 0 and due to the constraint of Eq. (3.24) the convergence value of $u^{(s^*)}$ is forced to decrease. Note that the constraint of Eq. (3.24) does not prevent $u^{(s^*)}$ from saturating to its upper boundary from below. In Fig. 3.7a the converged values of $\tilde{t}^{(s^*)}$, extracted immediately from the same data of Fig. 3.6a, are plotted in parametric space.

While to extract the value of $V_c[\beta, \delta]$ is a non-trivial analytic problem due to the insurgence of elliptic integrals and the smearing effects of temperature, for the high-β half-filled case it is immediate from Eq. (3.27) to see that

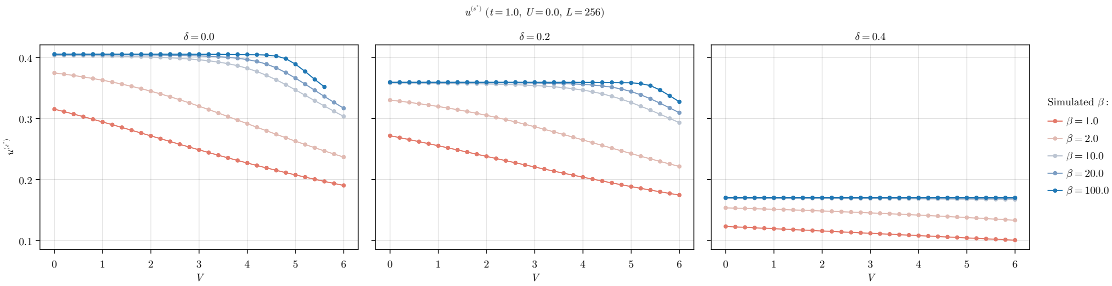
$$V_c[\beta \gg 1, 0] = \frac{2t}{(2/\pi)^2} \approx 4.93t \tag{3.28}$$

which makes the real interesting point of this section and the whole bands-shrinking effect. At half-filling, the effective bands amplitude reduction is so strong that already around relatively small values of V (compared to t) electronic kinetics is nullified. In Fig. 3.7b we report, superimposed, the convergence values of the RMP $\tilde{t}^{(s^*)}$ and $V_c[\beta = 100, \delta]$ computed with the simple procedure described. As is evident from the plot itself, even if a little rough, our prediction of the critical value for V gives reasonably a sort of phase boundary delimiting a region where bands are flattened and electrons are localized.

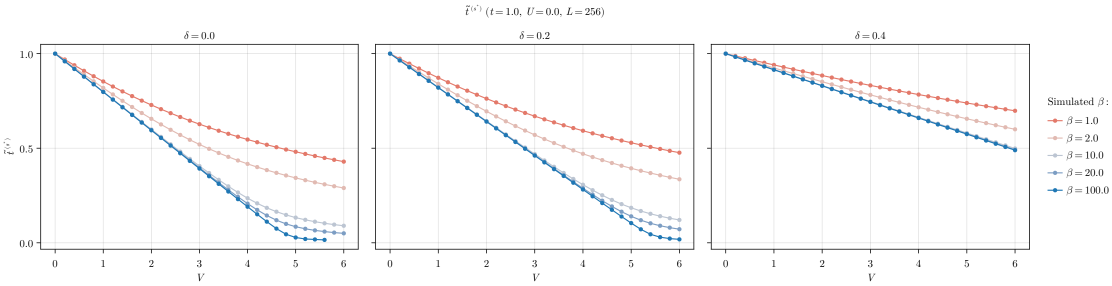
This phenomenon is largely studied in solid-state physics and falls within the large class of Mott physics. [Blabla about Mott physics...]

Increasing temperature, predictably, Mott localization is suppressed. In Fig. 3.8a we report results for the HFP $u^{(s^*)}$ computed at decreasingly lower temperature and three example doping levels,

$$\beta = 1.0, 2.0, 10.0, 20.0, 100.0 \qquad \delta = 0.0, 0.2, 0.4$$



a | Temperature dependence of $u^{(s^*)}$ plots at various dopings.



b | Temperature dependence of $\tilde{t}^{(s^*)}$ plots at various dopings.

Figure 3.8 | Comparative plots of the HFP $u^{(s^*)}$ (top panel) and RMP $\tilde{t}^{(s^*)}$ (bottom panel) in null-doped (left plots), "medium-doped" ($\delta = 0.2$, central plots) and "heavily-doped" ($\delta = 0.4$, right plots) contexts. Increasing temperature progressively destroys Mott localization, reducing the bands shrinking effect.

[Evaluate inserting more temperatures.] At high temperature the V-independence region completely disappears, making $u^{(s^*)}$ a monotonically descendent function. This is no surprise, since all our derivation was performed at low temperature. In Fig. 3.8a the same dataset is plotted with focus on the RMP $\tilde{t}^{(s^*)}$. Temperature increase disrupts localization effects, at least at small values of V. Interestingly enough, however, the bands shrinking effect is still relevant at $\beta = 1.0$, the hopping parameter approximately halving at $V \approx 5 \div 6$.

3.6.2 Solution free energy and entropy densities

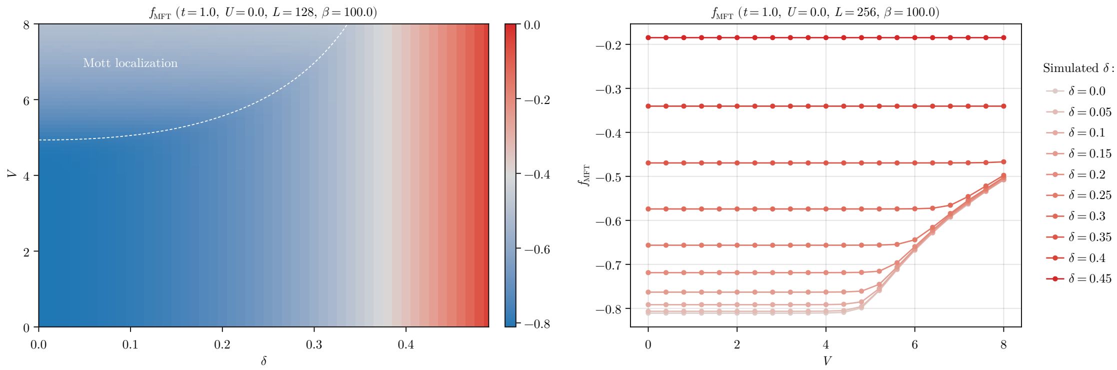
The plots of Fig. 3.9 report our findings on the Normal phase free energy, computed based on Eq. (3.22). Doping induced energy increase is simply due to the simultaneous entropy decrease and band filling (the first being a consequence of the second, because less and less room is available for electron scattering). By looking the data of Fig. 3.9a, at fixed δ free energy density remains pretty much constant up to the Mott boundary. This is simply because as long as a band with non-zero amplitude is present, the HFP correction of Eq. (3.22) compensates the free energy increase due to bands shrinking,

$$
\begin{aligned}
-V\left|u^{(s^*)}\right|^2 &= -\frac{V u^{(s^*)*}}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} \left(\epsilon_{\mathbf{k}} - \tilde{\epsilon}_{\mathbf{k}}\right) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})
\end{aligned}
$$

having we used in the second passage

$$
\tilde{\epsilon}_{\mathbf{k}} = \tilde{\epsilon}_{\mathbf{k}} + u^{(s^*)*} V (\cos k_x + \cos k_y)
$$

Recalling the first point of the derivation of Sec. 3.4, to add said correction to free energy is the same as removing the *tilde* sign from single-state energies, thus using the free model ($V = 0$) free energy density expression.



a | Heatmap of f_{MFT} in δV plane.   b | V-dependency of f_{MFT} varying δ.

Figure 3.9 | Free energy density of the normal phase. On the left-side panel, the dashed white line has the same meaning as the dotted line of Fig. 3.7b, indicating the Mott transition.

From Fig. 3.9b, energy cost of Mott localization is made evident. The lower the doping, the higher the energetic cost for total bands shrinking. At low doping levels, hopping renormalization is the largest, as discussed in previous sections; at low temperatures the band is full below $\tilde{\mu}$, thus by shrinking it we just "move up" single-state energies, but as long as a FS is well-defined (which is, as long as the band is not flattened enough) the Fermi sea is essentially the same. Including the aforementioned correction, correctly, gives a total free energy density that is indistinguishable from the one of the Normal phase for the HM. On a flat band a FS is not well defined, and because of this energy must rise: in the Mott-localized region the Fermi-Dirac factors of Eq. (3.22) degenerate to

$$\text{Mott localized} \quad \Longrightarrow \quad \lim_{\beta\to\infty} f(0;\beta,\tilde{\mu}(\delta)) = \frac{1}{2}$$

since for the chemical potential $\tilde{\mu}(\delta) = 0$. From a free, single band Fermi gas the contribution to entropy given by a state with occupation f is [1]

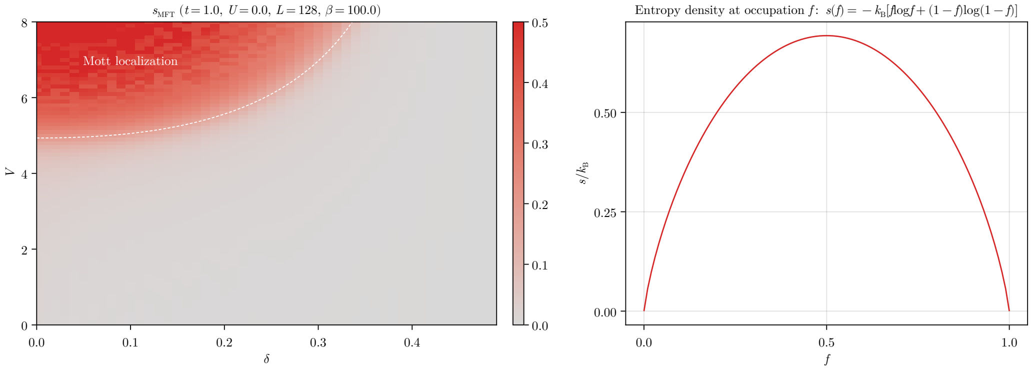
$$s(f) = -k_{\mathrm{B}}\left[f\log f + (1-f)\log(1-f)\right]$$

represented in Fig. 3.10b. The maximum is precisely located at $f = 1/2$, which from the limit above we know is the limiting expression for the Fermi-Dirac population of each single-particle state for a flat band. This implies that the key free energy non-trivial contribution is entropic, represented in Fig. 3.10a. Correctly entropy is low all around the Mott insulating phase, which is coherent with a normal Fermi gas at low temperature, for which the FS interface (the region where the Fermi-Dirac factor varies continuously from ~ 1, inside, to ~ 0, outside) is narrow. For said model almost all single-particle state are approximately full or empty, which as is represented in Fig. 3.10b are two symmetric low-entropy conditions. The Mott insulating region can be seen as a degeneration of the FS interface to the entire BZ, a process that maximises entropy.

All of this holds, as all the rest, at low temperature: by lowering β, finite-temperature effects become more and more relevant and the statements above need to be corrected.

3.6.3 Convergence problems and mixing fine-tuning

Although extremely simple and characterised by a single HFP, this framework is ideal for us to describe some of the typical convergence problems arising in HF iterative schemes anticipated in Sec. 1.3.2. Bands flattening is a source of algorithmic instability: numeric fluctuations can lead to values of $u^{(s^*)}$, during iterations, slightly out of bounds with respect to Eq. (3.24). Even if the numeric effect is small, the consequence can be vast. To heal this effect and extract converged values, it is necessary to pay a little care with the choice of the mixing parameter $g \in [0,1]$. We defined it in Sec. 1.3.2, and operatively in Eq. (1.21). This parameter controls deterministically how, from an initializer (the HFPs vector we plug in the algorithm) and an output (the HFPs vector we get back from the algorithm) the new initializer for the following step is generated. As a general rule, it is pretty safe to use $g \approx 0.5$; this rule of thumb, however, breaks in situations



a | Heatmap of s_{MFT} in δV plane. b | V-dependency of f_{MFT} varying δ.

Figure 3.10 | Entropy density of the normal phase (left panel) and theoretical contribution to entropy for a state with occupation f. On the left-side panel, the dashed white line has the same meaning as the dotted line of Fig. 3.7b, indicating the Mott transition. [Perhaps use a lighter gray in heatmap.]

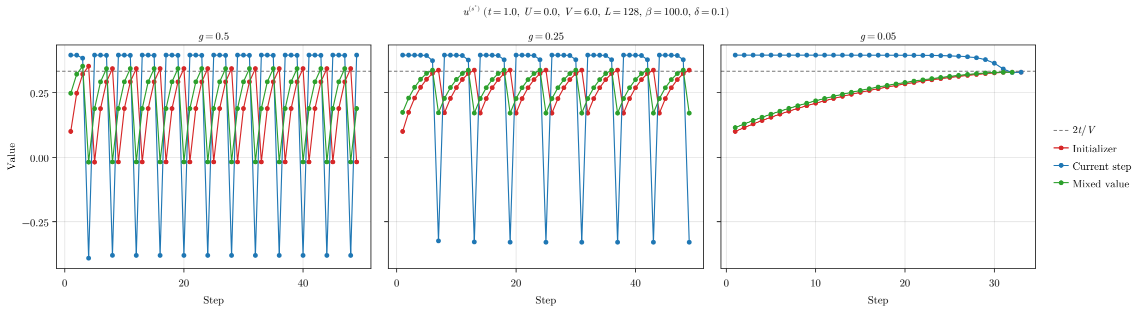


Figure 3.11 | Comparison of three example tracks for the single-valued HFPs vector of the normal phase, simulated at a "Mott-deep" point.

like the one plotted comparatively in Fig. 3.11, representing the evolutions at different mixing parameters of a simulation at

$$V = 6.0 \qquad \delta = 0.1 \qquad \beta = 100.0$$

As can be checked in Fig. 3.6a this is a point well inside the Mott region. Localization, as we see in Fig. 3.7, is already pretty intense and to sustain bands flattening $u^{(s^*)}$ needs to converge at $2t/V = t/3$. Let us follow the evolution of the algorithm starting from the arbitrary initializer

$$\text{Step 1} \quad : \quad u^{(s^*)} = 0.1$$

In each panel of Fig. 3.11, the red track represents the value of $u^{(s^*)}$ we plug in the self-consistency equations, the blue one the result, the green one the mixed value of the two based on different gs. The gray dashed line is the theoretical convergence target. In the first two panels, there is an evident infinite loop occurring whenever the green line crosses the threshold $2t/V$. As we know the generated green point is out of bounds with respect to Eq. (3.24). When this happens, the next red point (initializer of following step) gives a band with inverted sign, a feature that "breaks" the self-consistency equation and generates a blue point with no physical significance.

To overcome this problem it is necessary to move "slowly". As is shown in the right panel, the choice of a small enough mixing parameter ensures convergence, implementing the idea that – within a given tolerance – even if the blue line is systematically above the convergence value (in the region of band sign inversion) our mixing can be made careful enough to make up for this, and generate a green track systematically below the convergence value. This kind of problems can turn out to be pretty common for iterative HF schemes, and to understand how to knob the algorithmic parameters without sacrificing accurateness and precision, as well as avoiding computationally costly and frustrating infinite loops, was an essential part of this algorithm design.

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